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OPTIMIZATION

Metaheuristic optimization algorithms: a comprehensive overview and classification of benchmark test functions

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Abstract
This review aims to exploit a study on different benchmark test functions used to evaluate the performance of Meta-Heuristic (MH) optimization techniques. The performance of the MH optimization techniques is evaluated with the different sets of mathematical benchmark test functions and various real-world engineering design problems. These benchmark test functions can help to identify the strengths and weaknesses of newly proposed MH optimization techniques. This review paper presents 215 mathematical test functions, including mathematical equations, characteristics, search space and global minima of the objective function and 57 real-world engineering design problems, including mathematical equations, constraints, and boundary conditions of the objective functions carried out from the literature. The MATLAB code references for mathematical benchmark test functions and real-world design problems, including the Congress of Evolutionary Computation (CEC) and Genetic and Evolutionary Computation Conference (GECCO) test suite, are presented in this paper. Also, the winners of CEC are highlighted with their reference papers. This paper also comprehensively reviews the literature related to benchmark test functions and real-world engineering design challenges using a bibliometric approach. This bibliometric analysis aims to analyze the number of publications, prolific authors, academic institutions, and country contributions to assess the field's growth and development. This paper will inspire researchers to innovate effective approaches for handling inequality and equality constraints.

Keywords Benchmark test functions · Real-world engineering design problems · Metaheuristic optimization techniques · MATLAB codes

1 Introduction

The Benchmark test function plays a vital role in the field of optimization techniques. These functions, often referred to as test functions or test suites in the existing literature, play a vital role in evaluating and validating the performance of optimization techniques. These benchmark functions are essentially artificial optimization problems with known solutions designed for the purpose of assessing optimization algorithm performance. The literature suggests that sets of benchmark test functions are utilized to validate and compare various optimization techniques. The underlying assumption is that an optimization technique demonstrating high performance on benchmark problems is likely to show similar effectiveness in real-world scenarios. Furthermore, investigating the effectiveness of new or hybrid algorithms, or a combination of algorithms, across a range of challenging scenarios with diverse characteristics allows designers to gain a comprehensive understanding of the strengths and limitations of the algorithm (Jamil et al. 2013).

The significance of benchmark test functions stems from multiple factors. First, they provide a means to facilitate the comparison of the effectiveness of various optimization techniques. Employing the same method across a standardized set of benchmark functions enables the evaluation of performance across diverse problem types, aiding in algorithm selection for specific applications. Second, benchmark test functions serve as a tool for assessing the performance of novel optimization algorithms. When a new algorithm is developed, subject to the multiple

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benchmark functions, it allows for performance assessment and potential algorithm enhancement or determination of its suitability for specific applications.

Additionally, using benchmark test functions allows for the analysis of performance characteristics of optimization algorithms. Running an algorithm on a standardized set of evaluation-focused functions offers insight into how the algorithm operates and reaches optimal solutions (Function 2014, 2021). Various research works have been proposed to analyze the performance of the newly presented algorithm based on different benchmark functions. In Civicioglu (2012), the author developed an MH optimization algorithm (Differential Search Algorithm), which has been compared and evaluated with 40 benchmark test functions. In Zheng (2015), the author presented water wave optimization, which can be compared and evaluated with 30 benchmark test functions. The vortex search algorithm (Doğan and Ölmez 2015) has been proposed by Berat Doğan, and Tamer Olmez, which can be compared and evaluated with 50 well-known mathematical benchmark test functions. In Baykasoğlu and Akpinar (2017), the author proposed the Weighted Superposition Attraction algorithm, tested with 71 well-known mathematical benchmark test functions. Seyedali Mirjalili has proposed an algorithm (Mirjalili 2015a) which has evaluated and compared with 29 well-known benchmark test functions and 9 real-world design problems. In Savsani and Savsani (2016), the author proposed an algorithm technique which has evaluated and compared with 13 real-life engineering design problems and 13 benchmark test functions. Water Evaporation Optimization (Kaveh and Bakhshpoori 2016) has been evaluated with 40 mathematical benchmark test functions and 3 real-world engineering design problems. Dynamic Virtual Bats Algorithm (Topal and Altun 2016) has also been evaluated with 30 mathematical test functions. In Meng and Pan (2016), the author proposed the Monkey King Evolutionary algorithm, which can be tested and compared with the 28 test functions. The optimal Foraging Algorithm (2016) has been tested with 26 benchmark test functions (Zhu and Zhang 2017). In reference Nematollahi et al. (2017), the author proposed an algorithm which can be tested and compared with 29 test functions and 6 engineering design problems. In Bouchekara (2020), the author presented the Most Valuable Player Algorithm, which has been compared and tested with 100 mathematical test functions. Human mental search (Mousavirad and Ebrahimpour-Komleh 2017) has compared and tested with 57 benchmark test functions and 3 engineering problems. The atom search optimization algorithm has also been tested with 37 test functions. Kaur et al. (2020) presented the Sandpiper optimization algorithm, which has been tested and compared with 44 test functions and 7 real-world engineering design problems. The seagull optimization algorithm has also been compared and tested with 7 real-world engineering problems and 49 benchmark test functions (Dhiman and Kumar 2019). In Shabani et al. (2019), the author proposed an MH algorithm which has been tested and compared with 3 engineering problems and 55 test functions. Chameleon Swarm Algorithm has been tested and compared with 5 real-world engineering problems and 67 benchmark test functions (Braik 2021). In Talatahari et al. (2021), the author proposed Crystal Structure Algorithm, which can be evaluated and compared with 239 mathematical benchmark test functions and engineering design problems.

In recent years, many researchers are currently working on the evaluation of novel or hybrid optimization algorithms. These algorithms are being tested across a diverse range of benchmark test suites with numerous distinct challenges. This is mainly due to the vast number of benchmark functions that have been documented in the existing literature. This study aims to examine the advancements in the field of benchmark functions within the context of optimization techniques. The current study analyzes many benchmark functions, offering valuable insights and rationale that might greatly help the optimization methods community. The main objective of the present article is to provide a comprehensive overview of various benchmark test functions that are important to optimization techniques. The main contributions are outlined below:

- This review paper shows the contribution of prolific authors, academic institutions, continents, and the quantitative analysis and statistics on the trends of publications.
- Review the different benchmark test functions, including their MATLAB code references related to the optimization techniques.
- Presents the review of the real-world engineering design test functions to evaluate the performance of optimization techniques.
- To provide a valuable reference for new researchers in this area and assist them in continuous improvement in the field.

The remainder of this review paper is organized as follows: Section 2 illustrates the bibliometric map for the classical benchmark test function in optimizations. Section 3 reviews the different mathematical benchmark test functions and their MATLAB code references. Section 4 illustrates the bibliometric map for the engineering design problems in optimization. Section 5 reviews real-world engineering design problem functions and their MATLAB code references. The final section of this review paper is the conclusion and future scope. Figure 1 represents the structural framework of the review article.

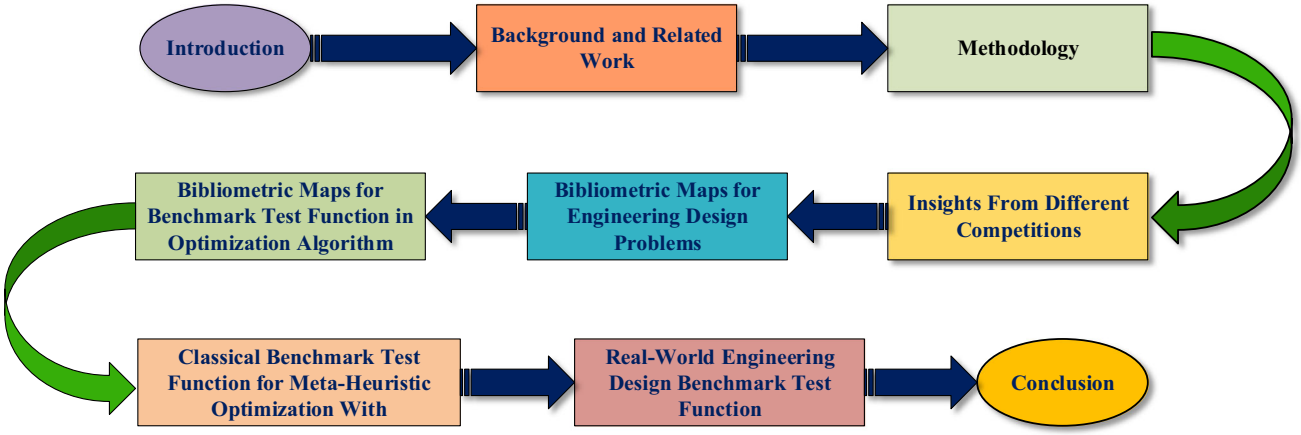

Fig. 1 Outline of the review article

1.1 Background and related work

The Background of benchmark test functions can be traced back to the early stages of numerical optimization. In the 1960s, many basic test functions were introduced, including the Rosenbrock and Rastrigin functions. The functions mentioned above were used to evaluate the effectiveness of novel optimization algorithms and facilitate comparisons between various algorithms regarding their performance. Over the years, many other benchmark test functions have been presented. The complexity, number of variables, as well as multimodality of these functions show variation. The Ackley, Rastrigin, Himmelblau's, Rosenbrock, Shekel, and Schwefel functions are widely recognized and frequently employed benchmark test functions in optimization.

The present collection of literature related to benchmark test functions is vast as well as continually expanding. Many academic articles have been published describing how to create, choose, and implement benchmark test functions. Several significant surveys have been conducted on benchmark test functions, including the publication by Jamil and Yang (2013), and the study by Fong et al. (2015), Garden and Engelbrecht (2014) and Hussain et al. (2017). The surveys presented in this study provide an analysis of many categories of benchmark test functions, examining their fundamental features and exploring their practical uses. Additionally, the authors provide a comprehensive assessment of the existing benchmark test functions and highlight the need to develop novel test functions. Recent research has mostly concentrated on the development of novel benchmark test functions that show increased difficulty and better reflect real-world optimization challenges (D'Angelo et al. 2022, 2023; D'Angelo and Palmieri 2023).

The selection of a benchmark test function is vital in evaluating optimization methods. The benchmark test function must accurately reflect the real-world optimization challenge for which the technique is specifically designed. Additionally, the challenges must provide a sufficient level of difficulty to evaluate the algorithm's capabilities effectively. In addition to the progress achieved with the development of benchmark functions, there has been a continuous discussion over the identification of suitable metrics for assessing algorithms. This paper comprehensively summarizes the extensive research on benchmark test functions. It traces the evolution of function design principles and underscores their pivotal role in advancing optimization algorithm enhancements.

1.2 Methodology

This study utilized a systematic overview of different approaches, which offers advantages over traditional reviews by aiding in identifying limitations in studies and providing insights into extensively explored areas (Xiao and Watson 2019; Ghosh 2023). The investigation involved various relevant databases, such as Google Scholar, Science Direct, and Web of Science to explore published literature within the past few decades. The Scopus database, administered by Elsevier, encompasses citations as well as abstract details of peer-reviewed literature encompassing scientific journals, books, and conference proceedings. In contrast, Google Scholar, another academic search engine, often produces results not covered by Scopus. The title of this paper indicates its focus on the research and advancement of benchmark test functions, particularly those related to optimization algorithms.

Inclusion criteria: Inclusivity criteria were defined, incorporating search terms like "Benchmark Functions" AND "Optimization Algorithms", as well as "Engineering

Design Problems". Additionally, the terms Benchmark Functions, Optimization Algorithms, real-world challenges, and different competitions related to optimization have emerged as common descriptors in recent articles. Given the diverse terminology, supplementary phrases like bibliometric maps and Matlab code references related to the test function were also employed to ensure comprehensive review coverage. The temporal scope primarily encompassed recent research from the past decade (1990–2022), although important older works were also considered.

Quality as well as eligibility assessment: Regarding the assessment of quality as well as eligibility, a careful evaluation of journal articles and books from reputable publishers was conducted. The review process involved an in-depth examination of the full content of these materials to measure their eligibility, overall quality, as well as leading to the inclusion of a select number of high-quality reports.

2 Bibliometric maps for benchmark test function in optimization algorithm

A bibliometric study systematically provides additional evidence to identify global trends in research in a relevant area using the results of a scholarly articles database such as Scopus. The Scopus database has been used to conduct data gathering from April 14th. The main objective of this bibliometric study is to research manuscripts containing "Benchmark Functions" AND "Optimization Algorithms" in the manuscript title, abstracts, and keywords. Furthermore, according to Falagas et al. (2008), Google Scholar is not used in the literature search because of the poor accuracy of the results. A total of 2998 articles were revealed at the end of the search. From 1990 to 2022, the annual number of scientific publications on "Benchmark Functions" AND "Optimization Algorithms" indexed in the Scopus database is presented in Fig. 2. The highest number of articles was published (443) in 2021.

The data obtained from the Scopus database, Journal research articles constituted 66% (1974), journal review articles 0.43% (13), conference research articles 32% (967), conference review articles 0.13% (4), and book chapter 1.16% (35). Figure 3 shows the number of scientific publications on Benchmark Functions indexed in the Scopus database from 1990 to 2022.

The authorship analysis related to Benchmark Functions indexed in the Scopus database from 1990 to 2022. A total of 159 authors were revealed at the end of the search. However, only 12 authors have at least 15 publications. Among these authors, Jeng ShyangPan (Shandong University of Science and Technology) has the highest number of publications with 33 articles [most cited article (Tsai et al. 2009) 215 citation], followed by Huiling Chen (Wenzhou University) with 27 articles [most cited article (Xu et al. 2019) 135 citation] and Juan Zhao from Jingchu University of Technology has published 26 articles while most cited article (GAO et al. 2019) with 10 citations. Figure 4 compares the authors with the publications (at least 15) related to Benchmark Functions indexed in the Scopus database from 1990 to 2022.

A total of 160 institutes were revealed at the end of the search. However, only 10 institutes have more than 30 publications among these institutes, as illustrated in Fig. 5. Ministry of Education China has published the highest number of (86) publications [most cited article (Sun et al. 2012) 250 citations], followed by the Chinese Academy of Sciences with 62 articles [Most cited article (Zhu and Kwong 2010) 1004 citations] and Universiti Sains Malaysia published 40 articles [most cited article (Lim and Mat Isa 2014) 60 citations].

A total of 76 countries contribute their research in the field of Benchmark Functions worldwide. However, only 28 nations have at least 20 publications among these nations. Continental, Asia has the most active continent (77% of total publishing), followed by North America

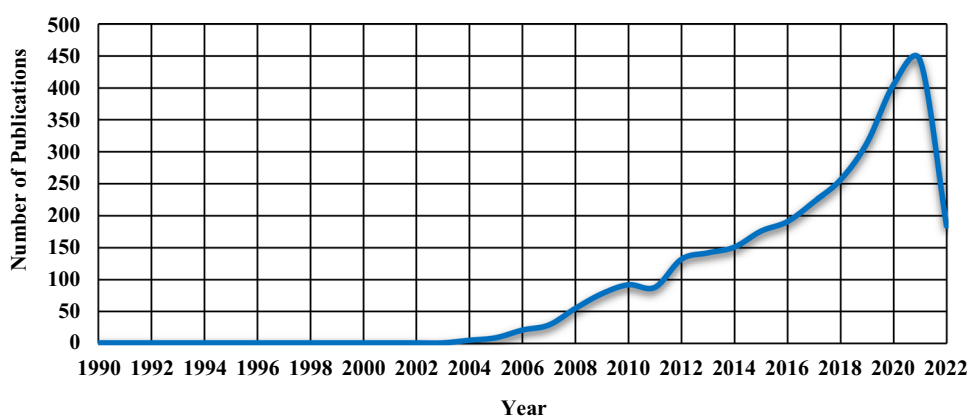

Fig. 2 The annual number of scientific publications on benchmark functions indexed in the Scopus database from 1990 to 2022

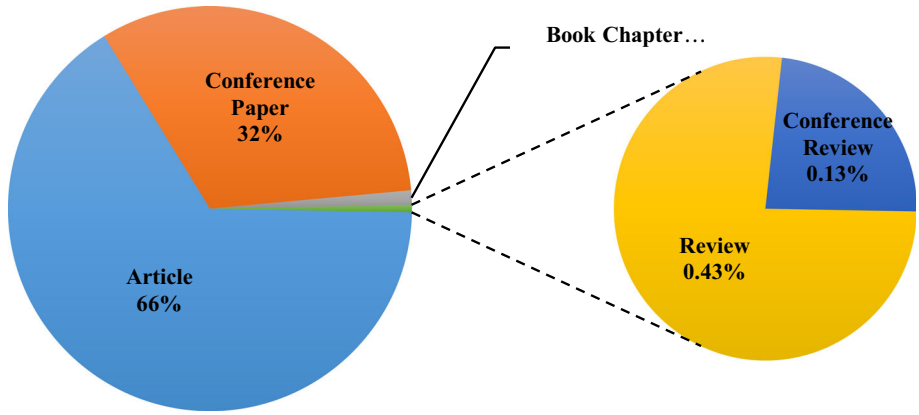

Fig. 3 The number of scientific publications on benchmark functions indexed in the Scopus database from 1990 to 2022

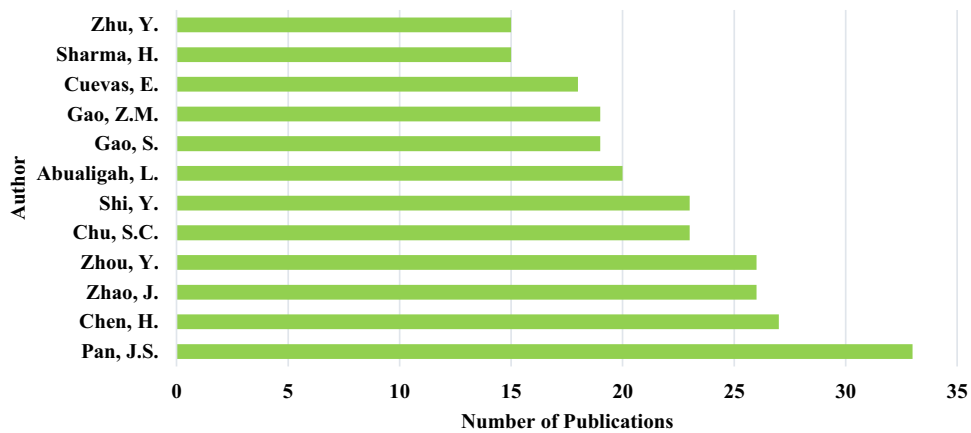

Fig. 4 Compare the authors with the number of publications related to benchmark functions indexed in the Scopus database from 1990 to 2022

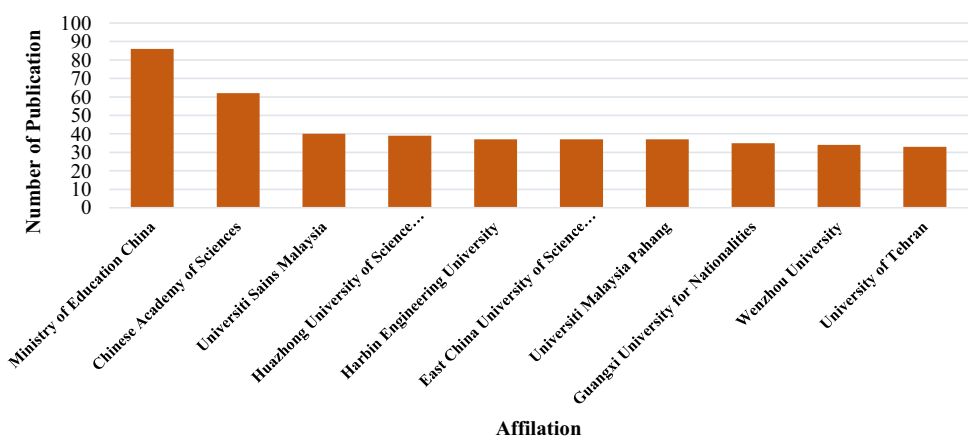

Fig. 5 Compare the document of different affiliation benchmark functions indexed in the Scopus database from 1990 to 2022

(6%), while Africa, Australia, and South America are the least active continents (4%, 2%, and 1%) as presented in Fig. 6. Also, Europe has contributed 10% in the Benchmark Functions area.

Table 1 illustrates the Journal name, cite score, publisher, most cited article related to the Benchmark Functions area, the total number of citations, the Category of the journal, Open Access, and the number of publications related to the Benchmark Functions area that have more than 20 publications. The highest number of papers was published in Lecture Notes in Computer Science (120), followed by IEEE Access has published 95 articles. According to the cite score, only 6 journals have a cite score of more than 10.0, 6 journals have a cite score of more than 5, and 6 journals have less than 5 cite score. The most cited article (1855 citations) (Mirjalili 2015a) has been published in Knowledge-Based Systems Journal (Elsevier), whose cite score is 11.3, followed by Applied Mathematics and Computation (6.8) journal (Elsevier)

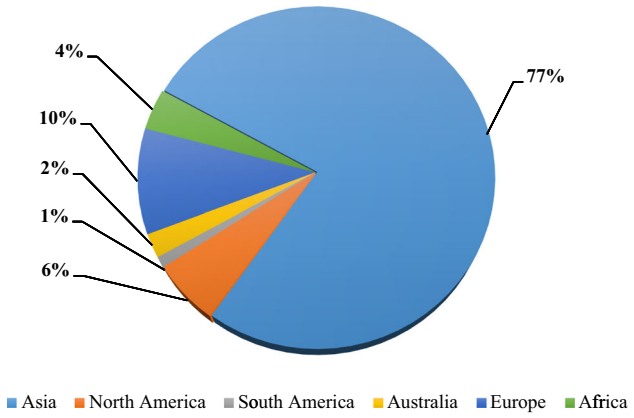

Fig. 6 Participation of continents in terms of percentage of publications, and benchmark functions are indexed in the Scopus database from 1990 to 2022

(Zhu and Kwong 2010) 1004 citations as presented in Table 1.

3 Classical benchmark test function for meta-heuristic optimization

From the literature, it is evident that the performance of the MH optimization was evaluated and compared by mathematical benchmark test functions. These functions have been developed that include a wide range of characteristics that can be evaluated to test the different capabilities of optimization algorithms, such as finding global minima, avoiding local minima, and scaling to high-dimensional challenges. Table 2 summarises benchmark test functions'

Table 1 Journal and conference (number of publications more than 20) benchmark functions indexed in the Scopus database from 1990 to 2022

S. No	Journal/conference name	Cite score	Publisher	Most-cited article reference	Cited	Category	Full open access	Number of publication
1	Lecture Notes in Computer Science	1.8	Springer Nature (Book Series)	(Shi 2011)	430	–	Yes	120
2	IEEE Access	4.8	IEEE	(Ling et al. 2017)	210	Q1	Yes	95
3	Soft Computing	5.1	Springer Nature	(Xue et al. 2018)	218	Q2	No	86
4	Applied Soft Computing Journal	11.2	Elsevier	(Rajabioun 2011)	722	Q1	No	79
5	Advances in Intelligent Systems and Computing	0.9	Springer Nature	(Pan et al. 2015)	35	–	No	60
6	Expert Systems with Applications	12.7	Elsevier	(Cuevas et al. 2013)	381	Q1	Yes	57
7	Neural Computing and Applications	7.3	Springer Nature	(Wang et al. 2014)	195	Q2	No	55
8	Information Sciences	12.1	Elsevier	(Cheng and Jin 2015)	416	Q1	No	48
9	Knowledge-Based Systems	11.3	Elsevier	(Mirjalili 2015a)	1855	Q1	No	43
10	Kongzhi yu Juece/Control and Decision	1.5	Northeastern University	(Qi-chang 2011)	16	Q4	Yes	41
11	Mathematical Problems in Engineering	1.8	Hindawi	(Wu and Zhang 2014)	61	Q3	Yes	38
12	Applied Intelligence	6.8	Springer Nature	(Cuevas et al. 2014)	121	Q2	Yes	37
13	Engineering Applications of Artificial Intelligence	9.0	Elsevier	(Zhao et al. 2020)	210	Q1	No	32
14	Communications in Computer and Information Science	0.8	Springer Nature	(Mishra et al. 2012)	9	Q4	No	27
15	Engineering with Computers	8.4	Springer Nature	(Ba et al. 2022)	24	Q2	No	27
16	Applied Soft Computing Journal	11.2	Elsevier	(Pahnehkolaei et al. 2017)	55	Q1	No	26
17	Swarm and Evolutionary Computation	12.7	Elsevier	(Mousa et al. 2012)	106	Q1	No	25
18	Applied Mathematics and Computation	6.8	Elsevier	(Zhu and Kwong 2010)	1004	Q1	No	23

Table 2 Benchmark test function with MATLAB code reference

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
1	Ackley 1	$f_1(x) = -20e^{-0.02\sqrt{D^{-1}\sum_{i=1}^{D} x_i^2}} - e^{D^{-1}\sum_{i=1}^{D}\cos(2\pi x_i)} + 20 + e$	Multi-modal	$-35 \le x_i \le 35$	0	(Jamil and Yang 2013)(Dieterich and Hartke 2012)	(Greene 2021)
2	Sphere	$f_2(x) = \sum_{i=1}^{D} x_i^2$	Multimodal	$-5.12 \le x_i \le 5.12$	0	(Bäck and Schwefel 1993)(Tayarani-N. and Akbarzadeh-T. 2014)	(Baldea and Daoutidis 2012)
3	Ackley 4 or Modified Ackley	$f_3(x) = \sum_{i=1}^{D}(e^{-0.2}\sqrt{x_i^2 + x_{i+1}^2} + 3(\cos(2x_i) + \sin(2x_{i+1})))$	Multimodal	$-35 \le x_i \le 35$	$-5.901X10^{14}$	(Ronkkonen 2009)	(Thevenot 2020)
4	Adjiman	$f_4(x) = \cos(x_1)\sin(x_2) - \frac{x_1}{x_2^2+1}$	Multimodal	$-5 \le x_i \le 5$	− 5	(Adjiman et al. 1998)	(Houssem 2020)
5	Alpine 1	$f_5(x) = \sum_{i=1}^{D} \lvert x_i \sin(x_i) + 0.1x_i \rvert$	Multimodal	$-10 \le x_i \le 10$	0	(Jamil et al. 2013)	(Lima 2020)
6	Alpine 2	$f_6(x) = \prod_{i=1}^{D} \sqrt{x_i}\sin(x_i)$	Multimodal	$[-0 \le x_i \le 100]^2$	− 7.8856	(Clerc 1999)	(Thevenot 2020)
7	Brad	$f_7(x) = \sum_{i=1}^{15} \left[\frac{y_i - x_1 - u_i}{v_i x_2 + w_i x_3}\right]^2$	Multimodal	$-0.25 \le x_1 \le 0.25$; $0.01 \le x_i \le 2.5$	0.00821487	(Jamil and Yang 2013)	(Luclaurent and Leinweber)
8	Bartels Conn	$f_8(x) = \lvert x_1^2 + x_2^2 + x_1x_2 \rvert + \lvert \sin(x_1) \rvert + \lvert cos(x_2) \rvert$	Multimodal	$-500 \le x_i \le 500$	1	(Jamil and Yang 2013)	(Al-Roomi 2021)
9	Beale	$f_9(x) = (1.5 - x_1 + x_1x_2)^2 + (2.25 - x_1 + x_1x_2^2)^2 + (2.625 - x_1 + x_1x_2^3)^2$	Unimodal	$-4.5 \le x_i \le 4.5$	0	(Razmjooy et al. 2016)	(Oldenhuis 2021)
10	Biggs EXP 2	$f_{10}(x) = \sum_{i=1}^{10}(e^{-t_ix_1} - 5e^{-t_ix_2} - y_i)^2$	Multimodal	$0 \le x_i \le 20$	0	(Jamil and Yang 2013)(BIGGS 1971)	(Al-Roomi 2021)
11	Biggs EXP 3	$f_{11}(x) = \sum_{i=1}^{10}(e^{-t_ix_1} - x_3e^{-t_ix_2} - y_i)^2$	Multimodal	$0 \le x_i \le 20$	0	(BIGGS 1971; Jamil and Yang 2013)	(Al-Roomi 2021)
12	Biggs EXP 4	$f_{12}(x) = \sum_{i=1}^{10}(x_3e^{-t_ix_1} - x_4e^{-t_ix_2} - y_i)^2$	Multimodal	$0 \le x_i \le 20$	0	(BIGGS 1971)	(Al-Roomi 2021)
13	Biggs EXP 5	$f_{13}(x) = \sum_{i=1}^{11}(x_3e^{-t_ix_1} - x_4e^{-t_ix_2} + 3e^{-t_ix_5} - y_i)^2$	Multimodal	$-20 \le x_i \le 20$	0	(Jamil and Yang 2013)(BIGGS 1971)	(Al-Roomi 2021)
14	Biggs EXP 6	$f_{14}(x) = \sum_{i=1}^{11}(x_3e^{-t_ix_1} - x_4e^{-t_ix_2} + x_6e^{-t_ix_5} - y_i)^2$	Multimodal	$-20 \le x_i \le 20$	0	(Jamil and Yang 2013)(BIGGS 1971)	(2022a)
15	Rosenbrock 1	$f_{15}(x) = \sum_{i=1}^{D}\left\{100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2\right\}$	Unimodal	$-30 \le x_i \le 30$	0	(Razmjooy et al. 2016) (Rahnamayan et al. 2007)	(Sam 2018)
16	Rosenbrock Modified	$f_{16}(x) = 74 + 100(x_2 - x_1^2)^2 + (1 - x)^2 - 400e^{-\frac{(x_1+1)^2+(x_2+1)^2}{0.1}}$	Multimodal	$-2 \le x_i \le 2$	0	(Jamil and Yang 2013)	(Al-Roomi 2021)
17	Rastrigin's	$f_{17}(x) = \sum_{i=0}^{D-1} x_i^2 - 10\cos(2\pi x_i) + 20$	Multimodal	$-5.12 \le x_i \le 5.12$	0	(Pohlheim 2005) (Horst and Tuy 1996)	(Oldenhuis 2021)
18	Booth	$f_{18}(x) = (x_1 + 2x_2 - 7)^2 + (2x_1 + x_2 - 5)^2$	Unimodal	$-10 \le x_i \le 10$	0	(Arora and Singh 2019)	(Mirjalili 2021)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
19	Cubic	$f_{19}(x) = \sum_{i=1}^{D} x_i^2$	Multimodal	$-2 \le x_i \le 2$	0	(Razmjooy et al. 2016)	(Montalvo 2021)
20	Ackley 2	$f_{20}(x) = -200e^{-0.2\sqrt{x_1^2+x_2^2}}$	Multimodal	$-32 \le x_i \le 32$	− 200	(Hashim et al. 2019)(Ackley 1987)	(Thevenot 2020)
21	Ackley 3	$f_{21}(x) = 200e^{-0.2\sqrt{x_1^2+x_2^2}} + 5e^{\cos(3x_1)+\sin(3x_2)}$	Unimodal	$-32 \le x_i \le 32$	− 219.1418	(Ackley 1987)	(Thevenot 2020)
22	Box-Betts	$f_{22}(x) = \sum_{i=1}^{k}\left(e^{-0.1(i+1)x_1} - e^{-0.1(i+1)x_2}\left[\left(e^{-0.1(i+1)}\right) - e^{-0.1(i+1)x_3}\right]^2\right)$	Multimodal	$0.9 \le x_1, x_3 \le 1.2; 9 \le x_2 \le 11.2$	0	(Salawudeen et al. 2021)	(Al-Roomi 2021)
23	Bird	$f_{23}(x) = sin(x_1)e^{(1-\cos(x_2))^2} + \cos(x_2)e^{(1-\sin(x_1))^2} + (x_1 - x_2)^2$	Multimodal	$-2\pi \le x_i \le 2\pi$	− 106.76453	(Jamil and Yang 2013)	(Oldenhuis 2021)
24	Bohachevsky 1	$f_{24}(x) = x_1^2 + 2x_2^2 - 0.3\cos(3\pi x_1) - 0.4\cos(4\pi x_2) + 0.7$	Multimodal	$-100 \le x_i \le 100$	0	(Kaveh and Zolghadr 2016)	(Leong 2016)
25	Bohachevsky 2	$f_{25}(x) = x_1^2 + 2x_2^2 - 0.3\cos(3\pi x_1) * 0.4\cos(4\pi x_2) + 0.3$	Multimodal	$-100 \le x_i \le 100$	0	(Kaveh and Zolghadr 2016)	(2021a)
26	Bohachevsky 3	$f_{26}(x) = x_1^2 + 2x_2^2 - 0.3\cos(3\pi x_1 + 4\pi x_2) + 0.3$	Multimodal	$-100 \le x_i \le 100$	0	(Hashim et al. 2019)	(Math 2022)
27	Camel-Three Hump	$f_{27}(x) = 2x_1^2 + 1.05x_1^4 + x_1^6/6 + x_1x_2 + x_2^2$	Multimodal	$-5 \le x_i \le 5$	0	(Hashim et al. 2019)	(Oldenhuis 2021)
28	Brown	$f_{28}(x) = \sum_{i=1}^{n-1}\left((x_i^2)^{(x_{i+1}^2+1)} + (x_{i+1}^2)^{(x_i^2+1)}\right)$	Multimodal	$-1 \le x_i \le 4$	0	(Begambre and Laier 2009)	(Lima 2020)
29	Cigar	$f_{29}(x) = x_i^2 + 10^6 \sum_{i=2}^{n} x_i^2$	Multimodal	$-100 \le x_i \le 100$	0	(Hashim et al. 2019)	(Al-Roomi 2021)
30	Cone	$f_{30}(x) = \sqrt{\sum_{i=1}^{n} x_i^2}$	Unimodal	$-100 \le x_i \le 100$	–	(Bassel et al. 2020)	(Oldenhuis 2021)
31	Schwefel 2.22	$f_{31}(x) = \sum_{i=1}^{n}\lvert x_i\rvert + \prod_{i=1}^{n}\lvert x_i\rvert$	Unimodal	$-10 \le x_i \le 10$	0	(Gandomi and Alavi 2012)	(Baldea and Daoutidis 2012)
32	Exponential	$f_{32}(x) = -\exp(-0.5\sum_{i=1}^{n} x_i^2)$	Multimodal	$-1 \le x_i \le 1$	− 1	(Hashim et al. 2019)(Meng et al. 2014)	(Lima 2020)
33	Chung Reynolds	$f_{33}(x) = \left(\sum_{i=1}^{n} x_i^2\right)^2$	Unimodal	$-100 \le x_i \le 100$	0	(Yapici and Cetinkaya 2019)	(Hussien 2022)
34	Quadric	$f_{34}(x) = \sum_{i=1}^{n}\left(\sum_{j=1}^{i} x_j\right)^2$	Unimodal	$-1.28 \le x_i \le 1.28$	–	(Bassel et al. 2020)	(Evers 2011)
35	Griewank	$f_{35}(x) = \frac{1}{4000}\sum_{i=1}^{n} x_i^2 + \prod \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$	Multimodal	$-100 \le x_i \le 100$	0	(Shareef et al. 2015)	(Oldenhuis 2021)
36	Weierstrass	$f_{36}(x) = \sum_{i=1}^{n}\left[\sum_{k=0}^{10} a^k\cos(2\pi b^k(x_i + 0.5)) - n\sum_{k=0}^{10} a^k\cos(\pi b^k)\right]$	Multimodal	$-10 \le x_i \le 10$	0	(Sharafi et al. 2016)	(Qinqin 2014)
37	Powell Singular 1	$f_{37}(x) = \sum_{i=1}^{D/4}(x_{4i-3} - 10x_{4i-2})^2 + 5(x_{4i-1} - x_{4i})^2 + (x_{4i-2} - x_{4i-1})^4 + 10(x_{4i-3} - x_{4i})^4$	Unimodal	$-4 \le x_i \le 5$	0	(Hashim et al. 2019)	(2022b)
38	Powell Singular 2	$f_{38}(x) = \sum_{i=1}^{D-2}(x_{i-1} - 10x_i)^2 + 5(x_{i+1} - x_{i+2})^2 + (x_i - 2x_{i+1})^4 + 10(x_{i-1} - x_{i+2})^4$	Unimodal	$-4 \le x_i \le 5$	0	(Hashim et al. 2022)	(2022b)
39	Powell Sum	$f_{39}(x) = \sum_{i=1}^{D}\lvert x_i\rvert^{i+1}$	Unimodal	$-1 \le x_i \le 1$	0	(Hashim et al. 2022)	(2022b)
40	Step 1	$f_{40}(x) = \sum_{i=1}^{n}\lfloor\lvert x_i\rvert\rfloor$	Unimodal	$-100 \le x_i \le 100$	0	(Hashim et al. 2019)	(Al-Roomi 2021)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
41	Step 2	$f_{41}(x) = \sum_{i=1}^{n} (\lfloor x_i + 0.5 \rfloor)^2$	Unimodal	$-100 \le x_i \le 100$	0	(Yapici and Cetinkaya 2019)	(Math 2022)
42	Step 3	$f_{42}(x) = \sum_{i=1}^{n} (\lfloor x_i^2 \rfloor)$	Unimodal	$-100 \le x_i \le 100$	0	(Jamil and Yang 2013)	(Al-Roomi 2021)
43	Schwefel	$f_{43}(x) = \left(\sum_{i=1}^{D} x_i^2\right)^{\alpha}$	Unimodal	$-100 \le x_i \le 100$	0	(Jamil and Yang 2013)	(2022c)
44	Schwefel 1.2	$f_{44}(x) = \sum_{i=1}^{D} \left(\sum_{j=1}^{i} x_j\right)^2$	Unimodal	$-100 \le x_i \le 100$	0	(Cuevas et al. 2013)	(Baldea and Daoutidis 2012)
45	Schwefel 2.4	$f_{45}(x) = \sum_{i=1}^{D} (x_i - 1)^2 + (x_1 - x_i^i)^2$	Multimodal	$-100 \le x_i \le 100$	0	(Jamil and Yang 2013)	(2022c)
46	Schwefel 2.6	$f_{46}(x) = max(\lvert x_1 + 2x_2 - 7 \rvert, \lvert 2x_1 + x_2 - 5 \rvert)$	Unimodal	$-100 \le x_i \le 100$	0	(Jamil and Yang 2013)	(2022c)
47	Schwefel 2.20	$f_{47}(x) = -\sum_{i=1}^{D} \lvert x_i \rvert$	Unimodal	$-100 \le x_i \le 100$	0	(Hashim et al. 2022)	(Diep 2022)
48	Schwefel 2.21	$f_{48}(x) = max_{1 \le i \le D} \lvert x_i \rvert$	Unimodal	$-100 \le x_i \le 100$	0	(Asef et al. 2021)	(Baldea and Daoutidis 2012)
49	Schwefel 2.23	$f_{49}(x) = \sum_{i=1}^{D} x_i^{10}$	Unimodal	$-10 \le x_i \le 10$	0	(Hashim et al. 2022)	(Thevenot 2020)
50	Schwefel 2.25	$f_{50}(x) = (x_i - 1)^2 + (x_1 - x_i^2)^2$	Multimodal	$0 \le x_i \le 10$	0	(Hashim et al. 2019)	(2022c)
51	Schwefel 2.26	$f_{51}(x) = -\frac{1}{D} \sum_{i=1}^{D} x_i sin\sqrt{\lvert x_i \rvert}$	Multimodal	$-500 \le x_i \le 500$	− 418.983	(Gandomi and Alavi 2012)	(Baldea and Daoutidis 2012)
52	Schwefel 2.36	$f_{52}(x) = x_1 x_2 (72 - 2x_1 - 2x_2)$	Multimodal	$0 \le x_i \le 500$	− 3456	(Jamil and Yang 2013)	(2022c)
53	Drop Wave	$f_{53}(x) = -\frac{1+\cos(12\sqrt{x_1^2+x_2^2}}{(0.5(x_1^2+x_2^2)+2)}$	Multimodal	$-5.2 \le x_i \le 5.2$	− 1	(Asef et al. 2021)	(Martínez-Cagigal 2016)
54	Goldstein Price	$f_{54}(x) = \Big[1 + (x_1 + x_2 + 1)^2 (19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 + 3x_2^2)\Big] X \Big[30 + (2x_1 - 3x_2^2)^2 (18 - 3x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 + 27x_2^2)\Big]$	Multimodal	$-2 \le x_i \le 2$	3	(Asef et al. 2021)	(Kelly 2016)(Kelly)
55	Xin-She Yang 1	$f_{55}(x) = \sum_{i=1}^{D} \in_i \lvert x_i \rvert^i$	Multimodal	$-5 \le x_i \le 5$	0	(Asef et al. 2021)	(Layeb 2021)
56	Xin-She Yang 2	$f_{56}(x) = \left(\sum_{i=1}^{D} \lvert x_i \rvert\right) exp\left[-\sum_{i=1}^{D} \sin(x_i^2)\right]$	Multimodal	$-2\pi \le x_i \le 2\pi$	0	(Yang 2010)	(Layeb 2021)
57	Xin-She Yang 3	$f_{57}(x) = \left[e^{-\sum_{i=1}^{D} (x_i/15)^{10}} - 2e^{-\sum_{i=1}^{D} (x_i)^2} . \prod_{i=1}^{D} cos^2(x_i)\right]$	Multimodal	$-20 \le x_i \le 20$	0	(Yang 2010)	(Layeb 2021)
58	Xin-She Yang 4	$f_{58}(x) = \left[e^{-\sum_{i=1}^{D} (x_i/15)^{10}} - 2e^{-\sum_{i=1}^{D} (x_i - \pi)^2} . \prod_{i=1}^{D} cos^2(x_i)\right]$	Multimodal	$-10 \le x_i \le 10$	− 1	(Yang 2010)	(Layeb 2021)
59	Salomon	$f_{59}(x) = 1 - \cos\left(2\pi\sqrt{\sum_{i=1}^{D} x_i^2}\right) + 0.1\sqrt{\sum_{i=1}^{D} x_i^2}$	Multimodal	$-100 \le x_i \le 100$	0	(Asef et al. 2021)	(Lima 2020)
60	Mishra 1	$f_{60}(x) = \left(1 + m - \sum_{i=1}^{m-1} x_i\right)^{m - \sum_{i=1}^{m-1} x_i}$	Multimodal	$0 \le x_i \le 1$	2	(Mishra 2011)	(Al-Roomi 2015)
61	Mishra 2	$f_{61}(x) = \left(1 + m - \sum_{i=1}^{m-1} \frac{x_i + x_{i+1}}{2}\right)^{m - \sum_{i=1}^{m-1} \frac{x_i + x_{i+1}}{2}}$	Multimodal	$0 \le x_i \le 1$	2	(Mishra 2011)	(Al-Roomi 2015)
62	Mishra 3	$f_{62}(x) = \left[\left\lvert cos\sqrt{\lvert x_1^2 + x_2 \rvert}\right\rvert^{0.5} + (x_1 + x_2)/100\right]$	Multimodal	$-10 \le x_i \le 10$	− 0.18466	(Mishra 2006a)	(Al-Roomi 2015)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
63	Mishra 4	$f_{63}(x) = \left[\left\lvert sin\sqrt{\lvert x_1^2 + x_2\rvert}\right\rvert^{0.5} + (x_1 + x_2)/100\right]$	Multimodal	$-10 \le x_i \le 10$	− 0.199441	(Mishra 2006a)	(Al-Roomi 2015)
64	Mishra 5	$f_{64}(x) = \left[\left\{(\sin((\cos(x_1) + \cos(x_2))^2)^2 - (\cos((\sin(x_1) + \sin(x_2))^2)^2\right\} + x_1\right]^2 + 0.01(x_1 + x_2)$	Multimodal	$-10 \le x_i \le 10$	− 1.01983	(Mishra 2006a)	(Al-Roomi 2015)
65	Mishra 6	$f_{65}(x) = -\ln\left[\left[\left\{(\sin((\cos(x_1) + \cos(x_2))^2)^2 - (\cos((\sin(x_1) + \sin(x_2))^2)^2\right\} + x_1\right]^2\right] + [(x_1 - 1)^2 + (x_2 - 1)^2]/10$	Multimodal	$-10 \le x_i \le 10$	2.28395	(Mishra 2006a)	(Al-Roomi 2015)
66	Quintic	$f_{66}(x) = \sum_{i=1}^{D} \lvert x_i^5 - 3x_i^4 + 4x_i^3 + 2x_i^2 - 10x_i - 4\rvert$	Multimodal	$-10 \le x_i \le 10$	0	(Mishra 2006a)	(Al-Roomi 2021)
67	Needle-eye	$f_{67}(x) = \sum_{i=1}^{D} (100 + \lvert xi\rvert t_i$	Multimodal	–	–	(Mishra 2006a)	(Al-Roomi 2021)
68	Mishra 7	$f_{68}(x) = \left[\prod_{i=1}^{D} x_i - N!\right]^2$	Multimodal	–	0	(Jamil and Yang 2013)	(Al-Roomi 2015)
69	Mishra 11	$f_{69}(x) = \left[\frac{1}{D}\sum_{i=1}^{D} \lvert x_i\rvert - \left(\prod_{i=1}^{D} \lvert x_i\rvert^{\frac{1}{N}}\right)\right]^2$	Multimodal	$0 \le x_i \le 10$	0	(Hashim et al. 2019)	(Al-Roomi 2015)
70	Quartic	$f_{70}(x) = \sum_{i=1}^{D} ix_i^4 - random[0, 1]$	Multimodal	$-1.28 \le x_i \le 1.28$	0	(Hashim et al. 2019)	(Baldea and Daoutidis 2012)
71	Zakharov	$f_{71}(x) = \sum_{i=1}^{D} x_i^2 + \left(\frac{1}{2}\sum_{i=1}^{D} ix_i^2\right)^2 + \left(\frac{1}{2}\sum_{i=1}^{D} ix_i^2\right)^4$	Multimodal	$-5 \le x_i \le 10$	0	(Rahnamayan et al. 2007)	(2021b)
72	Chichinadze	$f_{72}(x) = x_1^2 + 12x_1 + 11 + 10\cos\left(\frac{\pi x_1}{2}\right) + 8\sin\left(\frac{5\pi x_1}{2}\right) - \sqrt{\frac{1}{5}}exp\left(-0.5(x_2 - 0.5)^2\right)$	Multimodal	$-30 \le x_i \le 30$	− 43.3159	(Bouchekara 2020)	(Oldenhuis 2021)
73	AMGM	$f_{73}(x) = \left(\frac{1}{n}\sum_{i=1}^{n} x_i - \sqrt[n]{\prod_{i=1}^{n} x_i}\right)^2$	Multimodal	–	–	(Bouchekara 2020)	(2022c)
74	Axis Parallel Hyper-Ellipsoid	$f_{74}(x) = \sum_{i=1}^{n} ix_i^2$	Multimodal	$-5.12 \le x_i \le 5.12$	0	(Rahnamayan et al. 2007)	(2022c)
75	Matyas	$f_{75}(x) = 0.26\left(x_1^2 + x_2^2\right) - 0.48x_1x_2$	Unimodal	$-10 \le x_i \le 10$	0	(Kumar et al. 2021c)	(Chen)
76	Perm	$f_{76}(x) = \sum_{k=1}^{n} \left[\sum_{i=1}^{n} (i^k + 0.5)((\frac{1}{i}x_i)^k - 1)\right]^2$	Multimodal	$-n \le x_i \le n$	0	(Rahnamayan et al. 2007)	(Mirjalili 2021)
77	Kowalik's	$f_{77}(x) = \sum_{i=1}^{11} \left[a_i - \frac{x_1(b_i^2 + b_ix_2)}{b_i^2 + b_ix_3 + x_4}\right]^2$	Multimodal	$-5 \le x_i \le 5$	0.0003075	(Rahnamayan et al. 2007)	(Dogan 2020)
78	Becker and Lago	$f_{78}(x) = (\lvert x_1\rvert - 5)^2 + (\lvert x_2\rvert - 5)^2$	Multimodal	$-10 \le x_i \le 10$	0	(Ali et al. 2005)	(Leong 2016)
79	Inverted Cosine Wave	$f_{79}(x) = -\sum_{i=1}^{n-1} \left(\exp\left(\frac{-(x_i^2 + x_{i+1}^2 + 0.5x_ix_{i+1})}{8}\right)\right) \times \cos\left(4\sqrt{x_i^2 + x_{i+1}^2 + 0.5x_ix_{i+1}}\right)$	Multimodal	$-10 \le x_i \le 10$	− n + 1		(Al-Roomi 2021)
80	Egg Crate	$f_{80}(x) = x_1^2 + x_2^2 + 25(sin^2(x_1) + sin^2(x_2))$	Multimodal	$-5 \le x_i \le 5$	0	(Hashim et al. 2019)	(Khayou 2020)
81	Wolfe	$f_{81}(x) = \frac{4}{3}(x_1^2 + x_2^2 - x_1x_2)^{0.75} + x_3$	Multimodal	$0 \le x_i \le 2$	0	(Kumar et al. 2021c)	([CSL STYLE ERROR: reference with no printed form.])
82	Test Tube Holder	$f_{82}(x) = -4\left\lvert\sin(x_1)\cos(x_2)e^{\lvert\cos\left(x_1^2 + x_2^2\right)/200\rvert}\right\rvert$	Multimodal	$-10 \le x_i \le 10$	− 10.8723	(Mishra 2006b)	(Oldenhuis 2021)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
83	Holder Table 1	$f_{83}(x) = -\left\lvert \cos(x_1)\cos(x_2)e^{\left\lvert 1-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert$	Multimodal	$-10 \le x_i \le 10$	− 26.920336	(Mishra 2006b)	(2022c)
84	Holder Table 2	$f_{84}(x) = -\left\lvert \sin(x_1)\cos(x_2)e^{\left\lvert 1-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert$	Multimodal	$-10 \le x_i \le 10$	− 19.2085	(Mishra 2006b)	(Chen)
85	Carrom Table	$f_{85}(x) = -\left[\left\{\cos(x_1)\cos(x_2)e^{\left\lvert 1-\frac{\left[(x_1^2+x_2^2)^{0.5}\right]}{\pi} \right\rvert}\right\}^2\right]/30$	Multimodal	$-10 \le x_i \le 10$	− 24.1568155	(Mishra 2006b)	(Mishra 2006b)
86	Cross in Tray	$f_{86}(x) = -0.0001\left[\left\lvert \sin(x_1)sine^{\left\lvert 100-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert + 1\right]^{0.1}$	Multimodal	$-10 \le x_i \le 10$	− 2.06261218	(Mishra 2006b)	(Alamsyah 2022)
87	Crowned Cross	$f_{87}(x) = 0.0001\left[\left\lvert \sin(x_1)sine^{\left\lvert 100-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert + 1\right]^{0.1}$	Multimodal	$-10 \le x_i \le 10$	0	(Mishra 2006b)	(Al-Roomi 2021)
88	Cross	$f_{88}(x) = \left[\left\lvert \sin(x_1)sine^{\left\lvert 100-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert + 1\right]^{0.1}$	Multimodal	–	0	(Mishra 2006b)	(Al-Roomi 2021)(Oldenhuis 2021)
89	Cross-leg table	$f_{89}(x) = -\left[\left\lvert \sin(x_1)sine^{\left\lvert 100-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert + 1\right]^{0.1}$	Multimodal	$-10 \le x_i \le 10$	− 1	(Mishra 2006b)	(Oldenhuis 2021)
90	Pen Holder	$f_{90}(x) = -exp\left[-\left\lvert \cos(x_1)\cos(x_2)e^{\left\lvert 1-\left[(x_1^2+x_2^2)^{0.5}\right]/\pi \right\rvert} \right\rvert^{-1}\right]$	Multimodal	$-11 \le x_i \le 11$	− 0.96354	(Mishra 2006b)	(Oldenhuis 2021)
91	Modified Schaffer #1	$f_{91}(x) = 0.5 + \frac{sin^2\left(x_1^2+x_2^2\right)-0.5}{\left[1+0.001\left(x_1^2+x_2^2\right)\right]^2}$	Multimodal	$-100 \le x_i \le 100$	0	(Mishra 2006b)	(Oldenhuis 2021)
92	Modified Schaffer #2	$f_{92}(x) = 0.5 + \frac{sin^2\left(x_1^2-x_2^2\right)-0.5}{\left[1+0.001\left(x_1^2+x_2^2\right)\right]^2}$	Multimodal	$-100 \le x_i \le 100$	0	(Mishra 2006b)	(Chen)
93	Modified Schaffer #3	$f_{93}(x) = 0.5 + \frac{sin^2\left[cos\left[\left\lvert x_1^2-x_2^2 \right\rvert\right]\right]-0.5}{\left[1+0.001\left(x_1^2+x_2^2\right)\right]^2}$	Multimodal	$-100 \le x_i \le 100$	0.00156685	(Mishra 2006b)	(Oldenhuis 2021)
94	Modified Schaffer #4	$f_{94}(x) = 0.5 + \frac{cos^2\left[sin\left[\left\lvert x_1^2-x_2^2 \right\rvert\right]\right]-0.5}{\left[1+0.001\left(x_1^2+x_2^2\right)\right]^2}$	Multimodal	$-100 \le x_i \le 100$	0.292579	(Mishra 2006b)	(Chen)
95	Zettle	$f_{95}(x) = \left(x_1^2 + x_2^2 - 2x_1\right)^2 + 0.25x_1$	Multimodal	$-5 \le x_i \le 5$	− 0.003791	(Mishra 2006b)	(Oldenhuis 2021)
96	Giunta	$f_{96}(x) = 0.6 + \sum_{i=1}^{2}\left[\sin\left(\frac{16}{15}x_i - 1\right) + sin^2\left(\frac{16}{15}x_i - 1\right) + \frac{1}{50}\sin\left(4\left(\frac{16}{15}x_i - 1\right)\right)\right]$	Multimodal	$-1 \le x_i \le 1$	0.0602472184	(Mishra 2006b)	(Oldenhuis 2021)
97	Leon	$f_{97}(x) = 100\left(100x_2 - x_1^2\right)^2 + (1 - x_1)^2$	Unimodal	$-1.2 \le x_i \le 1.2$	0	(Mishra 2006b)	(Oldenhuis 2021)
98	Styblinski–Tang	$f_{98}(x) = \frac{1}{2}\sum_{i=1}^{2}(x_i^4 - 16x_i^2 + 5x_i)$	Multimodal	$-5 \le x_i \le 5$	− 78.332	(Raouf and Hezam 2017)	(Kelly 2016)(Kelly)
99	Venter Sobieszczanski-Sobieski	$f_{99}(x) = x_1^2 - 100\cos(x_1)^2 - 100\cos\left(\frac{x_1^2}{30}\right) + x_2^2 - 100\cos(x_2)^2 - 100\cos\left(\frac{x_2^2}{30}\right)$	Multimodal	$-50 \le x_i \le 50$	− 400	(Jamil and Yang 2013)	(Al-Roomi 2021)
100	Trecanni	$f_{100}(x) = 4 - 4x_1^3 + 4x_1 + x_2^2$	Multimodal	$-5 \le x_i \le 5$	0	(Hashim et al. 2022)	(Al-Roomi 2021)
101	Pathological	$f_{101}(x) = \sum_{i=1}^{D-1}\left(0.5 + \frac{sin^2\sqrt{100x_i^2+x_{i+1}^2}-0.5}{1+0.001\left(x_i^2-2x_ix_{i+1}+x_{i+1}^2\right)}\right)$	Multimodal	$-100 \le x_i \le 100$	0	(Raouf and Hezam 2017)	(Luclaurent and Leinweber)
102	Sine envelope	$f_{102}(x) = -\sum_{i=1}^{n-1}\left[\frac{sin^2\sqrt{x_i^2+x_{i+1}^2}-0.5}{\left(0.001\left(x_i^2+x_{i+1}^2\right)+1\right)^2} + 0.5\right]$	Multimodal	$-100 \le x_i \le 100$	0	(Raouf and Hezam 2017)	(Layeb 2021)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
103	Colville	$f_{103}(x) = 100(x_1 - x_2^2)^2 + (1 - x_1)^2 + 90(x_4 - x_3^2)^2 + (1 - x_3)^2 + 10.1((x_2 - 1)^2 + \left(x_4 - 1)^2\right) + 19.8(x_2 - 1)(x_4 - 1)$	Multimodal	$-10 \le x_i \le 10$	0	(Kumar et al. 2021c)	(Mirjalili 2021)
104	Easom	$f_{104}(x) = -\cos(x_1)\cos(x_2).\exp(-(x_1 - \pi)^2 - (x_2 - \pi)^2)$	Multimodal	$-100 \le x_i \le 100$	− 1	(Sacco and De Oliveira 2005)	(Saghafi 2015)
105	Rosenbrock's	$f_{105}(x) = \sum_{i=1}^{D-1}\left[100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2\right]$	Unimodal	$-2.048 \le x_i \le 2.048$	0	(Alatas 2011)	(Oldenhuis 2021)
106	Michalewicz 2	$f_{106}(x) = -\sum_{i=1}^{D}\sin(x_i)(\sin(\frac{ix_i^2}{\pi}))^{20}$	Multimodal	$-0 \le x_i \le \pi$	− 1.8013	(Doğan and Ölmez 2015)	(Layeb 2021)
107	Stepint	$f_{107}(x) = 25 + \sum_{i=1}^{5}\lceil xi \rceil$	Unimodal	$-5.12 \le x_i \le 5.12$	0	(Doğan and Ölmez 2015)	(Math 2022)
108	Sum squares	$f_{108}(x) = \sum_{i=1}^{D} ix_i^2$	Unimodal	$-10 \le x_i \le 10$	0	(Doğan and Ölmez 2015)	(2021b)
109	Trid 6	$f_{109}(x) = \sum_{i=1}^{D}(x_i - 1)^2 - \sum_{i=2}^{D} x_i x_i - 1$	Unimodal	$-36 \le x_i \le 36$	− 50	(Doğan and Ölmez 2015)	(2021b)
110	Trid 10	$f_{110}(x) = \sum_{i=1}^{D}(x_i - 1)^2 - \sum_{i=2}^{D} x_i x_i - 1$	Multimodal	$-100 \le x_i \le 100$	− 200	(Doğan and Ölmez 2015)(FINDIK 2015)	(2021b)
111	Dixon–Price	$f_{111}(x) = (x_1 - 1)^2 + \sum_{i=2}^{D} i\left(2x_i^2 - x_i - 1\right)^2$	Unimodal	$-10 \le x_i \le 10$	0	(Ebrahimi and Khamehchi 2016)	(2021b)
112	Sphere Modifications and Alternate Forms	$f_{112}(x) = \frac{1}{899}\left(\sum_{i=1}^{6} x_i^2 2^i - 1745\right)$	Multimodal	$-0 \le x_i \le 1$	− 1.941389	(Picheny et al. 2013)	(2022c)
113	Camel Six Hump	$f_{113}(x) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3}\right) + x_1x_2 + (4x_2^2 - 4)x_2^2$	Multimodal	$-5 \le x_i \le 5$	− 1.0316	(Ebrahimi and Khamehchi 2016)	(Oldenhuis 2021)(2021c)
114	Shubert	$f_{114}(x) = \left(\sum_{i=1}^{5} icos(i+1)x_1 + 1)\right)\left(\sum_{i=1}^{5} icos(i+1)x_2 + 1)\right)$	Multimodal	$-10 \le x_i \le 10$	− 186.7309	(Yang 2012)	(2021b)(2021c)
115	Boachevsky 2	$f_{115}(x) = x_1^2 + 2x_2^2 - 0.3\cos(3\pi x_1)(4\pi x_2) + 0.3$	Multimodal	$-100 \le x_i \le 100$	0	(Kumar et al. 2021c)	(2021b)
116	Boachevsky 3	$f_{116}(x) = x_1^2 + 2x_2^2 - 0.3\cos((3\pi x_1) + (4\pi x_2)) + 0.3$	Multimodal	$-100 \le x_i \le 100$	0	(Kumar et al. 2021c)	(2021b)
117	Trefethen 4	$f_{117}(x) = e^{\sin(50x_1)} + \sin(60e^{x_2}) + \sin(70\sin(x_1) + \sin(80\sin(x_2) + \frac{1}{4}(x_1^2 + x_2^2)$	Multimodal	$-6.5 \le x_1 \le 6.5, -4.5 \le x_2 \le 4.5$	− 3.30686865	(Adorio 2005)	(Al-Roomi 2021)
118	Helical Valley	$f_{118}(x) = 100\left[(x_2 - 10\theta)^2 + \left(\sqrt{x_1^2 + x_2^2} - 1\right)\right] + x_3^2$	Multimodal	$-10 \le x_i \le 10$	0	(Fateen and Bonilla-Petriciolet 2014)	(Al-Roomi 2021)
119	Mishra Zero-Sum	$f_{119}(x) = \left(10000\left\lvert\sum_{i=1}^{D} x_i\right\rvert\right)^{0.5}$	Multimodal	$-10 \le x_i \le 10$	0	(Function 2021)	NA
120	Wood	$f_{120}(x) = 100\left(x_1^2 - x_2\right)^2 + (x_1 - 1)^2 + (x_3 - 1)^2 + 90\left(x_3^2 - x_4\right)^2 + 10.1\left[(x_2 - 1)^2 + (x_4 - 1)^2\right] + 19.8(x_2 - 1)(x_4 - 1)$	Multimodal	$-1000 \le x_i \le 1000$	0	(Fateen and Bonilla-Petriciolet 2014)	NA
121	Katsuura	$f_{121}(x) = \frac{10}{D^2}\prod_{i=1}^{D}\left(1 + i\sum_{j=1}^{32}\frac{\left\lvert 2^j x_i - round(2^j x_i)\right\rvert}{2^j}\right)^{\frac{10}{D^{1.2}}} - \frac{10}{D^2}$	Multimodal	$0 \le x_i \le 100$	1	(Ding and Tan 2014)	([CSL STYLE ERROR: reference with no printed form.])

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
122	Hartman 3	$f_{122}(x) = -\sum_{i=1}^{4} c_i exp\left[-\sum_{j=1}^{3} a_{ij}(x_j - p_{ij})^2\right]$	Multimodal	$0 \le x_i \le 1$; $j \in \{1, 2, 3\}$	− 3.862782	(Civicioglu and Besdok 2019)	(GeoMath 2019a)
123	Miele Cantrell	$f_{123}(x) = (e^{-x_1} - x_2)^4 + 100(x_2 - x_3)^6 + (tan(x_3 - x_4))^4 + x_1^8$	Multimodal	$-1 \le x_i \le 1$	0	(Baykasoğlu and Akpinar 2017)(Chowdhury 2008)	(Al-Roomi 2021)
124	McCormick	$f_{124}(x) = x_1 + 2x_2 + (x_1 - x_2)^2 + \sin(x_1 + x_2) + 1$	Multimodal	$-1.5 \le x_1 \le 1.5$; $-3 \le x_2 \le 3$	− 1.9133	(Baykasoğlu and Akpinar 2017)(Bouchekara 2020)	(Oldenhuis 2021)
125	Keane	$f_{125}(x) = \frac{sin^2(x_1 - x_2) + sin^2(x_1 + x_2)}{\sqrt{x_1^2 + x_2^2}}$	Multimodal	$0 \le x_i \le 10$	− 0.673668	(Jamil and Yang 2013)	(Al-Roomi 2021)
126	Jennrich–Sampson	$f_{126}(x) = \sum_{i=1}^{10} (2 + 2i - (e^{ix_1} + e^{ix_2}))^2$	Multimodal	$-1 \le x_i \le 1$	124.3612	(Saghafi 2015)	(Al-Roomi 2021)
127	2-D sine curve	$f_{127}(x) = sin\left(7.5\sqrt{(x_1 - 2.5)^2 + (x_2 - 2.5)^2}\right)$	Multimodal	$0 \le x_i \le 5$	0.999	(Formato 2007)	(Al-Roomi 2021)
128	Modulated R^2	$f_{128}(x) = -[(x_1^2 + x_2^2) + sin\left(10\pi\sqrt{x_1^2 + x_2^2}\right)]$	Multimodal	$-5 \le x_i \le 5$	0.977	(Formato 2007)	(Al-Roomi 2021)
129	Stretched V sine wave	$f_{129}(x) = \sum_{i=1}^{D-1} (x_{i+1}^2 + x_i^2)^{0.25}\left[sin^2\left\{50(x_{i+1}^2 + x_i^2)^{0.1}\right\} + 0.1\right]$	Unimodal	$-10 \le x_i \le 10$	0	(Jamil and Yang 2013)	
130	Tripod	$f_{130}(x) = p(x_2)(1 + p(x_1)) + \lvert x_1 + 50p(x_2)(1 - p(x_1))\rvert + \lvert x_2 + 50(1 - 2p(x_2))\rvert$	Multimodal	$-100 \le x_i \le 100$	0	(Rahnamayan et al. 2007)	(Birge 2006)
131	Branins's	$f_{131}(x) = a(x_2 - bx_1^2 + cx_1 - d)^2 + e(1 - f)\cos(x_1) + e$	Multimodal	$-5 \le x_1 \le 10, 0 \le x_2 \le 15$	0.3979	(Rahnamayan et al. 2007)	(Ates 2020)
132	Brent	$f_{132}(x) = (x_1 + 10)^2 + (x_2 + 10)^2 + e^{-x_1^2 - x_2^2}$	Unimodal	$-10 \le x_i \le 10$	0	(Jamil and Yang 2013)	(Al-Roomi 2021)
133	Venter Sobieszczanski-Sobieski	$f_{133}(x) = x_1^2 - 100\cos(x_1)^2 - 100\cos\left(\frac{x_1^2}{30}\right) + x_2^2 - 100\cos(x_2)^2 - 100\cos\left(\frac{x_2^2}{30}\right)$	NA	$-50 \le x_i \le 50$	− 400	(Begambre and Laier 2009)	(Al-Roomi 2021)(2022c)
134	Discus	$f_{134} = (x_1 - 1)^2 + \sum_{i=2}^{D} i(2x_i^2 - x_{i-1})^2$	Unimodal	$-100 \le x_i \le 100$	_	(Oyelade and Ezugwu 2021)	NA
135	Holzman	$f_{135}(x) = \sum_{i=0}^{99}\left[e^{\frac{1}{x_1}(u_i - x_2)^{x_3}} - 0.1(i + 1)\right]$	NA	-	-	(Bouchekara 2020)	(Luclaurent and Leinweber)
136	Holzman 2	$f_{136}(x) = \sum_{i=1}^{n} ix_i^4$	NA	$-100 \le x_i \le 100$	_	(Oyelade and Ezugwu 2021)	(Al-Roomi 2021)
137	HGBat	$f_{137}(x) = \left\lvert(\sum_{i=1}^{D} x_i^2)^2 - (\sum_{i=1}^{D} x_i)^2\right\rvert^{\frac{1}{2}} + (0.5\sum_{i=1}^{D} x_i^2 + \sum_{i=1}^{D} x_i)/D + 0.5$	Multimodal	$-100 \le x_i \le 100$	_	(Oyelade and Ezugwu 2021)	(Al-Roomi 2021)
138	HappyCat	$f_{138}(x) = \left\lvert(\sum_{i=1}^{D} x_i^2)^2 - D\right\rvert^{\frac{1}{4}} + (0.5\sum_{i=1}^{D} x_i^2 + \sum_{i=1}^{D} x_i)/D + 0.5$	NA	$-2 \le x_i \le 2$	0	(Awad et al. 2016a)	(Al-Roomi 2021)
139	High conditioned elliptic	$f_{139}(x) = \sum_{i=1}^{D} (10^6)^{\frac{i-1}{D-1}} x_i^2$	NA	$-100 \le x_i \le 100$	_	(Oyelade and Ezugwu 2021)	(Hamish 2022)
140	Wavy 1	$f_{140}(x) = \sum_{i=1}^{n} x_i^2 + (\sum_{i=1}^{n} 0.5ix_i)^2 + (\sum_{i=1}^{n} 0.5ix_i)^4$	Multimodal	$-100 \le x_i \le 100$	0	(Oyelade and Ezugwu 2021)	(Luclaurent and Leinweber)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
141	Noise	$f_{141}(x) = \sum_{i=1}^{n} x_i^4 + random[0,1]$	NA	$-1.28 \le x_i \le 1.28$	–	(Oyelade and Ezugwu 2021)	NA
142	Levy and Montalo	$f_{142}(x) = 0.1(sin^2(3\pi x_1)) + \sum_{i=1}^{n}(x_i - 1)^2 [1 + sin^2(3\pi x_{i+1})] + (x_n - 1)^2(1 + sin^2(2\pi x_n))$	Multimodal	$-5 \le x_i \le 5$	0	(Oyelade and Ezugwu 2021)	(2022c)
143	Sargan	$f_{143}(x) = \sum_{i=1}^{n} n\left[x_i^2 + 0.4\sum_{j\neq 1} x_i x_j\right]$	Multimodal	$-100 \le x_i \le 100$	0	(Avramenko et al. 2021)	(Al-Roomi 2015)
144	Egg Holder	$f_{144}(x) = \sum_{i=1}^{m-1}[-(x_{i+1} +47)sin\sqrt{\lvert x_{i+1} + x_i/2 + 47\rvert} - x_i sin\sqrt{\lvert x_i - (x_{i+1} + 47\rvert}]$	Multimodal	$-512 \le x_i \le 512$	959.64	(Jamil and Yang 2013)	(Chen)
145	Levy	$f_{145}(x) = \sum_{i=1}^{n}(x_i - 1)^2[sin^2(3\pi x_{i+1})] + sin^2(3\pi x_1) + \lvert x_n - 1\rvert(1 + sin^2(3\pi x_n))$	Multimodal	$-10 \le x_i \le 10$	0	(Oyelade and Ezugwu 2021)	(GeoMath 2019b)
146	Lévy 3	$f_{146}(x) = \sum_{i=1}^{n-1}\left[0.5 + \frac{sin^2\left(\sqrt{100x_i^2 + x_{i+1}^2}\right) - 0.5}{1 + 0.001(x_i^2 - 2x_i x_{i+1} + x_{i+1}^2)^2}\right]$	NA	$-10 \le x_i \le 10$	_	(Oyelade and Ezugwu 2021)	(2022c)
147	Salomon	$f_{147}(x) = 1 - cos\left(2\pi\sqrt{\sum_{i=1}^{n} x_i^2}\right) + 0.1\sqrt{\sum_{i=1}^{n} x_i^2}$	Multimodal	$-100 \le x_i \le 100$	0	(Oyelade and Ezugwu 2021)	(Lima 2020)
148	Fox holes	$f_{148}(x) = \left[\frac{1}{500} + \sum_{j=1}^{25} \frac{1}{j + \sum_{i=1}^{2}(x_i - a_{i,j})^6}\right]^{-1}$	Multimodal	$-65.536 \le x_i \le 65.536$	1	(Xin Yao et al. 1999)	(GeoMath 2019b)
149	Freudenstein–Roth	$f_{149}(x) = (x_1 - 13 + ((5 - x_2)x_2 - 2)x_2)^2 + (x_1 - 29 + ((x_2 + 1)x_2 - 14)x_2)^2$	Multimodal	$-10 \le x_i \le 10$	0	(Dolinsky 2001)	(2021d)
150	Giunta	$f_{150}(x) = 0.6 + \sum_{i=1}^{2}[\sin(16/15x_i - 1) + \sin^2(16/15x_i - 1) + \frac{1}{50}\sin(4(16/15x_i - 1))]$	Multimodal	$-1 \le x_i \le 1$	0.060477	(Mishra 2006b)	(Oldenhuis 2021)
151	Aluffi-Pentiny	$f_{151}(x) = \frac{1}{4}x_1^4 - \frac{1}{2}x_2^2 + 0.1x_1 + 0.5x_2^2$	Multimodal	$-10 \le x_i \le 10$	-0.352386	(Kaveh and Talatahari 2010a)	(Bossek)
152	Cb3	$f_{152}(x) = 2x_1^2 - 1.05x_1^5 + \frac{1}{6}x_1^6 + x_1x_2 + x_2^2$	Multimodal	$-5 \le x_i \le 5$	0	(Kaveh and Talatahari 2010a)	NA
153	Cosine mixture	$f_{153}(x) = \sum_{i=1}^{n} x_i^2 - 0.1\sum_{i=1}^{n}\cos(5\pi x_i)$	Multimodal	$-1 \le x_i \le 1$	0.4	(Kaveh and Talatahari 2010a)	(Al-Roomi 2021)
154	Branin RCOS 2	$f_{154}(x) = \left(x_2 - \frac{5.1x_1^2}{4\pi^2} + \frac{5x_1}{\pi} - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos(x_1)\cos(x_2)\ln\left(x_1^2 + x_2^2 + 1\right) + 10$	Multimodal	$-5 \le x_i \le 15$	5.559037	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Baldea and Daoutidis 2012)
155	Bukin 2	$f_{155}(x) = 100\left(x_2 - 0.01x_1^2 + 1\right) + 0.01(x_1 + 10)^2$	Multimodal	$-15 \le x_1 \le -5$; $-3 \le x_2 \le 3$	0	(Jamil and Yang 2013)	(Luclaurent and Leinweber)
156	Bukin 4	$f_{156}(x) = 100x_2^2 + 0.1\lvert\lvert x_1 + 10\rvert\rvert$	Multimodal	$-15 \le x_1 \le -5$; $-3 \le x_2 \le 3$	0	(Bouchekara 2020)	(Luclaurent and Leinweber)
157	Bukin 6	$f_{157}(x) = 100\sqrt{\lvert\lvert x_2 - 0.01x_1^2\rvert\rvert} + 0.01\lvert\lvert x_1 + 10\rvert\rvert$	Multimodal	$-15 \le x_1 \le -5$; $-3 \le x_2 \le 3$	0	(Harifi et al. 2019)	(Franco 2018)
158	Chen Bird	$f_{158}(x) = \frac{0.001}{\left\lfloor(0.001)^2 + (x_1 - 0.4x_2 - 0.1)^2\right\rfloor} - \frac{0.001}{\left\lfloor(0.001)^2 + (2x_1 + x_2 - 1.5)^2\right\rfloor}$	Multimodal	$-500 \le x_i \le 500$	− 2000	(Hashim et al. 2022)	(Al-Roomi 2021)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
159	Chen V	$f_{159}(x) = -\frac{0.001}{\left[(0.001)^2+\left(x_1^2+x_2^2-1\right)^2\right]} - \frac{0.001}{\left[(0.001)^2+\left(x_1^2+x_2^2-0.5\right)^2\right]} - \frac{0.001}{\left[(0.001)^2+\left(x_1^2-x_2^2\right)^2\right]}$	Multimodal	$-500 \le x_i \le 500$	− 2000	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
160	Zirilli	$f_{160}(x) = 0.5x_1^2 + 0.5(1-\cos(2x_1)) + x_2^2$	Multimodal	$-500 \le x_i \le 500$	− 2000	(Hashim et al. 2022)	(Al-Roomi 2021)
161	Sawtoothxy	$f_{161}(x) = \left[\sin(r) - \frac{Sin(2r)}{2} + \frac{Sin(3r)}{3} + \frac{Sin(4r)}{4}\right]\left(\frac{r^2}{r+1}\right)$ $[0.5\cos(2t-0.5)+\cos(t)+2]$	Multimodal	$-20 \le x_i \le 20$	0	(Hashim et al. 2019)	(Al-Roomi 2021)
162	Damavandi	$f_{162}(x) = \left[1 - \left\lvert\frac{sin[\pi(x_1-2)]sin[\pi(x_2-2)]}{\pi^2(x_1-2)(x_2-2)}\right\rvert\right]$ $X\left[2+(x_1-7)^2+2+(x_2-7)^2\right]$	Multimodal	$0 \le x_i \le 14$	0	(Hashim et al. 2022)	(Al-Roomi 2021)
163	Csendes	$f_{163}(x) = \sum_{i=1}^{D} x_i^6\left[2+\sin\left(\frac{1}{x_i}\right)\right]$	Multimodal	$-1 \le x_i \le 1$	0	(Hashim et al. 2022)	(Optimization and Mathematica 2006)
164	Whitley	$f_{164}(x) = \sum_{i=1}^{D}\sum_{j=1}^{D}\left[\frac{(100\left(x_i^2-x_j\right)^2+(1-x_j)^2)^2}{4000}\right.$ $\left.-\cos(100\left(x_i^2-x_j\right)^2+\left(1-x_j\right)^2+1)\right]$	Multimodal	–	–	(Jamil and Yang 2013)	(Baldea and Daoutidis 2012)
165	*W/Wavy*	$f_{165}(x) = 1 - \frac{1}{D}\sum_{i=1}^{D} cos(kx_i)e^{-\frac{x_i^2}{2}}$	Multimodal	$-\pi \le x_i \le \pi$	0	(Jamil and Yang 2013)	(Al-Roomi 2021)
166	Wayburn Seader 1	$f_{166}(x) = (x_1^6 + x_2^4 - 17)^2 + (2x_1 + x_2 - 4)^2$	Unimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
167	Wayburn Seader 2	$f_{167}(x) = \left[1.613 - 4(x_1-0.3125)^2 - 4(x_2-1.625)^2\right]^2 + (x_2-1)^2$	Unimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
168	Wayburn Seader 3	$f_{168}(x) = \frac{2x_1^3}{3} - 8x_1^2 + 33x_1 - x_1x_2 + 5 + \left[(x_1-4)^2 + (x_2-5)^2 - 4\right]^2$	Unimodal	$-500 \le x_i \le 500$	21.35	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
169	Watson	$f_{169}(x) = \sum_{i=0}^{29}\left[\sum_{j=0}^{4}((j-1)a_i^j x_{j+1} - \left[\sum_{j=0}^{5} a_i^j x_{j+1}\right]^2 - 1\right]^2 + x_1^2$	Unimodal	$\le x_i \le 10$	0.002288	(Jamil and Yang 2013)	(Al-Roomi 2021)
170	Ursem 1	$f_{170}(x) = -\sin(2x_1 - 0.5\pi) - 3\cos(x_2) - 0.5(x_1)$	Multimodal	$-2.5 \le x_1 \le 3;$ $-2 \le x_2 \le 2$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
171	Ursem 3	$f_{171}(x) = -\sin(2.2\pi x_1 + 0.5\pi).\frac{2-\lvert x_2\rvert}{2}.\frac{3-\lvert x_1\rvert}{2}$ $-\sin\left(0.5\pi x_2^2 + 0.5\pi\right).\frac{2-\lvert x_2\rvert}{2}.\frac{2-\lvert x_1\rvert}{2}$	Multimodal	$-2 \le x_1 \le 2;$ $-1.5 \le x_2 \le 1.5$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
172	Ursem 4	$f_{172}(x) = -\sin(0.5\pi x_1 + 0.5\pi).\frac{2-\sqrt{x_1^2+x_2^2}}{4}$	Multimodal	$-2 \le x_i \le 2$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
173	Ursem Waves	$f_{173}(x) = -0.9x_1^2 + \left(x_2^2 - 4.5x_2^2\right)x_1x_2$ $+4.7\cos\left(3x_1 - x_2^2(2+x_1)\right)\sin(2.5\pi x_1)$	Multimodal	$-0.9 \le x_1 \le 1.2$ $-1.2 \le x_2 \le 1.2$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
174	Cube	$f_{174}(x) = 100(x_2 - x_1^3)^2 + (1-x_1)^2$	Unimodal	$-10 \le x_i \le 10$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
175	Deb 1	$f_{175}(x) = -\frac{1}{D}\sum_{i=1}^{D} sin^6(5\pi x_i)$	Multimodal	$-1 \le x_i \le 1$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA
176	Deb 3	$f_{176}(x) = -\frac{1}{D}\sum_{i=1}^{D} sin^6(5\pi(x_i^{3/4} - 0.05))$	Multimodal	$-1 \le x_i \le 1$	–	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA
177	Deckkers Aarts	$f_{177}(x) = 10^5 x_1^2 + x_2^2 - (x_1^2 + x_2^2)^2 + 10^{-5}(x_1^2 + x_2^2)^4$	Multimodal	$-20 \le x_i \le 20$	− 24,777	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA
178	deVilliers Glasser 1	$f_{178}(x) = \sum_{i=1}^{24} \left[x_1 x_2^{t_i} \sin(x_3 t_i + x_4) - y_i\right]^2$	Multimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA
179	deVilliers Glasser 2	$f_{179}(x) = \sum_{i=1}^{16} \left[x_1 x_2^{t_i} \tanh[x_3 t_i + \sin(x_4 t_i)]\cos(t_i e^{x_5}) - y_i\right]^2$	Multimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Layeb 2021)
180	Trigonometric 1	$f_{180}(x) = \sum_{i=1}^{D} \left[D - \sum_{j=1}^{D} cosx_j + i(1 - \cos(x_i) - \sin(x_i))\right]^2$	Multimodal	$-0 \le x_i \le \pi$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
181	Trigonometric 2	$f_{181}(x) = 1 + \sum_{i=1}^{D} 8sin^2\left[7(x_i - 0.9)^2\right] + 6sin^2\left[14(x_i - 0.9)^2\right] + (x_i - 0.9)^2$	Multimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
182	Shubert 3	$f_{182}(x) = \left(\sum_{i=1}^{D}\sum_{j=1}^{5} jsin((j+1)x_i + j)\right)$	Multimodal	$-10 \le x_i \le 10$	− 29.6733337	(Jamil and Yang 2013)	(Al-Roomi 2021)
183	Shubert 4	$f_{183}(x) = \left(\sum_{i=1}^{D}\sum_{j=1}^{5} jcos((j+1)x_i + j)\right)$	Multimodal	$-10 \le x_i \le 10$	− 25.740858	(Jamil and Yang 2013)	(Al-Roomi 2021)
184	Dolan	$f_{184}(x) = (x_1 + 1.7x_2)\sin(x_1) - 1.5x_3 - 0.1x_4\cos(x_4 + x_5 - x_1) + 0.2x_5^2 - x_2 - 1$	Multimodal	$-100 \le x_i \le 100$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
185	El-Attar-Vidyasagar-Dutta	$f_{185}(x) = (x_1^2 + x_2 - 10)^2 + (x_1 + x_2^2 - 7)^2 + (x_1^2 + x_2^3 - 1)^2$	Unimodal	$-500 \le x_i \le 500$	0.470427	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Al-Roomi 2021)
186	EX 1	$f_{186}(x) = 0.1(1 - x_1^2) + 0.1\sin(10x_1) + (11 - x_2)^2 + \sin(10x_2)$	Multimodal	$x_i \in [0, 2] \cap [10, 12]$	− 1.28186	(Jamil and Yang 2013)	NA
187	Gulf Research	$f_{187}(x) = \sum_{i=1}^{99} \left[exp\left(-\frac{(u_i - x_2)^{x_3}}{x_i}\right) - 0.01i\right]^2$	Multimodal	$-0.1 \le x_1 \le 100$; $0 \le x_2 \le 25.6$; $0 \le x_3 \le 5$	0.470427	(Jamil and Yang 2013) (Baykasoğlu and Akpinar 2017) (Bouchekara 2020)	(Al-Roomi 2021)
188	Schumer Steiglitz	$f_{188}(x) = \sum_{i=1}^{D} x_i^4$	Multimodal	–	0.470427	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Lima 2020)
189	Schmidt Vetters	$f_{189}(x) = \frac{1}{1+(x_1 - x_2)^2} + \sin\frac{(\pi x_2 + x_3)}{2} + e^{\left(\frac{x_1 + x_2}{x_2} - 2\right)^2}$	Multimodal	–	3	(Function 2014)	(Al-Roomi 2021)
190	Schaffer F6	$f_{190}(x) = \sum_{i=1}^{D} 0.5 + \frac{sin^2\sqrt{x_i^2 + x_{i+1}^2} - 0.5}{[1 + 0.001(x_i^2 + x_{i+1}^2)]^2}$	Multimodal	$-100 \le x_i \le 100$	0	(Hashim et al. 2019)	(Lima 2020)
191	Rump	$f_{191}(x) = (333.75 - x_1^2)x_2^6 + x_1^2(11x_1^2x_2^2 - 121x_2^4 - 2) + 5.5x_2^8 + \frac{x_1}{2x_2}$	Multimodal	$-500 \le x_i \le 500$	0	(Function 2014)	NA

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
192	Ellipse	$f_{192}(x) = \sum_{i=1}^{n} (10^{(\frac{4(i-1)}{n-1})}) x_i^2$	Unimodal	$-10 \le x_i \le 10$	0	(Sharafi et al. 2016)	NA
193	Rotated Ellipse	$f_{193}(x) = 7x_1^2 - 6\sqrt{3}x_1x_2 + 13x_2^2$	Multimodal	$-500 \le x_i \le 500$	0	(Function 2014)	NA
194	Rotated Ellipse 2	$f_{194}(x) = x_1^2 - x_1x_1 + x_2^2$	Multimodal	$-500 \le x_i \le 500$	0	(Function 2014)	NA
195	Ripple 1	$f_{195}(x) = \sum_{i=1}^{2} -e^{-2ln2\left(\frac{x_i-0.1}{0.8}\right)^2} (sin^6(5\pi x_i) + 0.1cos^2(500\pi x_i))$	Multimodal	$0 \le x_i \le 1$	− 2.2	(Talatahari et al. 2021)(Function 2014)	NA
196	Ripple 25	$f_{196}(x) = \sum_{i=1}^{2} -e^{-2ln2\left(\frac{x_i-0.1}{0.8}\right)^2} (sin^6(5\pi x_i))$	Multimodal	$0 \le x_i \le 1$	− 2	(Jamil and Yang 2013)(Talatahari et al. 2021)	NA
197	Rana	$f_{197}(x) = \sum_{i=1}^{n} (x_{i+1}+1)cos\left(\sqrt{\lvert\lvert x_{i+1} - x_i + 1\rvert\rvert}\right) sin\left(\sqrt{\lvert\lvert x_{i+1} + x_i + 1\rvert\rvert}\right) + x_i cos\left(\sqrt{\lvert\lvert x_{i+1} + x_i + 1\rvert\rvert}\right) Sin\left(\sqrt{\lvert\lvert x_{i+1} - x_i + 1\rvert\rvert}\right)$	Multimodal	$-500 \le x_i \le 500$	–	(Bouchekara 2020)	NA
198	Quadratic	$f_{198}(x) = -3803.84 - 138.08x_1 - 232.92x_2 + 128.08x_1^2 + 203.64x_2^2 + 182.25x_1x_2$	NA	$-10 \le x_i \le 10$	− 3873.7243	(Function 2014)	NA
199	Qing	$f_{199}(x) = \sum_{i=1}^{D} (x_i^2 - i)^2$	Multimodal	$-500 \le x_i \le 500$	0	(Jamil and Yang 2013)(Talatahari et al. 2021)	(Wang 2018)
200	Price 1	$f_{200}(x) = (\lvert x_1\rvert - 5)^2 + (\lvert x_2\rvert - 5)^2$	Multimodal	$-500 \le x_i \le 500$	0	(Bouchekara 2020)(Talatahari et al. 2021)	(Luclaurent and Leinweber)
201	Price 2	$f_{201}(x) = 1 + sin^2(x_1) + sin^2(x_2) - 0.1e^{-x_1^2 - x_2^2}$	Multimodal	$-10 \le x_i \le 10$	0.9	(Bouchekara 2020)(Talatahari et al. 2021)	(Luclaurent and Leinweber)
202	Price 3	$f_{202}(x) = 100(x_2 - x_1^2)^2 + 6\left[6.4(x_2 - 0.5)^2 - x_1 - 0.6\right]^2$	Multimodal	$-500 \le x_i \le 500$	0	(Bouchekara 2020)(Talatahari et al. 2021)	(Luclaurent and Leinweber)
203	Price 4	$f_{203}(x) = (2x_1^3 - x_2^3)^2 + (6x_1 - x_2^2 + x_2)^2$	Multimodal	$-500 \le x_i \le 500$	0	(Bouchekara 2020)(Talatahari et al. 2021)	(Luclaurent and Leinweber)
204	Hansen	$f_{204}(x) = \sum_{i}^{4} (i+1)\cos(ix_1 + i + 1) \sum_{j=0}^{4} (j+1)\cos((j+2)x_2 + j + 1)$	Multimodal	$-10 \le x_i \le 10$	–	(Bouchekara 2020)(Talatahari et al. 2021)(Function 2014)	(Luclaurent and Leinweber)
205	Himmelblau	$f_{205}(x) = \frac{1}{n}\sum_{i=1}^{n} (x_i^4 - 16x_i^2 + 5x_i)$	Multimodal	$-10 \le x_i \le 10$	–	(Kiran 2015)	(Pilario 2020)
206	Hosaki	$f_{206}(x) = (1 - 8x_1 + 7x_1^2 - \frac{7}{3x_1^3} + \frac{1}{4x_1^4})x_2^2 e^{-x_2}$	Multimodal	$0 \le x_1 \le 5, 0 \le x_2 \le 6$	− 2.3458	(Bouchekara 2020)(Talatahari et al. 2021)(Function 2014)	(Shrestha 2010)

Table 2 (continued)

S. no	Function name	Objective function	Characteristics of test functions	Search space	Global min (f_{min}(x))	References	MATLAB code reference
207	Parsopoulos	$f_{207}(x) = cos(x_1)^2 + sin(x_2)^2$	Multimodal	$-5 \le x_i \le 5$	–	(Parsopoulos and Vrahatis 2002)	(Al-Roomi 2021)
208	Levy No. 5	$f_{208}(x) = \sum_{i=1}^{5}[icos((i-1)x_1+i)]\sum_{j=1}^{5}[jcos((j+1)x_2+j)]$	Multimodal	$-10 \le x_i \le 10$	− 176.1375	(Parsopoulos and Vrahatis 2002)	(Luclaurent and Leinweber)
209	Levy No. 8	$f_{209}(x) = sin^2(\pi y_1) + \sum_{i=1}^{n-1}(y_i-1)^2[1 +10sin^2(\pi y_{i+1})] + (y_n-1)^2$	Multimodal	$-10 \le x_i \le 10$	0	(Parsopoulos and Vrahatis 2002)	NA
210	Paviani	$f_{210}(x) = \sum_{i=1}^{10}\left[(\ln(x_i-2))^2 + (\ln(10-x_i))^2\right] - \left(\prod_{i=1}^{10} x_i\right)^{0.2}$	Multimodal	$-2.001 \le x_i \le 10$	− 45.778	(Jamil and Yang 2013)(Baykasoğlu and Akpinar 2017)(Talatahari et al. 2021)	(Luclaurent and Leinweber)
211	Pinter	$f_{211}(x) = \sum_{i=1}^{D} ix_i^2 + \sum_{i=1}^{D} 20sin^2(x_{i-1}sinx_i + sinx_{i+1}) + \sum_{i=1}^{D} ilog_{10}(1 + i(x_{i-1}^2 - 2x_i + 3x_{i+1} - cosx_i + 1)^2)$	Multimodal	$-10 \le x_i \le 10$	0	(Jamil and Yang 2013)	(Luclaurent and Leinweber)
212	Periodic	$f_{212}(x) = 1 + sin^2(x_1)^2 + sin^2(x_2)^2 - 0.1e^{-1(x_1^2+x_2^2)}$	Multimodal	$-10 \le x_i \le 10$	0.9	(Hashim et al. 2019)	(Leong 2016)
213	Styblinski Tang Noise	$f_{213}(x) = \frac{1}{2}\sum_{i=1}^{d} x_i^4 - 16x_i^2 + 5x_i$	Multimodal	$-5 \le x_i \le 5$	− 39.16599d	(Kim et al. 2022)(Örnek et al. 2022)	(Kelly 2016)(Kelly)
214	Bumped Bowl	$f_{214} = log\|x\|^2 + e^{-10\|x\|^2}$	Multimodal	$-4 \le x_i \le 4$	1.1	(Sanders et al. 2019)(Mirjalili and Lewis 2016)	NA
215	Hartman 6	$f_{122}(x) = -\sum_{i=1}^{4} c_i exp\left[-\sum_{j=1}^{6} a_{ij}(x_j - p_{ij})^2\right]$	Multimodal	$0 \le x_j \le 1$	− 3.32236	(Charilogis et al. 2022)	(Julie 2012)

Source code is not available (NA)

names, mathematical equations, and the characteristics of test functions (unimodal test functions and multimodal test functions) as well as references their MATLAB code from MathWorks of MATLAB Central, Github, research papers, and webpages.

3.1 Insights from different competitions

Various competitions are held to evaluate the performance of MH optimization techniques. These competitions provide different test suites to evaluate the performance of optimization methods. Among all those competitions, the Congress of Evolutionary Computation (CEC) has provided a different set of benchmark test functions almost every year. CEC held the evolutionary Real Parameter Single Objective Optimization (RPSOO) competition in 2005 and provided a test suite CEC-05 (Suganthan et al. 2005). In 2006, there was a contest on evolutionary constrained RPSOO and provided a test suite CEC-06 (Liang et al. 2013). In 2007, CEC held a competition on the performance assessment of Real Parameter Multi-Objective Optimization Algorithms (MOEAs) and provided 13 benchmark functions test suites (Huang et al. 2007). In 2008, the competition for large-scale single-objective global optimization with bound constraints was born (Tang et al. 2008). In 2009, CEC held a contest on dynamic optimization and Performance Assessment of real-parameter MOEAs and provided 30 test functions in their test suite (CEC-09) (Li et al. 2009; Zhang et al. 2009b). In 2010, evolutionary-constrained RPSOO was utilized in the CEC-2010 competition and provided 18 test functions (Mallipeddi and Suganthan 2010a). While in 2011, Testing Evolutionary Algorithms on Real-world Numerical Optimization Problems were used in the CEC-2011 competition (Das and Suganthan 2011).

Since 2013, CEC has held Special Sessions as well as Competitions on RPSOO, which have indeed been organized yearly to test the efficiency of new optimization algorithms. Every year, this competition examines algorithms that rely on their performance. A wide variety of benchmark functions may vary from season to season. Each function has to be computed for the dimensions of 10, 30, and 50 in the first year (2013). And from the year 2014, every function has to be computed for the dimensions of 100. In 2016 and 2018, no new benchmark test function was used, as presented in Table 3. In 2019, CEC 2019 provided a benchmark test function suite with 10 functions, and the competition was based on the 100-Digit Challenge on Single Objective Numerical Optimization. In addition, single objective benchmark problems are being used to test all new MH optimization Techniques. Table 3 illustrates the CEC benchmark test function and their reference and MATLAB code references, and the best top 3 algorithms from IEEE CEC with their reference. The result of the IEEE-CEC-2022 will be declared soon. In CEC-2006, CEC-2015 and CEC-2018, two algorithms got the third position, and both got the same score. In 2009, IEEE CEC organized two competitions: Performance Assessment of Constrained/Bound Constrained Multi-Objective Optimization Algorithms (23 test functions) (Zhang et al. 2009b), and Evolutionary Computation in Dynamic and Uncertain Environments (7 test functions) (Li et al. 2009).

From 2005, there was no other real parameter competition until 2009. In 2009, GECCO held a special session in which the Black Box Optimization Benchmark (BBOB) was proposed. The main difference is that BBOB has only two sets of functions: noisy and noiseless functions with different dimension values (Molina et al. 2018). In 2016, a bi-objective test suite was proposed. This test suite contains 55 noiseless multi-objective and 24 bi-objective test functions. The BBOB has a test suite with various dimensions with 24 noiseless test functions (COCO standards). The noisy test suit has 30 noisy test functions with various dimensions. Also 2017, 10 real word problems from various domains were added to this competition (2022f). Since 2020, CEC and GECCO Competitions have been declaring the results of the competition collectively. The drawback of the CEC Competition in recent years is that, in a given year, they have provided different sets of benchmark test functions and the results; thus far obtained may or may not be superior to the previous year's winner algorithms.

After 2020, the IEEE CEC and GECCO competitions declared results together. In 2020, GECCO and CEC held two completions: Real-World Single Objective Constrained Optimization and Real Parameter Multimodal Multi-objective Optimization (Awad et al. 2020; Suganthan et al. 2020). In 2021, CEC and GECCO held three competitions: Multi-modal Multi-objective Path Planning Optimization, Real-World Multi-objective Constrained Optimization, and Real Parameter Single Objective Bound Constrained Optimization. And the result of CEC 2022 has not yet been announced. Table 4 shows the test suite, test function, and code reference used in the GECCO-BBOB competition. And Table 5 presents the BBOB competition year, benchmark test functions, and the best top 3 algorithms from BBOB.

4 Bibliometric maps for engineering design problems

A bibliometric study systematically provides additional evidence to identify global trends in research in a relevant area using the results of a scholarly articles database such as Scopus. The Scopus database has been used to conduct data gathering from April 11th, 2022. The main objective

Table 3 IEEE-CEC code with benchmark test function, their Matlab code reference, and best top 3 algorithms (IEEE-CEC) with their reference

S No	Competition year	CEC benchmark function	Benchmark functions	MATLAB Code reference	1st position	2nd position	3rd position	References
1	2005	CEC-2005	25	(Suganthan 2022a)	G-CMA-ES (Auger and Hansen 2005)	NA	NA	(Suganthan et al. 2005)(Suganthan 2022a)
2	2006	CEC-2006	24	(Suganthan 2022b)	ε_DE (Takahama and Sakai 2006)	DMS-PSO (Liang and Suganthan 2006)	SaDE (Huang et al. 2006), and MDE (Mezura-Montes et al. 2006)	(Liang et al. 2013)(Suganthan 2022b)
3	2007	CEC-2007	13	(Huang et al. 2007) (Suganthan 2022c)	NSGAII_SBX (Sharma et al. 2007)	GDE3 (Kukkonen and Lampinen 2007)	NSGAII_PCX (Kumar et al. 2007)	(Suganthan 2022c) (Huang et al. 2007)
4	2008	CEC-2008	7	(Suganthan 2022d)	MTS (Lin-Yu Tseng and Chun Chen 2008)	LSEDA-gl (Yu Wang and Bin Li 2008)	jDEdynNP-F (Brest et al. 2008)	(Tang et al. 2008)(Suganthan 2022d)
5	2009	CEC-2009 (Dynamic Optimization)	6	(Suganthan 2022e) (Suganthan 2022f)	jDE (Brest et al. 2009)	NA	NA	(Li et al. 2009)(Zhang et al. 2009b)(Suganthan 2022e)
		CEC-2009 (Unconstrained multi-objective Problem)	10	(Suganthan 2022g)	MOEA/D (Zhang et al. 2009a)	LiuLi (Liu and Xueqiang Li 2009)	DMOEADD (Liu et al. 2009)	(Zhang et al. 2009b)
6	2010	CEC-2010	18	(Suganthan 2022h)	εDEg (Takahama and Sakai 2010)	ECHT (Mallipeddi and Suganthan 2010b)	jDEsoco (Brest et al. 2010)	(Suganthan 2022h) (Mallipeddi and Suganthan 2010a)
7	2011	CEC-2011	22	(Suganthan 2010)	GA-MPC (Elsayed et al. 2011a)	DE –ΛCr (Reynoso-Meza et al. 2011)	SAMODE (Elsayed et al. 2011b)	(Suganthan 2011) (Das and Suganthan 2011)
8	2013	CEC-2013	28	(Tian 2022)(Suganthan 2022i)	NBIPOPaCMA (Loshchilov 2013)	Icmaesils (Liao and Stützle 2013)	DRMA-LSCh-CMA (Lacroix et al. 2013)	(Loshchilov et al. 2013) (Suganthan et al. 2013)
9	CEC-2014	CEC-2014	30	(Suganthan 2022j)	MVMO (Erlich et al. 2014)	SOMODS (Muller et al. 2014)	Without-Approx (Md Asafuddoula 2014)	(Suganthan 2022k)
10	2015	CEC-2015	15	(Suganthan 2022k)	SPS-L-SHADE-EIG (Guo et al. 2015)	DEsPA (Awad et al. 2015)	MVMO (Rueda and Erlich 2015) and LSHADE-ND (Sallam et al. 2015)	(Liang et al. 2014)(2022d)
11	2016	CEC-2014	NA	NA	UMOEAII (Elsayed et al. 2016) and LSHADE_EpSin (Awad et al. 2016b)	NA	NA	(2022e)
		CEC-2015	NA	NA	MVMO (Rueda and Erlich 2016)	NA	NA	(2022e)

Table 3 (continued)

S No	Competition year	CEC benchmark function	Benchmark functions	MATLAB Code reference	1st position	2nd position	3rd position	References
12	2017	CEC-2017	29	(Yadav 2020)(Suganthan 2022l)	EBO with CMAR (Kumar et al. 2017)	jSO (Brest et al. 2017)	LSHADE-cnEpSin (Awad et al. 2017b)	(2017a) (Awad et al. 2017a)
13	2018	CEC-2017	NA	(Suganthan 2022m)	HS-ES (Zhang and Shi 2018)	LSHADE-RSP (Stanovov et al. 2018)	ELSHADE_SPACMA (Hadi et al. 2018)	([CSL STYLE ERROR: reference with no printed form.])
14	2019	CEC-2019	10	(K. V. Price, N. H. Awad, M. Z. Ali 2018)	jDE100 (Brest et al. 2019)	DISHchain1e + 12 (Zamuda 2019)	HyDE-DF (Lezama et al. 2019) and SOMA-T3A (Diep et al. 2020)	(K. V. Price, N. H. Awad, M. Z. Ali 2018)
15	2020	CEC-2020 (Single Objective Bound Constrained Numerical Optimization)	10	(Suganthan 2022n)	IMODE (Sallam et al. 2020)	AGSK (Mohamed et al. 2020)	j2020 (Brest et al. 2020)	(Suganthan 2022n)(Awad et al. 2020)
		CEC-2020 (Real World Single Objective Constrained Optimization)	57	(Suganthan 2022o)	SASS (Kumar et al. 2020a)	COLSHADE (Gurrola-Ramos et al. 2020)	sCMAgES (Kumar et al. 2020b)	(Suganthan et al. 2020)
16	2021	CEC-2021, GECCO-21 (Real Parameter Single Objective Bound Constrained Optimization)	10	(Suganthan 2022p)	IMODE(Mohamed et al. 2021)	MadDE (Biswas et al. 2021)	NL-SHADE-RSP (Stanovov et al. 2021)	(Suganthan 2022p) (Awad et al. 2020)
		CEC-2021, GECCO-21 (Real-World Multi-objective Constrained Optimization)	13	(Kumar et al. 2021b)	cNSGAIII (Jain and Deb 2014)	cARMOEA (Tian et al. 2018)	AnD (Liu et al. 2020)	(Kumar et al. 2021b)
		CEC-2021, GECCO-21 (Multi-modal Multi-objective Path Planning Optimization)	50	(Yue et al. 2021)	MOASTAR (Jin 2021)	MACOSX (Zhao et al. 2021)	MOEASES (Dong et al. 2021)	(Yue et al. 2021)
17	2022	CEC-2022	12	(Suganthan 2022q)	NA	NA	NA	(Kumar et al. 2021a)

Source code is not available (NA)

of this bibliometric study is to research manuscripts containing "Engineering Design Problems" in the manuscript title, manuscript abstracts, and manuscript keywords. A total of 1665 articles have been revealed at the end of the search. From 1953 to 2022, the annual number of scientific publications on "Engineering Design Problems" indexed in the Scopus database is presented in Fig. 7.

The data obtained from the Scopus database, Journal research articles constituted 60.3% (1004), journal review articles 0.78% (13), conference research articles 36.156% (602), conference review articles 0.541% (9), book 0.3% (5), book chapter 1.74% (29), editorial 0.06% (1), reports 0.06% (1), and unidentified articles 0.06% (1). Figure 8 shows the number of scientific publications on "Engineering Design Problems" indexed in the Scopus database from 1953 to 2022.

The authorship analysis related to "Engineering Design Problems" indexed in the Scopus database from 1953 to 2022. A total of 159 authors were revealed at the end of the search. However, only 18 authors have at least 10 publications. Among these authors, Kalyanmoy Deb (Michigan State University) has the highest number of publications with 40 articles (most cited article (Deb 2000) 2873 citation), followed by Yongquan Zhou (Guangxi University for Nationalities) with 28 articles (most cited article (Zhou et al. 2016) 80 citation) and Qifang Luo also from Guangxi University for Nationalities has published 19 articles while most cited article (Zhou et al. 2016) with 80 citations. Figure 9 compares the authors with the number of publications (at least 10) related to "Engineering Design Problems" indexed in the Scopus database from 1953 to 2022.

A total of 160 institutes were revealed at the end of the search. However, only 18 institutes have more than 15 publications among these institutes, as illustrated in Fig. 10, Guangxi University for Nationalities has published the highest number of publications (30), followed by Huazhong University of Science and Technology (29) and The Indian Institute of Technology Kanpur published 26 articles.

A total of 71 countries contribute their research in the field of engineering design problems worldwide. However, only 13 nations have at least 30 publications among these nations. Continental, Asia has the most active continent (47% of total publishing), followed by North America (31%), while Africa, Australia, and South America are the least active continents (4%, 3%, and 1%) as presented in Fig. 11. Also, Europe has contributed 4% in the engineering design problems' area.

Table 6 illustrates the Journal name, cite score, publisher, most cited article related to Engineering Design Problems, the total number of citations, the Category of the journal, Open Access, and the number of publications related to Engineering Design Problems have more than 11 publications. The highest number of papers was published in Proceedings Of The ASME Design Engineering Technical Conference (70), followed by Expert Systems with Applications and Lecture Notes In Computer Science, Including Subseries Lecture Notes In Artificial Intelligence And Lecture Notes In Bioinformatics has been published 39 articles. The highest cited paper has been published in Computer Methods in Applied Mechanics and Engineering (Elsevier publisher) with 2873 citations, followed by Computers and Structures journal (Elsevier publisher) with 998 citations. According to the cite score, only three journals have cite score of more than 10. Expert Systems with Applications journal has the highest cite score (12.7), while Advances in Intelligent Systems and Computing journal has a low cite score (0.9).

5 Real-world engineering design benchmark test function

The literature shows that the MH optimization performance was evaluated and compared with real-world engineering design problems. Real-world engineering design benchmark problems have been shown to demonstrate a higher level of complexity compared to the conventional benchmark functions that have been previously discussed. They may have multiple objectives, a variety of design variables, as well as constraints that show non-linear behaviour. The utilization of real-world engineering design benchmark problems provides significant importance in assessing optimization algorithms. By evaluating the performance of various algorithms across a diverse range of problem domains, it becomes feasible to identify the algorithms that demonstrate the highest efficacy for a particular application. Table 7 summarises 57 real-world engineering design problems as benchmark test functions and references their

Table 4 Test suite, test function, and code reference

S. no	Test suite	Function	Code reference	References
1	Noiseless	24	(2022 g)	(Doerr et al. 2009)
2	Noisy	30	(2022 g)	(2022 h)
3	Bi-objective	79	(2022 g)	(Hansen et al. 2021)(2022i)
4	Real word	10	(2022f)	(2022f)

Table 5 BBOB competition year, benchmark test functions, and the best top three algorithms from BBOB

S no	Competition year	BBOB benchmark function	1st position	2nd position	3rd position	References
1	2015	GECCO-2015	KNITRO	MVMO	Naive Multi-Scale Optimization	(2022f)(2022j)
2	2016	GECCO-Three Objective	General UB-MO-CMA-ES	Restarted local search + SMS-EMOA	Grays	(2022f)(2022j)
		GECCO-Expensive Two Objective	PA-DDS	Artelys Knitro	MO-SOO	(2022f) (2022j)
		GECCO-Two Objective	Restarted local search + SMS-EMOA	UB-MO-CMA-ES	Poly Montreal	(2022f) (2022j)
		GECCO-Expensive Single Objective	Artelys Knitro	Mickey	Simon Wessing	(2022f) (2022j)
		GECCO-Single Objective	Two-stage algorithms	NA	NA	(2022f) (2022j)
3	2017	EMO-Real World	Model-based HV-maximization	BMOO	PADDS-CHC	(2022f) (2017b)
		GEECO-Three Objective	Restarted local search + SMS-EMOA	Artelys Knitro	Custom algorithm	(2022f) (2017b)
		GEECO-Expensive Two Objective	Artelys Knitro	MBHVMAX + SMSEMOA	Emagin's Tomcat	(2022f) (2017b)
		GEECO-Two Objective	Restarted local search + SMS-EMOA	Artelys Knitro	NA	(2022f) (2017b)
		GEECO-Expensive Single Objective	DTS-CMA-ES or BOBYQA	RLMBO + LBFGSB + RS	Artelys Knitro	(2022f) (2017b)
		GEECO-Single Objective	AS-AC-CMA-ES	aDTS-CMA-ES	Two-stage algorithms	(2022f) (2017b)
4	2018	GEECO-Three Objective	jpsbook	Artelys	Research algorithm	(2022f) (2018)
		GEECO-Expensive Two Objective	Artelys	jpsbook	MBHVMAX + ALGSS	(2022f) (2018)
		GEECO-Two Objective	jpsbook	Artelys	Research algorithm	(2022f) (2018)
		GEECO-Expensive Single Objective	Gaussian processes + Local search	biteopt2017	Artelys	(2022f) (2018)
		GEECO- Single Objective	Feature Based Algorithm Selection	NA	NA	(2022f) (2018)
5	2019	GEECO- Three Objective	AS-AC-CMA-ES	Artelys	RBFMOpt	(2022f) (2019)
		GEECO-Expensive Two Objective	RBFMOpt	Artelys	AS-AC-CMA-ES	(2022f) (2019)
		GEECO-Two Objective	Nacim Belkhir	Artelys	RBFMOpt	(2022f) (2019)
		GEECO-Expensive Single Objective	Artelys	AS-AC-CMA-ES	biteopt2018	(2022f) (2019)
		GEECO-Single Objective	AS-AC-CMA-ES	biteopt2018	radka	(2022f) (2019)

Source code is not available (NA)

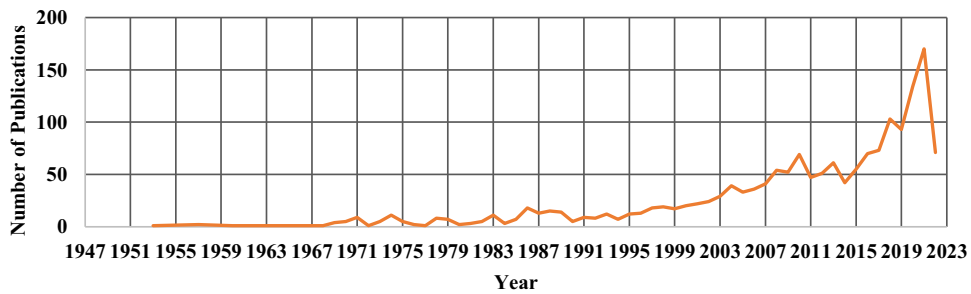

Fig. 7 The annual number of scientific publications on "engineering design problems" indexed in the scopus database from 1953 to 2022

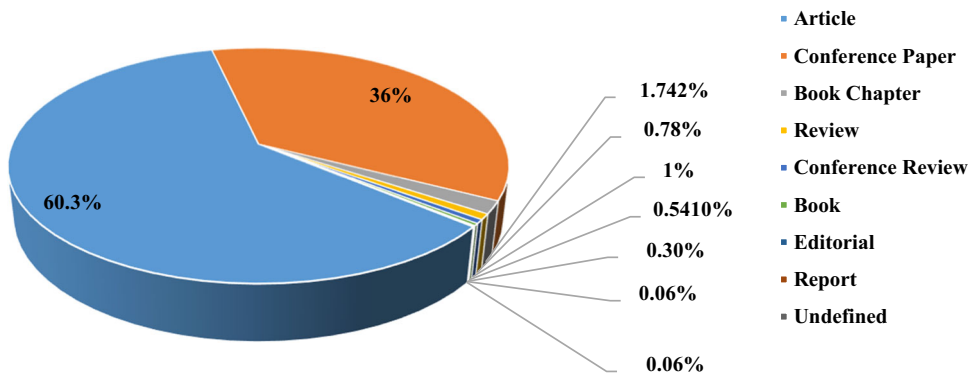

Fig. 8 The number of scientific publications on "engineering design problems" indexed in the scopus database from 1953 to 2022

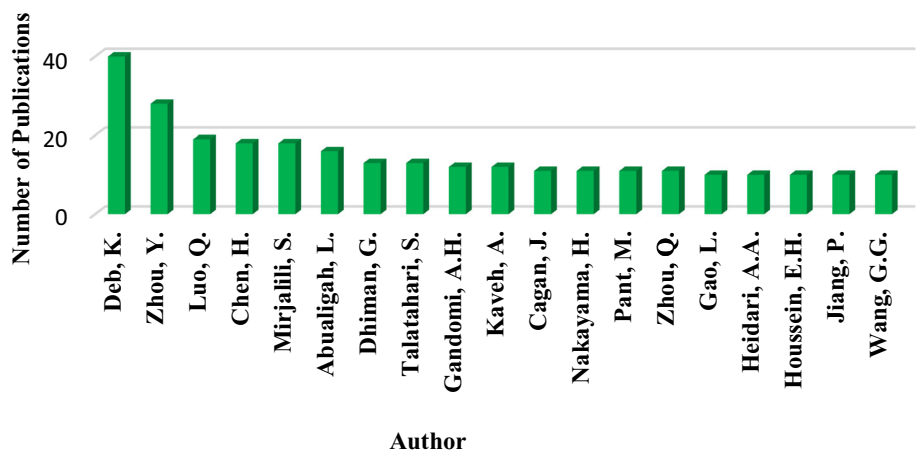

Fig. 9 Compare the authors with the number of publications related to "engineering design problems" indexed in the scopus database from 1953 to 2022

MATLAB code from Math Works of MATLAB central, research papers, GitHub, and webpages.

5.1 Pressure vessel

Many researchers used pressure vessels as a benchmark test function to evaluate the performance of the novel MH optimization techniques. This optimization problem has four key variables: Ts, Th, R, and L. The four variables under consideration can be mathematically represented by the following equation when employing algorithms to optimize the pressure vessel design problem. Some assumptions are Ts = X_1 is the thickness of the shell, Th = X_2 is the thickness of the head, $R = X_3$ is the inner radius, and $L = X_4$ is the length of the cylindrical sections (Güngör 2022; Dehghani et al. 2021)

$$f_1 = 0.6224x_1x_3x_4 + 17781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3.$$

The various constraints used for this objective function are

$$g_1(x) = -x_1 + 0.0193x_3 \leq 0,$$

$$g_2(x) = -x_2 + 0.00954x_3 \leq 0,$$

$$g_3(x) = -\pi x_3^2 x_4 - \frac{4}{3}\pi x_3^3 + 1296000 \leq 0,$$

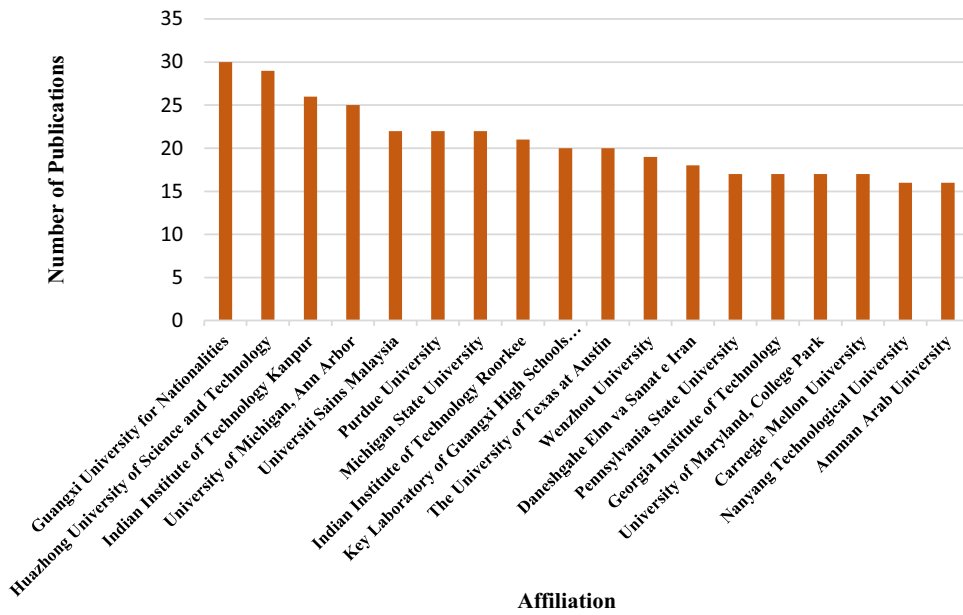

Fig. 10 Compare the document of different affiliation "engineering design problems" indexed in the scopus database from 1953 to 2022

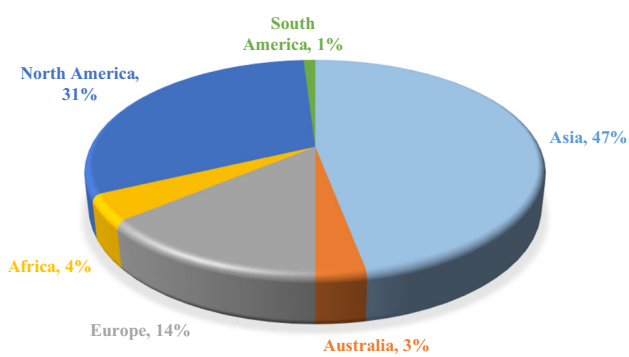

Fig. 11 Participation of continents in terms of percentage of publications "engineering design problems" indexed in the scopus database from 1953 to 2022

$$g_4(x) = x_4 - 240 \leq 0,$$

with

$$0 \leq x_1 \leq 99, 0 \leq x_2 \leq 99, 10 \leq x_3 \leq 200, 10 \leq x_4 \leq 200.$$

5.2 Welded beam

The literature shows that the welded beam design challenge is widely used to evaluate the performance of novel MH optimization techniques. Various assumptions are $x_1 = h$ is the thickness of the welds, $x_2 = l$ is the length of the welds, $x_3 = t$ is the height of the beam, and × 4 = b is the width of the beam (Got et al. 2022). In welded beam engineering design problem, there are some constraints, end deflection of the beam (δ), beam deflection (β), normal stress (α), allowable normal stress ($\alpha_{max} = 30000psi$) shear stress (τ), allowable shear stress ($\tau_{max} = 13600$psi) B load (6000 lb), and buckling load (BL) (Sayed and Hassanien 2018; Garg 2014)

$$f_2 = 1.1047x_1^2x_2 + 0.04811x_3x_4(14 + x_2),$$

Version 1.

The various constraints used for this objective function are

$$g_1(x) = \tau(x) - \tau_{max} \leq 0,$$

$$g_2(x) = \alpha(x) - \alpha_{max} \leq 0,$$

$$g_3(x) = x_1 - x_4 \leq 0,$$

$$g_4(x) = 0.125 - x_1 \leq 0,$$

$$g_5(x) = \delta(x) - 0.25 \leq 0,$$

$$g_6(x) = B - BL(x) \leq 0,$$

where

$$0.1 \leq x_1 \leq 2, 0.1 \leq x_2 \leq 10, 0.1 \leq x_3 \leq 10, 0.1 \leq x_4 \leq 2.$$

$$\tau(x) = \sqrt{\tau_1^2 + 2\tau_1\tau_2\left(\frac{x_2}{2R}\right) + \tau_2^2},$$

Primary stress: $\tau_1 = \frac{B}{\sqrt{2}x_1x_2}$.

Secondary stress: $\tau_{2=MR/J}$

$$M = B\left(L + \frac{x_2}{2}\right),$$

$$J(X) = 2\left\{\frac{x_1x_2}{\sqrt{2}}\left[\frac{x_2^2}{12} + \left(\frac{x_1 + x_3}{2}\right)^2\right]\right\},$$

$$R = \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2},$$

Table 6 Journal and conference (number of publications more than 10) "engineering design problems" indexed in the Scopus database from 1953 to 2022

S. no	Journal/conference name	Cite score	Publisher	Most cited article reference	Cited	Category	Full open access	Number of publication
1	Proceedings of The ASME Design Engineering Technical Conference	1	ASME-Conference Proceeding	(Finch and Ward 1997)	58	–	Yes	70
2	Expert Systems with Applications	12.7	Elsevier	(Faramarzi et al. 2020)	357	Q1	No	39
3	Lecture Notes In Computer Science Including Subseries Lecture Notes In Artificial Intelligence And Lecture Notes In Bioinformatics	1.8	Springer Book Series	(Bilchev and Parmee 1995)	275	–	Yes	39
4	Engineering with Computers	8.4	Springer Nature	(Yu et al. 2020a)	75	Q2	No	35
5	Journal Of Mechanical Design Transactions Of The ASME	6.4	ASME	(Wang 2003)	422	Q1	No	35
6	Engineering Optimization	4.6	Taylor & Francis	(Lee et al. 2005)	298	Q2	Yes	31
7	Engineering Applications of Artificial Intelligence	9	Elsevier	(He and Wang 2007)	728	Q1	Yes	25
8	Structural and Multidisciplinary Optimization	6.6	Springer Nature	(Liu et al. 2018)	169	Q1	No	25
9	Applied Soft Computing Journal	11.2	Elsevier	(Sadollah et al. 2013)	502	Q1	No	20
10	IEEE Access	4.8	IEEE	(Premkumar et al. 2021)	34	Q1	Yes	20
11	Soft Computing	5.1	Springer Nature	(Arora and Singh 2019)	421	Q2	No	19
12	Neural Computing and Applications	7.3	Springer Nature	(Mirjalili and Lewis 2014)	146	Q1	No	17
13	Applied Intelligence	6.8	Springer Nature	(Mirjalili et al. 2017)	282	Q2	No	16
14	Advances in Intelligent Systems and Computing	0.9	Springer Nature	(Dhiman and Kaur 2019)	57	–	No	14
15	SAE Technical Papers	1.0	SAE Conference Proceeding	(Sudjianto et al. 2005)	21	–	No	14
16	Applied Mathematical Modelling	7.5	Elsevier	(Chen et al. 2019)	135	Q1	No	13
17	Computer Methods in Applied Mechanics and Engineering	9.9	Elsevier	(Deb 2000)	2873	Q1	No	12
18	Computers and Structures	8.0	Elsevier	(Askarzadeh 2016)	998	Q1	No	11
19	Engineering Computations	2.3	Emerald	(Kaveh and Talatahari 2010b)	225	Q1	No	11
20	Journal of Intelligent Manufacturing	10.06	Springer Nature	(Akay and Karaboga 2012)	345	Q1	No	11

Table 7 Real-world engineering benchmark test function and their MATLAB code references

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
1	Pressure vessel	$f_1 = 0.6224x_1x_3x_4 + 17781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3$	$0 \le x_1 \le 99$; $0 \le x_2 \le 99$; $10 \le x_3 \le 200$; $10 \le x_4 \le 200$	(Örnek et al. 2022)(Yang et al. 2013)	(Chopra 2018)
2	Welded beam	$f_2 = 1.1047x_1^2x_2 + 0.04811x_3x_4(14 + x_2)$	$0.1 \le x_1 \le 2$; $0.1 \le x_2 \le 10$; $0.1 \le x_3 \le 10$; $0.1 \le x_4 \le 2$	(Abualigah et al. 2022)	(Pedramasl 2015) (2022 k)
3	Three-bar truss	$f_3 = \left(2\sqrt{2}x_1 + x_2\right)Xl$	$0 \le x_i \le 1$	(Askarzadeh 2016)	(Bayzidi 2022a)
4	Tension/compression	$f_4 = (x_3 + 2)x_2x_1^2$	$0.05 \le x_1 \le 2$; $0.25 \le x_2 \le 1.3$; $2 \le x_3 \le 15$	(Askarzadeh 2016)	(Yapici 2020)
5	Speed reducer	$f_5 = 0.7854x_1x_2^2\left(3.3333x_3^2 + 14.9334x_3 - 43.0934\right) -1.508x_1\left(x_6^2 + x_7^2\right) + 7.4777x_1\left(x_6^3 + x_7^3\right) +0.7854x_1\left(x_4x_6^2 + x_5x_7^2\right)$	$2.6 \le x_1 \le 3.6$; $0.7 \le x_2 \le 0.8$; $17 \le x_3 \le 28$; $7.3 \le x_4 \le 8.3$; $7.8 \le x_5 \le 8.3$; $2.9 \le x_6 \le 3.9$; $5 \le x_7 \le 5.5$	(Dehghani 2022)	(Bayzidi 2022a)
6	Gear train	$f_6 = \left(\left(\frac{1}{6.931}\right) - \left(\frac{x_3x_2}{x_1x_4}\right)\right)^2$	$12 \le x_i \le 60$	(Askarzadeh 2016)	(Bayzidi 2022a)
7	Cantilever beam	$f_7 = 0.0624(x_1 + x_2 + x_3 + x_4 + x_5)$	$0.01 \le x_i \le 100$	(Naruei and Keynia 2021)	(Yapici 2020)
8	Process flow sheeting	$f_8 = -0.7x_3 + 5(0.5 - x_1)^2 + 0.8$	$0.2 \le x_1 \le 1$; $-2.22554 \le x_2 \le 1$; $x_3 \in \{0, 1\}$;	(Peraza-Vázquez et al. 2021a)	(Suganthan 2022p)
9	Process synthesis	$f_9 = (1 - x_1)^2 + (2 - x_2)^2 + (3 - x_3)^2 +(1 - x_4)^2 + (1 - x_5)^2 + (1 - x_6)^2 - \ln(1 - x_7)$	$0 \le x_1, x_2, x_3 \le 1$; $(x_4, x_5, x_6, x_7 \in \{0, 1\})$	(Peraza-Vázquez et al. 2021a)	(Suganthan 2022p)
10	Tuning of a proportional-integral-derivative (PID) controller	$f_{10} = G(s) = \frac{1}{6s^1 + 11s^2 + 6s^3 + s^4}$	NA	(Peraza-Vázquez et al. 2021a)	(Razmjooy 2014)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
11	Multiple-disc clutch brake	$f_{11} = \pi\left(x_2^2 - x_1^2\right)x_3(x_5+1)\rho$	$60 \le x_1 \le 80$; $90 \le x_2 \le 110$; $1 \le x_3 \le 3$; $0 \le x_4 \le 1000$; $2 \le x_5 \le 9$	(Yildiz et al. 2021)	(Rather 2022)
12	Hydrostatic thrust bearing	$f_{12} = \left(\frac{QXP_0}{0.7}\right) + E_f$	$1 \le R, R_0$; $Q \le 16$	(Yildiz et al. 2021)	(Wong 2012)
13	Tubular column	$f_{13} = 5W + 2d,$ $= 5\rho l\pi dt + 2d,$ $= 9.8\text{dt} + 2\text{d}$	$2 \ll d \ll 14$; $0.2 \ll t \ll 0.8$	(Meng et al. 2017)	(Bayzidi 2022a)
14	I-beam	$f_{14} = \frac{5000}{\frac{x_3(x_2-2x_4)^3}{12} + \frac{x_4^3 x_1}{6} + 2x_4^3 x_1\left(\frac{x_2-x_4}{2}\right)^2}$	$10 \le x_1 \le 50, 10 \le x_2 \le 80, 0.9 \le x_3 \le 5, 0.9 \le x_4 \le 5$	(Yu et al. 2020b)	(Bayzidi 2022b)
15	Crashworthiness	$f_{15} = 4.90x_1 + 6.67x_2 + 6.98x_3 + 4.01x_4 + 1.78x_5 + 2.73x_7 + 1.98$	$0.5 \le x_1, x_2, x_3,$ $x_4, x_5, x_6, x_7 \le 1$; $x_8, x_9 \in .\{0.192, 0.345\}$ $-30 \le x_{10}, x_{11}, x_{12}, x_{13}, x_{14} \le 30$	(Kumar et al. 2020d)	NA
16	Industrial refrigeration system	$f_{16} = 63098.88x_2x_4x_{12} + 5441.5x_2^2x_{12} + 115055.5x_2^{1.664}x_6$ $+6172.27x_2^2x_6 + 63098.88x_1x_3x_{11}$ $+5441.5x_1^2x_{11} + 115055.5x_1^{1.664}x_5 + 6172.27x_1^2x_5$ $+140.53x_1x_{11} + 281.29x_3x_{11} + 70.26x_1^2$ $+281.29x_1x_3 + 281.29x_3^2 + 14437x_8^{1.8812}x_{12}^{0.3424}x_{10}x_{14}^{-1}x_1^2x_7x_9^{-1}$ $+20470.2x_{11}^{0.316}x_7^{2.893}x_1^2$	$0.001 \le x_i \le 5$	(Zamani et al. 2022)	NA
17	Gas transmission compressor	$f_{17} = 8.61X10^5x_2x_1^{1/2}x_3^{-2/3}x_4^{-1/2} + 3.69X10^4x_3$ $+7.72X10^8x_1^{-1}x_2^{0.219} - 765.43X10^6x_1^{-1},$	$20 \le x_1 \le 50$; $1 \le x_2 \le 10$; $20 \le x_3 \le 45$; $0.1 \le x_4 \le 60$	(Zamani et al. 2022)	NA
18	Corrugated bulkhead	$f_{18} = \frac{5.8884x_4(x_1+x_3)}{x_1+\sqrt{(x_3^2-x_2^2)}}$	$0 \le x_1, x_2, x_3 \le 100$; $0 \le x_4 \le 5$	(Gandomi et al. 2013)(Azizi et al. 2021)	(Bayzidi 2022a)
19	Piston lever	$f_{19} = \frac{1}{4}\pi x_3^2(L_2 - L_1)$	$\theta = 0to45^0$	(Gandomi et al. 2013)	(Bayzidi 2022a)
20	Step-cone pulley	$f_{20} = \rho w\left[d_1^2\left(11 + \left(\frac{N_1}{N}\right)^2\right) + d_2^2\left(1 + \left(\frac{N_2}{N}\right)^2\right) + +d_3^2\left(1 + \left(\frac{N_3}{N}\right)^2\right)\right.$ $\left. + + d_4^2\left(1 + \left(\frac{N_4}{N}\right)^2\right)\right]$	–	(Talatahari and Azizi 2020)(Naruei and Keynia 2021)	(2022 l)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
21	Planetary gear train	$f_{21_1} = \frac{m_n z_a}{cos\beta}(1 + u^H_{a-g})\frac{cos\alpha_t}{cos\alpha_{wt}}$, $f_{21_2} = \frac{1-n^H_{ab}u^H_{ab}}{1-u^H_{ab}}$, $f_{21_3} = \varepsilon_{\alpha(a-g)}(x)$, $f_{21_4} = \alpha_{w(a-g)}(x)$, $f_{21_5} = S_{F(a)}(x) = \frac{[\sigma_F]_{M(a)}}{\sigma_{F(a)}}$, $f_{21_6} = S_{H(a)}(x) = \frac{[\sigma_H]_{M(a)}}{\sigma_{H(a)}}$, $f_{21_7} = V(x)$, $f_{21_8} = S_{F(b)}(x) = \frac{[\sigma_F]_{M(b)}}{\sigma_{F(b)}}$, $f_{21_9} = D_{out}(x)$	–	(Rosic et al. 2011)	NA
22	Optimal power flow for minimizing fuel cost, voltage deviation, active and reactive power loss	$f_{22_1} = \sum_{i=1}^{N}(P_{g,i} - P_{l,i})$, $f_{22_2} = \sum_{i=1}^{N}(Q_{g,i} - Q_{l,i})$, $f_{22_3} = \sum_{i=1}^{N}(a_i + b_i P_{g,i} + c_i P^2_{g,i})$, $f_{22_4} = \sum_{i=1}^{N}(1 - \lvert V_i \rvert)$	$V_{min} \le V_k \le V_{max}$; $\delta_{min} \le \delta_k \le \delta_{max}$; $P_{min} \le P_{g,k} \le P_{max}$; $Q_{min} \le Q_{g,k} \le Q_{max}$	(Biswas et al. 2020)	(Suganthan 2022r)
23	Vibrating platform	$f_{23_1} = -\frac{\pi}{2L^2}\sqrt{\frac{EI}{\mu}}$, $f_{23_2} = 2bL(c_1 d_1 + c_2(d_2 - d_1) + c_3(d_3 - d_2))$	$0.05 \le d_1 \le 0.5$; $0.2 \le d_2 \le 0.5$; $0.2 \le d_3 \le 0.6$; $0.35 \le b \le 0.5$; $3 \le L \le 6$	(Narayanan and Azarm 1999)	(Suganthan 2022r)
24	Four-bar plane truss	$f_{24_1} = L(2x_1 + \sqrt{2x_2} + \sqrt{x_3} + x_4$, $f_{24_2} = \frac{FL}{E}\left[\frac{2}{x_2} + \frac{2\sqrt{2}}{x_2} - \frac{2\sqrt{2}}{x_3} + \frac{2\sqrt{2}}{x_4}\right]$	$(F/\sigma) \le x_1, x_4 \le 3(F/\sigma)$; $\sqrt{2}(F/\sigma) \le x_2,\ x_3 \le 3(F/\sigma)$	(Coello Coello and Pulido 2005)	(Suganthan 2022r)
25	Two-bar plane truss	$f_{25_1} = x_1\sqrt{16 + x_3^2} + x_2\sqrt{1 + x_3^2}$, $f_{25_2} = \frac{20\sqrt{16+x_3^2}}{x_3 x_1}$	$10^{-5} \le x_1, x_2 \le 100$; $1 \le x_3 \le 3$	(Coello Coello and Pulido 2005) (Tanabe and Oyama 2017)	(2022 m)
26	Bulk carrier	$f_{26_1} = \frac{C_v + C_r + C_c}{ac}$, $f_{26_2} = ls$, $f_{26_3} = -ac$	$150 \le L \le 274.32$; $20 \le B \le 32.31$; $13 \le D \le 25$; $10 \le T \le 11.71$; $14 \le V_k \le 18$; $0.63 \le C_B \le 0.75$	(Parsons and Scott 2004)	(Suganthan 2022r)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
27	Front rail	$f_{27_1} = \frac{E_a}{E}$, $f_{27_2} = \frac{F_a}{F}$	$56 \le w \le 68$; $136 \le hh \le 146$; $1.4 \le t \le 2.2$	(Fan et al. 2018)	(Suganthan 2022r)
28	Optimal droop setting for minimizing voltage deviation and active power loss	$f_{28_1} = \sum_{i=1}^{N} P_i$, $f_{28_2} = \sum_{i=1}^{N} \left(1 - \lvert V_i \rvert^2\right)$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $C_{pmin,k} \le C_{pk} \le C_{pmax,k}$; $C_{qmin,k} \le C_{qk} \le C_{qmax,k}$; $w_{min} \le w \le w_{max}$	(Kumar et al. 2019b)	(Suganthan 2022r)
29	Optimal power flow for minimizing fuel cost, active and reactive power loss	$f_{29_1} = \sum_{i=1}^{N} (P_{g,i} - P_{l,i})$, $f_{29_2} = \sum_{i=1}^{N} (Q_{g,i} - Q_{l,i})$, $f_{29_3} = \sum_{i=1}^{N} (a_i + b_i P_{g,i} + c_i P_{g,i}^2)$	$V_{min} \le V_k \le V_{max}$; $\delta_{min} \le \delta_k \le \delta_{max}$; $P_{min} \le P_{g,k} \le P_{max}$; $Q_{min} \le Q_{g,k} \le Q_{max}$;	(Biswas et al. 2020)	(Suganthan 2022r)
30	Power distribution system planning	$f_{30_1} = \sum_{i=1}^{6} (a_i + b_i x_i + c_i x_i^2)$, $f_{30_2} = \sum_{i=1}^{6} (\alpha_i + \beta_i x_i + \gamma_i x_i^2)$	–	(Rivas-Dávalos and Irving 2005)	(Suganthan 2022r)
31	Optimal droop setting for minimizing active and reactive power loss	$f_{31_1} = \sum_{i=1}^{N} P_i$, $f_{31_2} = \sum_{i=1}^{N} Q_i$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $C_{pmin,k} \le C_{pk} \le C_{pmax,k}$; $C_{qmin,k} \le C_{qk} \le C_{qmax,k}$; $w_{min} \le w \le w_{max}$	(Kumar et al. 2019a)	(Suganthan 2022r)
32	Optimal power flow for minimizing voltage deviation, active and reactive power loss	$f_{32_1} = \sum_{i=1}^{N} P_i$, $f_{32_2} = \sum_{i=1}^{N} \left(1 - \lvert V_i \rvert^2\right)$, $f_{32_3} = \sum_{i=1}^{N} Q_i$	$V_{min} \le V_{r,k} V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $P_{min,dg} \le P_{dg,k} \le P_{max,dg}$	(Biswas et al. 2020)	(Suganthan 2022r)
33	Optimal Sizing of Single Phase Distribution Generation with Reactive Power support for minimizing active and reactive power loss	$f_{33_1} = \sum_{i=1}^{N} P_i$, $f_{33_2} = \sum_{i=1}^{N} (1 - \lvert V_i \rvert)$, $f_{33_3} = \sum_{i=1}^{N} Q_i$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $P_{min,dg} \le P_{dg,k} \le P_{max}$	(Mishra et al. 2019)	(Suganthan 2022r)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
34	Optimal droop setting for minimizing voltage deviation, active, and reactive power loss	$f_{34_1} = \sum_{i=1}^{N} P_i$, $f_{34_2} = \sum_{i=1}^{N} \left(1 - \lvert V_i \rvert^2\right)$, $f_{34_3} = \sum_{i=1}^{N} Q_i$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $C_{pmin,k} \le C_{pk} \le C_{pmax,k}$; $C_{qmin,k} \le C_{qk} \le C_{qmax,k}$; $w_{min} \le w \le w_{max}$	(Kumar et al. 2020c)	(Suganthan 2022r)
35	Optimal power flow for minimizing fuel cost, voltage deviation, and active power loss	$f_{35_1} = \sum_{i=1}^{N} (P_{g,i} - P_{l,i})$, $f_{35_2} = \sum_{i=1}^{N} (1 - \lvert V_i \rvert)$, $f_{35_3} = \sum_{i=1}^{N} (a_i + b_i P_{g,i} + c_i P_{g,i}^2)$	$V_{min} \le V_k \le V_{max}$; $\delta_{min} \le \delta_k \le \delta_{max}$; $P_{min} \le P_{g,k} \le P_{max}$; $Q_{min} \le Q_{g,k} \le Q_{max}$	(Biswas et al. 2020)	(Suganthan 2022r)
36	Optimal power flow for minimizing voltage deviation, and active power loss	$f_{36_1} = \sum_{i=1}^{N} P_i$, $f_{36_2} = \sum_{i=1}^{N} (1 - \lvert V_i \rvert)$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $P_{min,dg} \le P_{dg,k} \le P_{max,dg}$;	(Biswas et al. 2020)	(Suganthan 2022r)
37	Optimal power flow for minimizing active and reactive power loss	$f_{37_1} = \sum_{i=1}^{N} P_i$, $f_{37_2} = \sum_{i=1}^{N} Q_i$	$V_{min} \le V_{r,k}, V_{m,k} \le V_{max}$; $P_{min} \le P_k \le P_{max}$; $Q_{min} \le Q_k \le Q_{max}$; $P_{min,dg} \le P_{dg,k} \le P_{max,dg}$	(Biswas et al. 2020)	(Suganthan 2022r)
38	Optimal sizing of single phase distribution generation with reactive power support for phase balancing at main transformer/grid and minimizing active and reactive power loss	$f_{38_1} = \sum_{i=1}^{N} \sum_{j \in (a,b,c)} P_i^j$, $f_{38_2} = \sum_{i=1}^{N} \sum_{j \in (a,b,c)} Q_i^j$	$V_{min} \le V_{r,k}^j, V_{m,k}^j \le V_{max}$; $P_{min} \le P_k^j \le P_{max}$; $Q_{min} \le Q_k^j \le Q_{max}$; $P_{dg,min} \le P_{dg,k}^j \le P_{dg,max}$; $Q_{dg,min} \le Q_{dg,k}^j \le Q_{dg,max}$	(Mishra et al. 2019)	(Suganthan 2022r)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
39	Optimal sizing of single phase distribution generation with reactive power support for phase balancing at main transformer/grid and minimizing reactive power loss	$f_{39_1} = (I^a_{r,1} + I^b_{r,1} + I^c_{r,1})^2 + (I^a_{m,1} + I^b_{m,1} + I^c_{m,1})^2 + (I^a_{r,1} - 0.5\left(I^b_{r,1} + I^c_{r,1}\right) - 0.5\sqrt{3}\left(I^b_{m,1} - I^c_{m,1}\right))^2 + (I^a_{m,1} - 0.5\left(I^b_{m,1} + I^c_{m,1}\right) + 0.5\sqrt{3}(I^b_{r,1} - I^c_{r,1}))^2$, $f_{39_2} = \sum_{i=1}^{N} \sum_{j\in(a,b,c)} Q^j_i$	$V_{min} \le V^j_{r,k}, V^j_{m,k} \le V_{max}$; $P_{min} \le P^j_k \le P_{max}$; $Q_{min} \le Q^j_k \le Q_{max}$; $P_{dg,min} \le P^j_{dg,k} \le P_{dg,max}$; $Q_{dg,min} \le Q^j_{dg,k} \le Q_{dg,max}$	(Mishra et al. 2019)	(Suganthan 2022r)
40	Synchronous Optimal Pulse-width Modulation of 13-level Inverters	$f_{40_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{\sqrt[6]{\sum_k (k^{-4})}}$, $f_{40_2} = \left[6m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	–	(Edpuganti and Rathore 2017)	(Suganthan 2022r)
41	Optimal Power Flow for Minimizing Fuel Cost, and Active Power Loss	$f_{41_1} = \sum_{i=1}^{N} (P_{g,i} - P_{l,i})$, $f_{41_2} = \sum_{i=1}^{N} (a_i + b_i P_{g,i} + c_i P^2_{g,i})$	–	(Biswas et al. 2020)	(Suganthan 2022r)
42	Synchronous optimal pulse-width modulation of 7-level inverters	$f_{42_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{\sqrt[3]{\sum_k (k^{-4})}}$, $f_{42_2} = \left[3m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	$0 < x_i < \frac{\pi}{2}$	(Edpuganti and Rathore 2014)	(Suganthan 2022r)
43	Synchronous Optimal Pulse-width Modulation of 11-level Inverters	$f_{43_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{\sqrt[5]{\sum_k (k^{-4})}}$, $f_{43_2} = \left[5m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	$0 < x_i < \frac{\pi}{2}$	(Edpuganti and Rathore 2017)	(Suganthan 2022r)
44	Synchronous optimal pulse-width modulation of 5-level inverters	$f_{44_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{\sqrt[2]{\sum_k (k^{-4})}}$, $f_{44_2} = \left[2m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	$0 < x_i < \frac{\pi}{2}$	(Rathore et al. 2010a)	(Suganthan 2022r)
45	Synchronous optimal pulse-width modulation of 3-level inverters	$f_{45_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{\sqrt[4]{\sum_k (k^{-4})}}$, $f_{45_2} = \left[m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	$0 < x_i < \frac{\pi}{2}$	(Rathore et al. 2010b)	(Suganthan 2022r)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
46	Synchronous optimal pulse-width modulation of 9-level inverters	$f_{46_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{4\sqrt{\sum_k (k^{-4})}}$, $f_{46_2} = \left[4m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2$	$0 < x_i < \frac{\pi}{2}$	(Kumar et al. 2021b)(Edpuganti et al. 2015)	(Suganthan 2022r)
47	Two-reactor problem	$f_{47} = 7.5x_5 + 5.5x_6 + 7x_1 + 6x_2 + 5(x_3 + x_4)$	$x_i \geq 0, i = \{1,2,3,4,5,6,7,8\}$ $x_i = \{0,1\}, i = \{5,6\}$	(Cantú et al. 2021)(Kocis and Grossmann 1989)	(Suganthan 2022r)
48	Heat exchanger network design	$f_{48_1} = 35x_1^{0.6} + 35x_2^{0.6}$, $f_{48_2} = 200x_1x_4 - x_3$, $f_{48_3} = 200x_1x_6 - x_5$	$0 \leq x_1 \leq 10$; $0 \leq x_2 \leq 200$; $0 \leq x_3 \leq 100$; $0 \leq x_4 \leq 200$; $1000 \leq x_5 \leq 2000000$; $0 \leq x_6 \leq 600$; $100 \leq x_7 \leq 600$; $100 \leq x_8 \leq 600$; $100 \leq x_9 \leq 900$	(Guillén-Gosálbez 2011)	(Suganthan 2022r)
49	Reactor network design (RND)	$f_{49_1} = -x_4$, $f_{49_2} = x_5^{0.5} + x_6^{0.5}$	$0 \leq x_4, x_3, x_2, x_1 \leq 4$; $0.00001 \leq x_6, x_5 \leq 16$	(Gujarathi et al. 2015)(Gujarathia et al. 2016)	(Suganthan 2022r)
50	Haverly's pooling (HP)	$f_{50_1} = -9x_1 - 15x_2 + 6x_3 + 16x_4$, $f_{50_2} = 10(x_5 + x_6)$	$0 \leq x_8, x_6, x_5, x_4, x_3, x_1 \leq 100$; $0 \leq x_9, x_7, x_2 \leq 200$	(Maier and Whiting 1998)	(Suganthan 2022r)
51	Multi-product batch plant	$f_{51_1} = \sum_{j=1}^{M} \alpha_j N_j V_j^{\beta_j}$, $f_{51_2} = 65\left[\frac{Q_1}{B_1} + \frac{Q_2}{B_2}\right] + 0.08Q_1 + 0.1Q_2$, $f_{51_3} = Q_1 \frac{T_{L1}}{B_1} + Q_2 \frac{T_{L2}}{B_2}$	$1 \leq N_i \leq N_j^u$; $V_j^l \leq V_j \leq V_j^u$; $T_{Li}^l \leq T_{Li} \leq T_{Li}^u$; $B_j^l \leq B_j \leq B_j^u$	(Angira and Babu 2006)	(Suganthan 2022r)
52	Process synthesis and design problem	$f_{52_1} = x_2 - x_3 + 2x_1$, $f_{52_2} = x_1x_3 - x_1^2 - x_2$	$0.5 \leq x_2, x_1 \leq 1.4$; $x_3 \varepsilon \{0,1\}$	(Angira and Babu 2006)	(Suganthan 2022r)
53	Parameter estimation of solar PV models	$f_{53_1} = I_{sc} - I_{ph} + I_o\left(\exp\left(\frac{I_{sc}R_s}{V_T}\right) - 1\right) + \frac{I_{sc}R_s}{R_p}$, $f_{53_2} = I_{ph} - I_o\left(\exp\left(\frac{V_{0c}}{V_T}\right) - 1\right) + \frac{V_{oc}}{R_p}$, $f_{53_3} = I_{MP} - I_{ph} + I_o\left(\exp\left(\frac{V_{MP}+I_{MP}R_s}{V_T}\right) - 1\right) + \frac{V_{MP}+I_{MP}R_s}{R_p}$, $f_{53_4} = I_{MP} - (V_{MP} - I_{MP}R_s)\left(\frac{I_o}{V_T}\exp\left(\frac{V_{MP}+I_{MP}R_s}{V_T}\right) + \frac{1}{R_p}\right)$, $f_{53_5} = \frac{R_s}{R_p} - \frac{I_o}{V_T}\exp\left(\frac{I_{sc}R_s}{V_T}\right)(R_p - R_s)$	$8 \leq I_{ph} \leq 8.6$; $7X10^{-10} \leq I_o \leq 1X10^{-7}$; $0.1 \leq R_s \leq 0.5$; $50 \leq R_p \leq 5000$; $0.1 \leq V_T \leq 2$	(Ayyarao and Kumar 2022)(Sharma et al. 2022a)(Sharma and Chinnappa Naidu 2022)	(V 2022)

Table 7 (continued)

S. no	Engineering design problem	Objective function	Range	References	MATLAB code references
54	Wind farm layout	$f_{54} = \sum_{i=1}^{N} E(P_i)$	$x + R \le x_i \le \overline{x} - R$; $y + R \le y_i \le \overline{y} - R$	(Wang et al. 2018)	(Suganthan 2022p)
55	10-Bar truss	$f_{55} = \sum_{i=1}^{10} L_i(x_i)\rho_i A_i$	$6.45X10^{-5} \le A_i \le 5X10^{-3}$	(Kumar et al. 2020d)	(Farshchin 2020)
56	Beef cattle	$f_{56} = \sum_{i=1}^{n} x_i c_i$	–	(Uyeh et al. 2018)	(Suganthan 2022p)
57	Dairy cattle	$f_{57} = \sum_{i=1}^{n} x_i c_i$,	–	(Uyeh et al. 2018)	(Suganthan 2022p)

Source code is not available (NA)

$$\delta(x) = \frac{4BL^3}{Ex_3^3x_4},$$

$$BL(x) = \frac{4.013\sqrt{\frac{EGx_3^2x_4^2}{36}}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right),$$

$G = 12X10^6$ Psi,

E = $30X10^6 psi$,

$L = 14in$,

Version 2.

$$g_4(x) = 0.10471x_1^2 + 0.04811x_3x_4(14.0 + x_2) - 5 \le 0,$$

$$\delta(x) = \frac{6BL^3}{Ex_3^3x_4},$$

$$BL(x) = \frac{4.013E\sqrt{\frac{x_3^2x_4^2}{36}}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right),$$

$$J(X) = 2\left\{\sqrt{2}x_1x_2\left[\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2\right]\right\}.$$

5.3 Three-bar truss

The literature shows that the three-bar truss real-world engineering design test function is utilized to evaluate the performance of the novel MH optimization techniques. Tapabrata Ray and Pankaj Saini first revealed this problem (Erdoğan Yildirim and Karci 2019). The various constants are P = 2 KN/cm^2, σ = 2kN/cm^2, and l = 100 cm (Sheikhi Azqandi et al. 2020). The real-world objective function (Three-Bar Truss design) can be expressed in the following equation:

$$f_3 = \left(2\sqrt{2}x_1 + x_2\right) * l.$$

The various constraints used for this objective function are

$$g_1(x) = \frac{\sqrt{2}x_1 + x_2}{\sqrt{2}x_1^2 + 2x_1x_2}P - \sigma \le 0,$$

$$g_2(x) = \frac{x_2}{\sqrt{2}x_1^2 + 2x_1x_2}P - \sigma \le 0,$$

$$g_3(x) = \frac{1}{x_1 + \sqrt{2}x_2}P - \sigma \le 0.$$

The value of the x lies between $0 \le x_i \le 1$.

5.4 Tension/compression spring

The literature shows that the tension/compression spring is used to evaluate the performance of the novel MH

optimization techniques. Various assumptions are $x_1 = d$ is the wire diameter, $x_2 = D$ is the coil diameter, and $x_3 = P$ is the number of the active coil. In this test function, there are three variable $\{x_1, x_2, x_3 = d, D, P\}$ (Kutlu Onay and Aydemir 2022). The main objective of this test function is to minimize the weight of the tension or compression spring (Meng et al. 2017)

$$f_4 = (x_3 + 2)x_2x_1^2.$$

The various constraints used for this objective function are

$$g_1(x) = 1 - \frac{x_2^3x_3}{71785x_1^4} \leq 0,$$

$$g_2(x) = \frac{4x_2^2 - x_1x_2}{12566(x_2x_1^3)} + \frac{1}{5108x_1^2} - 1 \leq 0,$$

$$g_3(x) = 1 - \frac{140.45x_1}{x_2^2x_3} \leq 0,$$

$$g_4(x) = \frac{x_1 + x_2}{1.5} - 1 \leq 0.$$

The values of the x lies between: $0.05 \leq x_1 \leq 2, 0.25 \leq x_2 \leq 1.3, 2 \leq x_3 \leq 15$.

5.5 Speed reducer

The literature shows that the speed reducer is used to evaluate the performance of the novel MH optimization techniques. This is a fascinating and challenging problem with 7 variables and 11 constraints. Various assumptions are x_1=b is the face width, x_2 =m is the module of teeth, x_3 =p is the number of the teeth on the pinion, x_4 =l_1 is the length of the 1st shaft between bearings, x_5=l_2 is the length of the 2nd shaft between bearings, x_6=d_1 is the 1st shaft diameter, and x_7 =d_2 is the 2nd shaft diameter. In this test function, there are three variables $\{x_1, x_2, x_3, x_4, x_5, x_6, x_7 = b, m, p, l_1, l_2, d_1, d_2\}$ (Lin et al. 2013). The mathematical formulation for the real-world speed reducer design challenges is as follows:

$$\begin{aligned} f_5 = {} & 0.7854x_1x_2^2\left(3.3333x_3^2 + 14.9334x_3 - 43.0934\right) \\ & - 1.508x_1\left(x_6^2 + x_7^2\right) + 7.4777x_1\left(x_6^3 + x_7^3\right) \\ & + 0.7854x_1\left(x_4x_6^2 + x_5x_7^2\right). \end{aligned}$$

The various constraints used for this objective function are

$$g_1(x) = \frac{27}{x_1x_2^2x_3} - 1 \leq 0,$$

$$g_2(x) = \frac{397.5}{x_1x_2^2x_3} - 1 \leq 0,$$

$$g_3(x) = \frac{1.93x_4^3}{x_2x_6^4x_3} - 1 \leq 0,$$

$$g_4(x) = \frac{1.93x_5^3}{x_2x_7^4x_3} - 1 \leq 0,$$

$$g_5(x) = \frac{1}{110x_6^3}\sqrt{\left(\frac{7.54x_4}{x_2x_3}\right)^2 + 16.9X10^6} - 1 \leq 0,$$

$$g_6(x) = \frac{1}{85x_7^3}\sqrt{\left(\frac{7.54x_5}{x_2x_3}\right)^2 + 157.5X10^6} - 1 \leq 0,$$

$$g_7(x) = \frac{x_2x_3}{40} - 1 \leq 0,$$

$$g_8(x) = \frac{5x_2}{x_1} - 1 \leq 0,$$

$$g_9(x) = \frac{x_1}{12x_2} - 1 \leq 0,$$

$$g_{10}(x) = \frac{1.5x_6 + 1.9}{x_4} - 1 \leq 0,$$

$$g_{11}(x) = \frac{1.1x_7 + 1.9}{x_5} - 1 \leq 0,$$

with boundary conditions:

$$2.6 \leq x_1 \leq 3.6, 0.7 \leq x_2 \leq 0.8, 17 \leq x_3 \leq 28,$$
$$7.3 \leq x_4 \leq 8.3; 7.8 \leq x_5 \leq 8.3, 2.9 \leq x_6 \leq 3.9, 5 \leq x_7 \leq 5.5.$$

5.6 Gear train

The literature shows that the gear train is used to evaluate the performance of the novel MH optimization techniques. The gear train problem is only contained boundary constraints. In this test function, there are four variables $\{T_d, T_b, T_a, T_f\} = \{x_3, x_2, x_1, x_4\}$, Ti is the teeth of the gear I, and the gear ratio (1/6.931) (Askarzadeh 2016). The mathematical formulation of the gear train test function is as follows:

$$f_6 = \left(\left(\frac{1}{6.931}\right) - \left(\frac{x_3x_2}{x_1x_4}\right)\right)^2.$$

The value of the x_i lies between: $12 \leq x_i \leq 60$.

5.7 Cantilever beam

The literature shows that a cantilever is used to evaluate the performance of the novel MH optimization techniques. In this test function, there are 5 variables (x_1, x_2, x_3, x_4, as well as x_5) and 1 constraint. The mathematical formulation as well as constraint functions of the cantilever beam real-world challenges are given below (Mirjalili 2015a, b)

$$f_7 = 0.0624(x_1 + x_2 + x_3 + x_4 + x_5).$$

The constraints used for this objective function are

$$g_1(x) = \frac{61}{x_1^3} + \frac{27}{x_2^3} + \frac{19}{x_3^3} + \frac{7}{x_4^3} + \frac{1}{x_5^3} - 1 \le 0,$$

with boundary condition

$$0.01 \le x_i \le 100.$$

5.8 Process flow sheeting

Many novel MH optimization techniques are tested with the process flow sheeting function. This test function has 3 inequality constraints and 3 variables (x_1, x_2, as well as x_3), which are expressed in the following equation (Peraza-Vázquez et al. 2021a; Kumar et al. 2020d):

$$f_8 = -0.7x_3 + 5(0.5 - x_1)^2 + 0.8.$$

The 3 inequality constraints used for this objective function are

$$g_1(x) = -e^{(x_1 - 0.2)} - x_2 \le 0,$$

$$g_2(x) = x_2 + 1.1x_3 \le -1.0,$$

$$g_3 = x_1 - x_3 \le 0.2,$$

with boundary condition

$$0.2 \le x_1 \le 1, -2.22554 \le x_2 \le 1, x_3 \in \{0, 1\}.$$

For this objective function, the best feasible solution is 1.0765430833.

5.9 Process synthesis

The literature shows that the process synthesis is used to evaluate the performance of the novel MH optimization techniques. The process synthesis test function contains 7 decision variables (x_1, x_2, x_3, x_4, x_5, x_6, as well as x_7) and 9 constraints (g_1, g_2, g_3, g_4, g_5, g_6, g_7, g_8, as well as g_9). The mathematical equation of this test function is represented as follows (Peraza-Vázquez et al. 2021a; Kumar et al. 2020d):

$$f_9 = (1 - x_1)^2 + (2 - x_2)^2 + (3 - x_3)^2 + (1 - x_4)^2 + (1 - x_5)^2 + (1 - x_6)^2 - \ln(1 - x_7).$$

The 9 inequality constraints used for this objective function are

$$g_1(x) = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \le 5,$$

$$g_2(x) = x_1^2 + x_2^2 + x_3^2 + x_6^2 \le 5.5,$$

$$g_3(x) = x_1 + x_4 \le 1.2,$$

$$g_4(x) = x_5 + x_2 \le 1.8,$$

$$g_5(x) = x_3 + x_6 \le 2.5,$$

$$g_6(x) = x_1 + x_7 \le 1.2,$$

$$g_7(x) = x_2^2 + x_5^2 \le 1.64,$$

$$g_8(x) = x_3^2 + x_6^2 \le 4.25,$$

$$g_9(x) = x_3^2 + x_5^2 \le 4.64,$$

with boundary condition

$$0 \le x_1, x_2, x_3 \le 1, (x_4, x_5, x_6, x_7 \in \{0, 1\}).$$

5.10 Tuning of a proportional-integral-derivative (PID) controller

The literature shows that the tuning of a proportional-integral-derivative (PID) controller test function is utilized to evaluate the performance of the novel MH optimization techniques. This test function has 5 design variables $\{s^1, s^2, s^3, s^4\}$ and is expressed as follows (Peraza-Vázquez et al. 2021b; Pareek et al. 2014):

$$f_{10} = G(s) = \frac{1}{6s^1 + 11s^2 + 6s^3 + s^4}.$$

5.11 Multiple-disc clutch brake

The literature shows that the multiple disc clutch brake is used to evaluate the performance of the novel MH optimization techniques. This test function has 5 variables $\{x_1, x_2, x_3, x_4, x_5\}$, where ($x_1$) inner radius, ($x_2$) outer radius, ($x_3$) disc thickness, ($x_4$) force of actuators, and the last variable (x_5) number of frictional surfaces. This test function also contains 8 constraints, and the mathematical representation of this test function is given below (Pira 2022; Che and He 2022)

$$f_{11} = \pi\left(x_2^2 - x_1^2\right)x_3(x_5 + 1)\rho.$$

The various constraints used for this objective function are

$$g_1(x) = -x_2 + x_1 + \Delta R \le 0,$$

$$g_2(x) = L_{max} - (x_5 + 1)(x_3 + \Delta) \le 0,$$

$$g_3(x) = p_{rz} - p_{max} \le 0,$$

$$g_4(x) = p_{rz}XV_{sr} - p_{max}XV_{sr,max} \le 0,$$

$$g_5(x) = V_{sr} - V_{sr,max} \le 0,$$

$$g_6(x) = T - T_{max} \le 0,$$

$$g_7(x) = SXM_s - M_h \le 0,$$

$g_8(x) = -T \leq 0,$

where

$$M_h = \frac{\frac{2}{3}\mu x_4 x_5 \left(x_2^3 - x_1^3\right)}{x_2^2 - x_1^2},$$

$$R_{sr} = \frac{\frac{2}{3}\left(x_2^3 - x_1^3\right)}{x_2^2 - x_1^2},$$

$$V_{sr} = \pi X R_{sr} X n/30,$$

$$A = \pi X\left(x_2^2 - x_1^2\right),$$

$$p_{rz} = \frac{x_4}{A},$$

$$w = \frac{\pi X n}{30},$$

$$T = \frac{I_z X w}{M_h + M_f},$$

where $\Delta R = 20mm$, $L_{max} = 30$, $\mu = 0.6$, $\Delta = 50$, $V_{sr,max} = 10\frac{m}{s}$, $\delta = 0.5mm$, $s = 1.5$, $T_{max} = 15$, $n = 250rpm$, $I_z = 55Kgm^2$, $M_s = 40Nm$, $M_f = 3Nm$, $\rho = 0.0000078$, ρ_{max}=1.

With boundary conditions:

$60 \leq x_1 \leq 80, 90 \leq x_2 \leq 110, 1 \leq x_3 \leq 3,$
$0 \leq x_4 \leq 1000$, and$2 \leq x_5 \leq 9$.

5.12 Hydrostatic thrust bearing

The literature shows that the hydrostatic thrust bearing is utilized to evaluate the performance of the novel MH optimization techniques. The hydrostatic thrust problem contains 7 constraints. In this test function, R is the bearing step radius, Q is the flow rate, R_0 is the recess radius, P_{max}= 1000, h_{min}= 0.001, ΔT_{max} = 50, ws = 45,804.99 kg, and γ= 0.0307 (Yildiz et al. 2021). The main aim of this optimization problem is to minimize the power loss during the operation of a hydrostatic plain bearing considering all constraints and is expressed as follows (Şahin et al. 2019):

$$f_{12} = \left(\frac{QXP_0}{0.7}\right) + E_f.$$

The various constraints used for this objective function are

$g_1(x) = w_s - w \leq 0,$

$g_2(x) = P_0 - P_{max} \leq 0,$

$g_3(x) = \Delta T - \Delta T_{max} \leq 0,$

$g_4(x) = h_{min} - h \leq 0,$

$g_5(x) = R_0 - R \leq 0,$

$$g_6(x) = \frac{\gamma}{gXP_0} X\left(\frac{Q}{2\pi XRXh}\right) - 0.001 \leq 0,$$

$$g_7(x) = \left(\frac{W}{\pi(R^2 - R_0^2)}\right) - 5000 \leq 0.$$

with boundary condition

$1 \leq R, R_0, Q \leq 16.$

5.13 Tubular column

The literature shows that the tubular column is used to evaluate the performance of the novel MH optimization techniques. In this test function, there are two variables {d, t}, where d is the mean diameter, and t is the thickness. The various constants are compressive load (P = 2500 kg), length of the column (L = 250 cm), σ_y is the yield stress, ρ is the weight density, E is the modulus of elasticity, and W is the weight (kg force) (Meng et al. 2017). The mathematical modelling of tubular column real-world challenge is given as

$f_{13} = 5W + 2d,$

$= 5\rho l\pi dt + 2d,$

= 9.8dt + 2d.

The various constraints used for this objective function are

$$g_1(x) = \frac{P}{\pi dt\sigma_y} - 1 \leq 0,$$

$$g_2(x) = \frac{8PL^2}{\pi^3 Edt(d^2 + t^2)} - 1 \leq 0.$$

The value of the d and t lies between $2 \ll d \ll 14, 0.2 \ll t \ll 0.8$.

5.14 I-beam design

In the literature, it is evident that the I-beam is utilized to evaluate the performance of the novel MH optimization techniques. In this test function, there are 4 variables {x_1, x_2, x_3, x_4} = {b, h, tw, tf}, and 1 constraint (Yu et al. 2020b). The mathematical representation of this test function is given as

$$f_{14} = \frac{5000}{\frac{x_{3(x_2-2x_4)^3}}{12} + \frac{x_4^3 x_1}{6} + 2x_4^3 x_1\left(\frac{x_2-x_4}{2}\right)^2}.$$

The various constraints used for this objective function are

$$g_1(x) = 2x_1x_3 + x_3(x_2 - 2x_4) \leq 0.$$

with boundary conditions

$$10 \leq x_1 \leq 50, 10 \leq x_2 \leq 80, 0.9 \leq x_3 \leq 5, 0.9 \leq x_4 \leq 5.$$

5.15 Crashworthiness design

In the literature, it is evident that crashworthiness is used to evaluate the performance of the novel MH optimization techniques. The key objective of the challenge is to minimize the vehicle's overall weight by optimizing 11 variables. Additionally, the vehicle's safety performance needs to adhere to the specified standards. In this test function, weight is minimized using 10 constraints as well as 11 design variables (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, as well as x_{11}). The car crashworthiness design challenge can be mathematically modelled as follows (Che and He 2022):

$$f_{15} = 4.90x_1 + 6.67x_2 + 6.98x_3 + 4.01x_4 + 1.78x_5 + 2.73x_7 + 1.98.$$

The various constraints used for this objective function are

$$g_1(x) = 1.16 - 0.3717x_2x_4 - 0.00931x_2x_{10} - 0.484x_3x_9 + 0.01343x_6x_{10} \leq 1,$$

$$g_2(x) = 0.261 - 0.0159x_1x_2 - 0.188x_1x_8 - 0.019x_2x_7 + 0.0144x_3x_5 + 0.0008757x_5x_{10} + 0.080405x_6x_9 + 0.00139x_8x_{11} + 0.00001575x_{10}x_{11} \leq 0.32,$$

$$g_3(x) = 0.214 + 0.00817x_5 - 0.131x_1x_8 - 0.0704x_1x_9 + 0.03099x_2x_6 - 0.018x_2x_7 + 0.0208x_3x_8 + 0.121x_3x_9 - 0.00364x_5x_6 + 0.0007715x_5x_{10} + 0.00121x_8x_{11} \leq 0.32,$$

$$g_4(x) = 0.074 - 0.061x_2 - 0.163x_3x_8 + 0.001232x_3x_{10} - 0.166x_7x_9 + 0.227x_2^2 \leq 0.32,$$

$$g_5(x) = 28.98 + 3.818x_3 - 4.2x_1x_2 + 0.0207x_5x_{10} + 6.63x_6x_9 - 7.7x_7x_8 + 0.32x_9x_{10} \leq 32,$$

$$g_6(x) = 33.86 + 2.95x_3 + 0.1972x_{10} - 5.057x_1x_2 - 11x_2x_8 - 0.0215x_5x_{10} - 9.98x_7x_8 + 22x_8x_9 \leq 32,$$

$$g_7(x) = 46.36 - 9.9x_2 - 12.9x_1x_8 + 0.1107x_3x_{10} \leq 32,$$

$$g_8(x) = 4.72 - 0.5x_4 - 0.19x_2x_3 - 0.0122x_4x_{10} + 0.009325x_6x_{10} + 0.000191x_{11}^2 \leq 4,$$

$$g_9(x) = 10.58 - 0.674x_1x_2 - 1.95x_2x_8 + 0.02054x_3x_{10} - 0.0198x_4x_{10} + 0.028x_6x_{10} \leq 9.9,$$

$$g_{10}(x) = 16.45 - 0.489x_3x_7 - 0.843x_5x_6 + 0.0432x_9x_{10} - 0.0556x_9x_{11} - 0.000786x_{11}^2 \leq 15.7,$$

with boundary conditions:

$$0.5 \leq x_1, x_2, x_3, x_4, x_5, x_6, x_7 \leq 1.5, x_8, x_9 \in \{0.192, 0.345\}, aswellas - 30 \leq x_{10}, x_{11,}x_{12}, x_{13}, x_{14} \leq 30.$$

5.16 Industrial refrigeration system

The literature shows that industrial refrigeration is utilized to evaluate the performance of novel MH optimization techniques. This test function has 15 inequality constraints and 14 variables (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, as well as x_{14}) (Kumar et al. 2020d)(Zamani et al. 2022). The industrial refrigeration system's real-world challenge is presented as follows:

$$f_{16} = 63098.88x_2x_4x_{12} + 5441.5x_2^2x_{12} + 115055.5x_2^{1.664}x_6 + 6172.27x_2^2x_6 + 63098.88x_1x_3x_{11} + 5441.5x_1^2x_{11} + 115055.5x_1^{1.664}x_5 + 6172.27x_1^2x_5 + 140.53x_1x_{11} + 281.29x_3x_{11} + 70.26x_1^2 + 281.29x_1x_3 + 281.29x_3^2 + 14437x_8^{1.8812}x_{12}^{0.3424}x_{10}x_{14}^{-1}x_1^2x_7x_9^{-1} + 20470.2x_{11}^{0.316}x_7^{2.893}x_1^2.$$

The various constraints used for this objective function are

$$g_1(x) = 1.5240x_7^{-1} \leq 1,$$

$$g_2(x) = 1.5240x_8^{-1} \leq 1,$$

$$g_3(x) = 0.07789x_1 - 2x_7^{-1}x_9 - 1 \leq 0,$$

$$g_4(x) = 7.05305x_9^{-1}x_{10}x_1^2x_8^{-1}x_2^{-1}x_1^{-1} - 1 \leq 0,$$

$$g_5(x) = 0.0833x_{14}x_{13}^{-1} \leq 0,$$

$$g_6(x) = 47.136x_{12}x_{10}^{-1}x_2^{0.333} - 1.333x_{13}^{2.1195}x_8 + 62.08x_{10}^{-1}x_8^{0.2}x_{12}^{-1}x_{13}^{2.1194} - 1 \leq 0,$$

$$g_7(x) = 0.04771x_{12}^{0.3424}x_8^{1.8812}x_{10} - 1 \leq 0,$$

$$g_8(x) = 0.0488x_{11}^{0.316}x_7^{1.893}x_9 - 1 \leq 0,$$

$$g_9(x) = 0.0099x_3^{-1}x_1 - 1 \leq 0,$$

$$g_{10}(x) = 0.0193x_4^{-1}x_2 - 1 \leq 0,$$

$$g_{11}(x) = 0.0298x_5^{-1}x_1 - 1 \leq 0,$$

$$g_{12}(x) = 0.056x_6^{-1}x_2 - 1 \leq 0,$$

$$g_{13}(x) = 2x_9^{-1} - 1 \leq 0,$$

$$g_{14}(x) = 2x_{10}^{-1} - 1 \leq 0,$$

$$g_{15}(x) = x_{12}x_{11}^{-1} - 1 \leq 0.$$

with boundary conditions

$0.001 \leq x_i \leq 5$. Where i = 1, 2, 3, 4… 14.

5.17 Gas transmission compressor

Many novel MH optimization techniques are tested with the gas transmission compressor. This mechanical test function has 4 variables (x_1, x_2, x_3, as well as x_4) and 1 constraint (g_1) (Che and He 2022; Sharma et al. 2022b). This challenge can be mathematically described as follows:

$$f_{17} = 8.61X10^5 x_2 x_1^{1/2} x_3^{-2/3} x_4^{-1/2} + 3.69X10^4 x_3 + 7.72X10^8 x_1^{-1} x_2^{0.219} - 765.43X10^6 x_1^{-1}.$$

The various constraints used for this objective function are

$$g_1(x) = x_2^{-2} x_4 + x_2^{-2} - 1 \le 0,$$

with boundary conditions

$$20 \le x_1 \le 50, 1 \le x_2 \le 10, 20 \le x_3 \le 45, 0.1 \le x_4 \le 60.$$

5.18 Corrugated bulkhead

Many novel MH optimization techniques are tested with the corrugated bulkhead. This test function has four variables $\{b, h, l, t\} = \{x_1, x_2, x_3, x_4\}$, where length ($l$), width ($b$), thickness ($t$), and depth ($h$), with 6 constraints. This problem is a minimization problem. The mathematical formulation of this test function can be defined as follows (Gandomi et al. 2013; Azizi et al. 2021): $f_{18} = \frac{5.8884 x_4 (x_1 + x_3)}{x_1 + \sqrt{(x_3^2 - x_2^2)}}$.

The various constraints used for this objective function are

$$g_1 = x_2 x_4 \left(0.4x_1 + \frac{x_3}{6}\right) - 8.94\left(x_1 + \sqrt{(x_3^2 - x_2^2)}\right) \ge 0,$$

$$g_2 = x_4 x_2^2 \left(0.2x_1 + \frac{x_3}{12}\right) - 2.2(8.94\left(x_1 + \sqrt{(x_3^2 - x_2^2)}\right))^{4/3} \ge 0,$$

$$g_3 = x_4 - 0.0156x_1 - 0.15 \ge 0,$$

$$g_4 = x_4 - 0.0156x_3 - 0.15 \ge 0,$$

$$g_5 = x_4 - 0.15 \ge 0,$$

$$g_6 = x_3 - x_2 \ge 0,$$

with boundary conditions

$$0 \le x_1, x_2, x_3 \le 100; 0 \le x_4 \le 5.$$

5.19 Piston lever

Many novel MH optimization techniques are tested with the piston lever. This test function has four variables length $\{H, B, D, X\} = \{x_1, x_2, x_3, x_4\}$ with 4 constraints. The piston is lifted (0 to 45 degrees). This test function is defined as follows (Gandomi et al. 2013):

$$min f_{19} = \frac{1}{4}\pi x_3^2 (L_2 - L_1).$$

The various constraints used for this objective function are

$$g_1 = QLcos\theta - RFat\theta = 45^o,$$

$$g_2 = Q(L - x_4) - M_{max} \le 0,$$

$$g_3 = 1.2(L_2 - L_1) - L_1 \le 0,$$

$$g_4 = x_3/2 - x_2 \le 0,$$

where

$$F = \pi P x_3^2/4,$$

$$R = \frac{|-x_4(x_4 sin\theta + x_1) + x_1(x_2 - x_4 cos\theta)|}{\sqrt{(x_4 - x_2)^2 + x_1^2}},$$

$$L_1 = \sqrt{(x_4 - x_2)^2 + x_1^2},$$

$$L_2 = \sqrt{(x_4 \sin 45 + x_1)^2 + (x_2 - x_4 cos 45)^2},$$

P = 10,000 lbs,
L = 240.

$$\theta = 0\, to\, 45^0.$$

5.20 Step-cone pulley

The primary objective of the step cone pulley test function is to minimize the weight of the step cone pulley using different constraints. Therefore, this test function has a single objective function, 5 variables, and 11 constraints. These variables consist of four diameters, each representing a step of the pulley, and one variable representing the width of the pulley (Naruei and Keynia 2021; Yusof and Kamaruddin 2021). The problem can be mathematically formulated in the following manner:

$$f_{20} = \rho w \left[d_1^2 \left(11 + \left(\frac{N_1}{N}\right)^2\right) + d_2^2 \left(1 + \left(\frac{N_2}{N}\right)^2\right) + + d_3^2 \left(1 + \left(\frac{N_3}{N}\right)^2\right) + + d_4^2 \left(1 + \left(\frac{N_4}{N}\right)^2\right) \right].$$

The various constraints used for this objective function are

$$g_1 = C_1 - C_2,$$
$$g_2 = C_1 - C_3,$$
$$g_3 = C_1 - C_4,$$
$$g_{(4,5,6,7),i=1,2,3,4} = R_i \geq 2,$$
$$g_{(8,9,10,11),i=1,2,3,4} = P_i + (0.75 * 745.6998) \geq 0,$$

where,

$$\rho = \frac{7200kg}{m^3}, a = 3m, u = 0.35, s = 1.75MPa, t = 8mm, \text{ and } i = 1,2,3,4.$$

$$C_i = \frac{\pi d_i}{2}\left[1 + \frac{N_i}{N}\right] + \frac{\left[\frac{N_i}{N} - 1\right]^2}{4a} + 2a,$$
$$R_i = exp\left\{\mu\left[\pi - 2sin^{-1}\left[\left(\frac{N_i}{N} - 1\right)\frac{d_i}{2a}\right]\right]\right\},$$
$$R_i = stw\{1 - exp\left\{-\mu\left[\pi - 2sin^{-1}\left[\left(\frac{N_i}{N} - 1\right)\frac{d_i}{2a}\right]\right]\right\}\frac{\pi d_i N_i}{60}.$$

5.21 Planetary gear train

This test function has 9 objective functions (f_{21_1}, f_{21_2}, f_{21_3}, f_{21_4}, f_{21_5}, f_{21_6}, f_{21_7}, f_{21_8}, as well as f_{21_9}), 10 design variables, and 8 constraints. The primary objective of this test function is to minimize the gear ratio using different constraints. The mathematical formulation of this test function is shown as follows (Rosic et al. 2011):

$$f_{21_1} = \frac{m_n z_a}{cos\beta}\left(1 + u^H_{a-g}\right)\frac{cos\alpha_t}{cos\alpha_{wt}},$$
$$f_{21_2} = \frac{1 - n^H_{ab}u^H_{ab}}{1 - u^H_{ab}},$$
$$f_{21_3} = \varepsilon_{\alpha(a-g)}(x),$$
$$f_{21_4} = \alpha_{w(a-g)}(x),$$
$$f_{21_5} = S_{F(a)}(x) = \frac{[\sigma_F]_{M(a)}}{\sigma_{F(a)}},$$
$$f_{21_6} = S_{H(a)}(x) = \frac{[\sigma_H]_{M(a)}}{\sigma_{H(a)}},$$
$$f_{21_7} = V(x),$$
$$f_{21_8} = S_{F(b)}(x) = \frac{[\sigma_F]_{M(b)}}{\sigma_{F(b)}},$$
$$f_{21_9} = D_{out}(x).$$

The various constraints used for this objective function are

$$g_{(1,2,3)} = \frac{[\sigma_F]_{M(a,g,b)}}{\sigma_{F(a,g,b)}} - S_F > 0,$$
$$g_4 = \frac{[\sigma_H]_{M(a,g)}}{\sigma_H} - S_H > 0,$$
$$g_5 = \frac{[\sigma_H]_{M(g,b)}}{\sigma_H} - S_H > 0,$$
$$g_6 = \Delta x > 0,$$
$$g_7 = 2\text{asin}\left(\frac{\pi}{n_w}\right) - f - d_{a-g} \geq 0,$$
$$g_8 = \frac{z_a z_b}{n_w D(z_g, z_b)} - INT,$$

where

$$\varepsilon_\alpha = \frac{0.5\left(\sqrt{d^2_{a(a)} - d^2_{a(b)}} + \sqrt{d^2_{a(g)} - d^2_{b(g)}}\right) - asin\alpha_{wt}}{\pi m_t cos\alpha_t},$$
$$inv\alpha_{wt} = 2\frac{x_a + x_g}{z_a + z_g}tan\alpha_n + inv\alpha_t,$$
$$\sigma_{F(a)} = \frac{F_t}{bm_n}Y_{Fa}Y_{Sa}Y_\epsilon Y_\beta K_A K_v K_\alpha K_\beta,$$
$$[\sigma_F]_M = \sigma_{Flim}Y_{ST}Y_{NT}Y_{\delta relT}Y_{RrelT}Y_X,$$
$$S_{F(a)} = \frac{[\sigma_F]_{(a)}}{\sigma_{F(a)}} \geq S_{Fmin},$$
$$\sigma_H = Z_H Z_E Z_\varepsilon Z_\beta\sqrt{\frac{F_t}{bd_{(a)}}\frac{u+1}{u}K_A K_v K_{H\alpha}K_{H\beta}},$$
$$\sigma_{Ha} = \sigma_{Hg} < [\sigma_H]_M = \min\{[\sigma_H]_{(a)}, \left[\sigma_H\right]_{(g)}\},$$
$$[\sigma_H]_M = \sigma_{H\lim,}Z_{NT}(Z_L Z_v Z_R)Z_v Z_X,$$
$$S_{Ha,g} = \frac{[\sigma_H]_{(a,g)}}{\sigma_H} \geq S_{Hmin},$$

where f_{21_1} represents centre distance, f_{21_2} represents efficiency, f_{21_3} denotes the contact ratio, f_{21_4} indicates pressure angle, f_{21_5} represents the safety factor for bending stress, f_{21_6} indicates the safety factor for contact stress, f_{21_7} represents volume of material used for gears, f_{21_8} denoted as safety factor for bending stress for ring gear, as well as f_{21_9} presents outer diameter of planetary gear train. Also, the transverse contact ratio is ε_α, the function of pressure angle is $inv\alpha_{wt}$, tooth root stress for the sun gear is $\sigma_{F(a)}$, critical root stress is $[\sigma_F]_M$, the factor of safety from breakage is $S_{F(a)}$, effective contact stress is σ_H, for the sun gear planet gear σ_{Ha}, critical contact stress $[\sigma_H]_M$, as well as the factor of safety from pitting $S_{Ha,g}$. Furthermore, it is

essential to include the functional limitations in the evaluation, such as $g_{(1,2,3)}$ presents factors of safety from bending, $g_{4,}andg_5$ indicates a factor of safety from pitting, g_6 represents radial interference, g_7 mean space requirements, as well as g_8 presents condition for assembly.

5.22 Optimal power flow for minimizing fuel cost, voltage deviation, and active and reactive power loss

This test function has 4 objective functions (f_{22_1},f_{22_2}, f_{22_3}, as well as f_{22_4}), 2 equality constraint (g_1, as well as g_2), and 4 inequality constraint (V_k, δ_k, $P_{g,k}$, as well as $Q_{g,k}$). The main objective of this problem is to minimize fuel cost, voltage deviation reactive and active power loss. This test function can be stated as (Biswas et al. 2020)

$$f_{22_1} = \sum_{i=1}^{N}\left(P_{g,i} - P_{l,i}\right),$$

$$f_{22_2} = \sum_{i=1}^{N}\left(Q_{g,i} - Q_{l,i}\right),$$

$$f_{22_3} = \sum_{i=1}^{N}(a_i + b_i P_{g,i} + c_i P_{g,i}^2),$$

$$f_{22_4} = \sum_{i=1}^{N}(1 - |V_i|).$$

The various equality constraints used for this objective function are

$$g_1 = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j(G_{i,j}\cos\left(\delta_i - \delta_j\right) + B_{i,j}\sin\left(\delta_i - \delta_j\right)),$$

$$g_2 = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j(G_{i,j}\sin\left(\delta_i - \delta_j\right) - B_{i,j}\cos\left(\delta_i - \delta_j\right)),$$

with boundary conditions

$$V_{min} \le V_k \le V_{max},$$

$$\delta_{min} \le \delta_k \le \delta_{max},$$

$$P_{min} \le P_{g,k} \le P_{max},$$

$$Q_{min} \le Q_{g,k} \le Q_{max},$$

where k = 1, 2, 3, 4,…..N, i = 1, 2, 3, …….N, cost coefficient of i-th bus (a_i,b_i, andc_i), the total number of buses in the system (N), and $P_i = \left(P_{g,i} - P_{l,i}\right)$, and $Q_i = \left(Q_{g,i} - Q_{l,i}\right)$, represents the active and reactive injected power.

5.23 Vibrating platform

This test function has 2 objective functions (f_{23_1}, as well as f_{23_2}), 5 variables, and 5 constraints. The motor offers a vibratory disturbance onto the beam, which has dimensions of length L and width b, and is symmetrical about its mid-plane. Variables d_1 and d_2 represent the positions of the contact points between materials 1 and 2, and materials 2 and 3, respectively. The variable d_3 is used to identify the location of the top of the beam. The parameters of interest for each material type include the mass density (ρ), Young's modulus of elasticity (E), and cost per unit volume (c). The mathematical model of this test function is as Narayanan and Azarm (1999)

$$f_{23_1} = -\frac{\pi}{2L^2}\sqrt{\frac{EI}{\mu}},$$

$$f_{23_2} = 2bL(c_1 d_1 + c_2(d_2 - d_1) + c_3(d_3 - d_2));$$

subject to.

$$g_1 = \mu L - 2800 \le 0,$$

$$g_2 = d_1 - d_2 \le 0,$$

$$g_3 = d_2 - d_1 - 0.15 \le 0,$$

$$g_4 = d_2 - d_3 \le 0,$$

$$g_5 = d_3 - d_2 - 0.01 \le 0;$$

with bounds

$$0.05 \le d_1 \le 0.5,$$

$$0.2 \le d_2 \le 0.5,$$

$$0.2 \le d_3 \le 0.6,$$

$$0.35 \le b \le 0.5,$$

$$3 \le L \le 6,$$

where $E_1 = 1.6$, $E_2 = 70$, $E_3 = 200$, $c_1 = 500$, $c_2 = 1500$, $c_3 = 800$, $\rho_1 = 100$, $\rho_2 = 2700$, $\rho_3 = 7780$.

And

$$EI = \frac{2b}{3}\left(E_1 d_1^3 + E_2\left(d_2^3 - d_1^3\right) + E_3\left(d_3^3 - d_2^3\right)\right),$$

$$\mu = 2b(\rho_1 d_1 + \rho_2(d_2 - d_1) + \rho_3(d_3 - d_2)).$$

5.24 Four-bar plane truss

The 4-bar plane truss test function has 2 objective functions and 4 variables. Therefore, this test function can be formulated as a (Coello Coello and Pulido 2005)

$$f_{24_1} = L(2x_1 + \sqrt{2x_2} + \sqrt{x_3} + x_4,$$

$$f_{24_2} = \frac{FL}{E}\left[\frac{2}{x_2} + \frac{2\sqrt{2}}{x_2} - \frac{2\sqrt{2}}{x_3} + \frac{2\sqrt{2}}{x_4}\right],$$

with boundary conditions

$(F/\sigma) \le x_1, x_4 \le 3(F/\sigma),$

$\sqrt{2}(F/\sigma) \le x_2, x_3 \le 3(F/\sigma),$

where σ =10 kN/cm^3; F = 10 kN, L = 200 cm, and E = 2 × 105 kN/cm^2.

5.25 Two-bar plane truss

This test function has 2 objective functions (f_{25_1}, as well as f_{25_2}), 3 variables (x_1, x_2, as well as x_3), and 3 constraints (g_1, g_2, and g_3). The mathematical model of this test function is as (Coello Coello and Pulido 2005)

$$f_{25_1} = x_1\sqrt{16 + x_3^2} + x_2\sqrt{1 + x_3^2},$$

$$f_{25_2} = \frac{20\sqrt{16 + x_3^2}}{x_3 x_1}.$$

The various constraints used for this objective function are

$$g_1 = x_1\sqrt{16 + x_3^2} + x_2\sqrt{1 + x_3^2} - 0.1 \le 0,$$

$$g_2 = \frac{20\sqrt{16 + x_3^2}}{x_3 x_1} - 10^5 \le 0,$$

$$g_3 = \frac{80\sqrt{1 + x_3^2}}{x_3 x_2} - 10^5 \le 0,$$

with bounds

$10^{-5} \le x_1, x_2 \le 100,$

$1 \le x_3 \le 3.$

5.26 Bulk carrier

This test function has 3 functions (f_{26_1}, f_{26_2}, as well as f_{26_3}), 6 variables (L, B, D, T, V_k, as well as C_B), and 9 constraints (g_1, g_2, g_3, g_4, g_5, g_6, g_7, g_8, as well as g_9). The mathematical model of this test function can be expressed by (Parsons and Scott 2004)

$$f_{26_1} = \frac{C_v + C_r + C_c}{ac},$$

$$f_{26_2} = ls,$$

$$f_{26_3} = -ac.$$

The various constraints used for this objective function are

$$g_1 = 6 - \frac{L}{B},$$

$$g_2 = \frac{L}{D} - 15,$$

$$g_3 = \frac{L}{T} - 19,$$

$$g_4 = T - 0.45D_{wt}^{0.31},$$

$$g_5 = T - 0.7D - 0.7,$$

$$g_6 = F_n - 0.32,$$

$$g_7 = 0.07B + (1 + 0.52D) - 0.53T - \frac{(0.085 * C_B - 0.002) * B^2}{(TC_B)},$$

$$g_8 = 3000 - D_{wt},$$

$$g_9 = D_{wt} - 500000,$$

with bounds

$150 \le L \le 274.32,$

$20 \le B \le 32.31,$

$13 \le D \le 25,$

$10 \le T \le 11.71,$

$14 \le V_k \le 18,$

$0.63 \le C_B \le 0.75,$

where

$$a = 4977.06C_B^2 - 8105.61C_B + 4456.51,$$

$$b = -10847.2C_B^2 + 12817C_B - 6960.32,$$

$$f_n = \frac{0.5144}{(9.8065L)^{0.5}},$$

$$P = \frac{(1.025LBTC_B)^{0.67}V_k^3}{a + bF_n},$$

$$W_s = 0.34L^{1.7}B^{0.6}D^{0.4}C_B^{0.5},$$

$$W_o = L^{0.8}B^{0.6}D^{0.3}C_B^{0.1},$$

$$W_m = 0.17P^{0.9},$$

$$l_s = W_s + W_0 + W_m,$$

$$D_{wt} = 1.025LBTC_B - l_s,$$

$$F_c = 4.56X10^{-5}P + 0.2,$$

$$D_{cwt} = D_{wt} - F_c\left[\frac{5000V_k}{24} + 5\right] - 2D_{wt}^{0.5},$$

$$R_{trp} = \frac{350}{\frac{5000*V_k}{24} + 2\left(\frac{D_{cwt}}{8000} + 0.5\right)},$$

$$s_d = \frac{5000 * V_k}{24},$$

$$ac = D_{cwt} R_{trp},$$

$$C_c = 0.26(2000W_s^{0.85} + 3500W_o + 2400P^{0.8},$$

$$c_r = 40000D_{wt}^{0.3},$$

$$C_v = \left(105F_c S_d + 6.3D_{wt}^{0.8}\right)R_{trp}.$$

5.27 Front rail

This test function has 2 objective functions (f_{27_1}, as well as f_{27_2}), 3 variables (w, hh, as well as t), and 3 constraints (g_1, g_2, as well as g_3). In this study, the design variables under consideration are the width (w), height (hh), and thickness (t) of the cross-section. The variables E_a and F_a represent the average value of energy absorption as well as the maximal collision force, respectively. The mathematical model of this test function is as (Fan et al. 2018)

$$f_{27_1} = \frac{E_a}{E},$$

$$f_{27_2} = \frac{F_a}{F}.$$

The various constraints used for this objective function are

$$g_1 = -(146 - hh)(hh - 146),$$

$$g_2 = -(66 - w)(w - 58),$$

$$g_3 = -(2.2 - t)(t - 1.4),$$

with bounds.

$$56 \le w \le 68,$$

$$136 \le hh \le 146,$$

$$1.4 \le t \le 2.2,$$

where $E_a = 14496.5$, $F_a = 234.9$.

And,

$$E = 958.656w - 70973.4 + 614.173hh - 3.827whh + 57.023wt + 63.274hht - 3.582w^2 - 1.4842hh.$$

5.28 Optimal droop setting for minimizing voltage deviation and active power loss

The main objective of this problem is to minimize voltage deviation and active power loss by tuning the droop parameters. This test function has 2 objective functions (f_{28_1}, as well as f_{28_2}), 7 variables ($V_{r,k}, V_{m,k}$, P_k, Q_k, C_{pk}, $C_{qk} aswellasw$), and 4 equality constraint (g_1, g_2, g_3 as well as g_4). This test function can be calculated below (Kumar et al. 2019b)

$$f_{28_1} = \sum_{i=1}^{N} P_i,$$

$$f_{28_2} = \sum_{i=1}^{N} \left(1 - |V_i|^2\right).$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N} (G_{k,i}V_{r,i} - B_{ki}V_{m,i}) - \frac{P_k V_{r,k} + Q_k V_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N} (B_{ki}V_{r,i} + G_{k,i}V_{m,i}) - \frac{P_k V_{m,k} - Q_k V_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - C_{pk}\left(w_k^* - w\right) + P_{l,k},$$

$$g_4 = Q_k - C_{qk}\left(V_k^* - \sqrt{(V_{r,k})^2 + (V_{m,k})^2}\right) + Q_{l,k},$$

with boundary conditions

$$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$$

$$P_{min} \le P_k \le P_{max},$$

$$Q_{min} \le Q_k \le Q_{max},$$

$$C_{pmin,k} \le C_{pk} \le C_{pmax,k},$$

$$C_{qmin,k} \le C_{qk} \le C_{qmax,k},$$

$$w_{min} \le w \le w_{max},$$

where k = 1, 2, 3, 4, …..N, P_i and Q_i represents active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.29 Optimal power flow for minimizing fuel cost, and active and reactive power loss

The main objective of this problem is to minimize fuel cost, as well as reactive and active power loss by tuning the droop parameters. This test function has 3 objective functions (f_{29_1}, f_{29_2} as well as f_{29_3}), 4 variables (V_k, δ_k, $P_{g,k}$, as well as $Q_{g,k}$), and 2 equality constraint (g_1 as well as g_2). The mathematical model of this test function can be calculated as below (Biswas et al. 2020)

$$f_{29_1} = \sum_{i=1}^{N} \left(P_{g,i} - P_{l,i}\right),$$

$$f_{29_2} = \sum_{i=1}^{N} \left(Q_{g,i} - Q_{l,i}\right),$$

$$f_{29_3} = \sum_{i=1}^{N}\left(a_i + b_i P_{g,i} + c_i P_{g,i}^2\right).,$$

The various equality constraints used for this objective function are

$$g_1 = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\cos(\delta_i - \delta_j) + B_{i,j}\sin(\delta_i - \delta_j)),$$

$$g_2 = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\sin(\delta_i - \delta_j) - B_{i,j}\cos(\delta_i - \delta_j)),$$

with boundary conditions

$$V_{min} \le V_k \le V_{max},$$

$$\delta_{min} \le \delta_k \le \delta_{max},$$

$$P_{min} \le P_{g,k} \le P_{max},$$

$$Q_{min} \le Q_{g,k} \le Q_{max},$$

where k = 1, 2, 3, 4,…..N, i = 1, 2, 3, …….N, cost coefficient of ith bus (a_i, b_i, and c_i), the total number of buses in the system (N), and $P_i = \left(P_{g,i} - P_{l,i}\right)$ and $Q_i = \left(Q_{g,i} - Q_{l,i}\right)$ represents the active and reactive injected power.

5.30 Power distribution system planning

This test function has 2 objective functions (f_{30_1}, as well as f_{30_2}), and 1 equality constraint (g_1). The mathematical model of this test function can be expressed by Rivas-Dávalos and Irving (2005)

$$f_{30_1} = \sum_{i=1}^{6}\left(a_i + b_i x_i + c_i x_i^2\right),$$

$$f_{30_2} = \sum_{i=1}^{6}\left(\alpha_i + \beta_i x_i + \gamma_i x_i^2\right).$$

Table 8 Values of a_i, b_i, c_i, α_i, β_i, and γ_i

i	a_i	b_i	c_i	α_i	β_i	γ_i
1	756.7988	38.5390	0.15247	13.8593	0.32767	0.00419
2	451.3251	46.1591	0.10587	13.8593	0.32767	0.00419
3	1243.5311	38.3055	0.03546	40.2669	− 0.54551	0.00683
4	1049.9977	40.3965	0.02803	40.2669	− 0.54551	0.00683
5	1356.6592	38.2704	0.01799	42.8955	− 0.51116	0.00461
6	1658.5696	36.3278	0.02111	42.8955	− 0.51116	0.00461

Table 9 Values of B_{ij}

B_{ij}	1	2	3	4	5	6
1	140	17	15	19	26	22
2	17	60	13	16	15	20
3	15	13	65	17	24	19
4	19	16	17	71	30	25
5	26	15	24	30	69	32
6	22	20	19	25	32	85

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{6}(x_i - PD - PL),$$

where the values of a_i, b_i, c_i, α_i, β_i, and γ_i and values of B_{ij} are presented in Tables 8 and 9.

PD = 12,000,

$$PL = \sum_{i=1}^{6}\sum_{j=1}^{6}\left(x_i x_j B_{ij} X10^{-6}\right).$$

5.31 Optimal droop setting for minimizing active and reactive power loss

The main objective of this problem is to minimize reactive and active power loss by tuning the droop parameters. This test function has 2 objective functions (f_{31_1} as well as f_{31_2}), 7 variables ($V_{r,k}$, $V_{m,k}$, P_k, Q_k, C_{pk}, C_{qk}, as well as w), and 4 equality constraints (g_1, g_2, g_3 as well as g_4). The mathematical model of this test function can be expressed as Kumar et al. (2019a)

$$f_{31_1} = \sum_{i=1}^{N} P_i,$$

$$f_{31_2} = \sum_{i=1}^{N} Q_i.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N}(G_{k,i}V_{r,i} - B_{ki}V_{m,i}) - \frac{P_k V_{r,k} + Q_k V_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N}(B_{ki}V_{r,i} + G_{k,i}V_{m,i}) - \frac{P_k V_{m,k} - Q_k V_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - C_{pk}\left(w_k^* - w\right) + P_{l,k},$$

$$g_4 = Q_k - C_{qk}\left(V_k^* - \sqrt{(V_{r,k})^2 + (V_{m,k})^2}\right) + Q_{l,k},$$

with boundary conditions

$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$

$P_{min} \le P_k \le P_{max},$

$Q_{min} \le Q_k \le Q_{max},$

$C_{pmin,k} \le C_{pk} \le C_{pmax,k},$

$C_{qmin,k} \le C_{qk} \le C_{qmax,k},$

$w_{min} \le w \le w_{max},$

where k = 1, 2, 3, 4, …..N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.32 Optimal power flow for minimizing voltage deviation, and active and reactive power loss

The main objective of this problem is to minimize voltage deviation, as well as reactive and active power loss. This test function has 3 objective functions (f_{32_1}, f_{32_2} as well as f_{32_3}), 5 variables ($V_{r,k}, V_{m,k}, P_k, Q_k, aswellasP_{dg,k}$), and 4 equality constraint (g_1, g_2, g_3 as well as g_4). The mathematical model of this test function can be calculated as Biswas et al. (2020)

$$f_{32_1} = \sum_{i=1}^{N} P_i,$$

$$f_{32_2} = \sum_{i=1}^{N}\left(1 - |V_i|^2\right),$$

$$f_{32_3} = \sum_{i=1}^{N} Q_i.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N}(G_{k,i}V_{r,i} - B_{ki}V_{m,i}) - \frac{P_kV_{r,k} + Q_kV_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N}(B_{ki}V_{r,i} + G_{k,i}V_{m,i}) - \frac{P_kV_{m,k} - Q_kV_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - P_{dg,k} + P_{l,k},$$

$$g_4 = Q_k + Q_{l,k},$$

with boundary conditions

$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$

$P_{min} \le P_k \le P_{max},$

$Q_{min} \le Q_k \le Q_{max},$

$P_{min,dg} \le P_{dg,k} \le P_{max,dg}.$

where k = 1, 2, 3, 4, …..,N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.33 Optimal sizing of single-phase distribution generation with reactive power support for minimizing active and reactive power loss

This test function has three objective functions with various constraints. This test function has 3 objective functions (f_{33_1}, f_{33_2} as well as f_{33_3}), 5 variables ($V_{r,k}, V_{m,k}, P_k, Q_k, aswellasP_{dg,k}$), and 4 equality constraint (g_1, g_2, g_3 as well as g_4). The mathematical model of this test function can be calculated as Mishra et al. (2019)

$$f_{33_1} = \sum_{i=1}^{N} P_i,$$

$$f_{33_2} = \sum_{i=1}^{N}(1 - |V_i|),$$

$$f_{33_3} = \sum_{i=1}^{N} Q_i.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N}(G_{k,i}V_{r,i} - B_{ki}V_{m,i}) - \frac{P_kV_{r,k} + Q_kV_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N}(B_{ki}V_{r,i} + G_{k,i}V_{m,i}) - \frac{P_kV_{m,k} - Q_kV_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - P_{dg,k} + P_{l,k},$$

$$g_4 = Q_k + Q_{l,k},$$

with boundary conditions

$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$

$P_{min} \le P_k \le P_{max},$

$Q_{min} \le Q_k \le Q_{max},$

$P_{min,dg} \le P_{dg,k} \le P_{max},$

where k = 1, 2, 3, 4, …..N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.34 Optimal droop setting for minimizing voltage deviation, and active and reactive power loss

The main objective of this problem is to minimize voltage deviation, as well as reactive and active power loss by tuning the droop setting. This test function has 3 objective functions (f_{34_1}, f_{34_2} as well as f_{34_3}), 7 variables ($V_{r,k}, V_{m,k}$, P_k, Q_k, C_{pk}, C_{qk}, as well as w), and 4 equality constraints (g_1, g_2, g_3 as well as g_4). The mathematical model of this test function can be calculated as Kumar et al. (2020c)

$$f_{34_1} = \sum_{i=1}^{N} P_i,$$

$$f_{34_2} = \sum_{i=1}^{N} \left(1 - |V_i|^2\right),$$

$$f_{34_3} = \sum_{i=1}^{N} Q_i.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N} (G_{k,i}V_{r,i} - B_{ki}V_{m,i}) - \frac{P_k V_{r,k} + Q_k V_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N} (B_{ki}V_{r,i} + G_{k,i}V_{m,i}) - \frac{P_k V_{m,k} - Q_k V_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - C_{pk}\left(w_k^* - w\right) + P_{l,k},$$

$$g_4 = Q_k - C_{qk}\left(V_k^* - \sqrt{(V_{r,k})^2 + (V_{m,k})^2}\right) + Q_{l,k},$$

with boundary conditions

$$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$$

$$P_{min} \le P_k \le P_{max},$$

$$Q_{min} \le Q_k \le Q_{max},$$

$$C_{pmin,k} \le C_{pk} \le C_{pmax,k},$$

$$C_{qmin,k} \le C_{qk} \le C_{qmax,k},$$

$$w_{min} \le w \le w_{max},$$

where k = 1, 2, 3, 4, …..N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.35 Optimal power flow for minimizing fuel cost, voltage deviation, and active power loss

The main objective of this problem is to minimize fuel cost, voltage deviation, and active power loss. This test function has 3 objective functions (f_{35_1}, f_{35_2} as well as f_{35_3}), 4 variables (V_k, δ_k, $P_{g,k}$, $aswellasQ_{g,k}$), and 2 equality constraint (g_1, as well as g_2). The mathematical model of this test function can be calculated as Biswas et al. (2020)

$$f_{35_1} = \sum_{i=1}^{N} \left(P_{g,i} - P_{l,i}\right),$$

$$f_{35_2} = \sum_{i=1}^{N} (1 - |V_i|),$$

$$f_{35_3} = \sum_{i=1}^{N} \left(a_i + b_i P_{g,i} + c_i P_{g,i}^2\right).$$

The various equality constraints used for this objective function are

$$g_1 = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\cos\left(\delta_i - \delta_j\right) + B_{i,j}\sin\left(\delta_i - \delta_j\right)),$$

$$g_2 = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\sin\left(\delta_i - \delta_j\right) - B_{i,j}\cos\left(\delta_i - \delta_j\right)),$$

with boundary conditions

$$V_{min} \le V_k \le V_{max},$$

$$\delta_{min} \le \delta_k \le \delta_{max},$$

$$P_{min} \le P_{g,k} \le P_{max},$$

$$Q_{min} \le Q_{g,k} \le Q_{max},$$

where k = 1, 2, 3, 4,…..N, i = 1, 2, 3, …….N, cost coefficient of ith bus (a_i, b_i, and c_i), the total number of buses in the system (N), and $P_i = \left(P_{g,i} - P_{l,i}\right)$ and $Q_i = \left(Q_{g,i} - Q_{l,i}\right)$ represents the active and reactive injected power.

5.36 Optimal power flow for minimizing voltage deviation, and active power loss

This test function has 2 objective functions (f_{36_1}, as well as f_{36_2}), 5 variables ($V_{r,k}, V_{m,k}, P_k$, Q_k, $aswellasP_{dg,k}$), and 4 equality constraint (g_1, g_2, g_3 as well as g_4). The main objective of this problem is to minimize voltage deviation

and active power loss. The mathematical model of this test function can be calculated as Biswas et al. (2020)

$$f_{36_1} = \sum_{i=1}^{N} P_i,$$

$$f_{36_2} = \sum_{i=1}^{N} (1 - |V_i|).$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N} (G_{k,i} V_{r,i} - B_{ki} V_{m,i}) - \frac{P_k V_{r,k} + Q_k V_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N} (B_{ki} V_{r,i} + G_{k,i} V_{m,i}) - \frac{P_k V_{m,k} - Q_k V_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - P_{dg,k} + P_{l,k},$$

$$g_4 = Q_k + Q_{l,k},$$

with boundary conditions

$$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$$

$$P_{min} \le P_k \le P_{max},$$

$$Q_{min} \le Q_k \le Q_{max},$$

$$P_{min,dg} \le P_{dg,k} \le P_{max,dg},$$

where k = 1, 2, 3, 4, …..,N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.37 Optimal power flow for minimizing active and reactive power loss

This test function has 2 objective functions (f_{37_1}, as well as f_{37_2}), 5 variables ($V_{r,k}, V_{m,k}, P_k, Q_k, aswellasP_{dg,k}$), and 4 equality constraint (g_1, g_2, g_3 as well as g_4). The main objective of this problem is to minimize reactive and active power loss. The mathematical model of this test function can be expressed as Biswas et al. (2020)

$$f_{37_1} = \sum_{i=1}^{N} P_i,$$

$$f_{37_2} = \sum_{i=1}^{N} Q_i.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N} (G_{k,i} V_{r,i} - B_{ki} V_{m,i}) - \frac{P_k V_{r,k} + Q_k V_{m,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_2 = \sum_{i=1}^{N} (B_{ki} V_{r,i} + G_{k,i} V_{m,i}) - \frac{P_k V_{m,k} - Q_k V_{r,k}}{(V_{r,k})^2 + (V_{m,k})^2},$$

$$g_3 = P_k - P_{dg,k} + P_{l,k},$$

$$g_4 = Q_k + Q_{l,k},$$

with boundary conditions

$$V_{min} \le V_{r,k}, V_{m,k} \le V_{max},$$

$$P_{min} \le P_k \le P_{max},$$

$$Q_{min} \le Q_k \le Q_{max},$$

$$P_{min,dg} \le P_{dg,k} \le P_{max,dg},$$

where k = 1, 2, 3, 4, …..,N, and P_i and Q_i represent active and reactive injected power. W is the operating frequency, and N represents the number of buses in the system.

5.38 Optimal sizing of single-phase distribution generation with reactive power support for phase balancing at main transformer/grid and minimizing active and reactive power loss

The main objective of this problem is to minimize reactive and active power loss. This test function has 2 objective functions (f_{38_1} and f_{38_2}). The mathematical model of this test function can be calculated as Mishra et al. (2019)

$$f_{38_1} = \sum_{i=1}^{N} \sum_{j \in (a,b,c)} P_i^j,$$

$$f_{38_2} = \sum_{i=1}^{N} \sum_{j \in (a,b,c)} Q_i^j.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{s \in (a,b,c)} \sum_{i=1}^{N} \left(G_{k,i}^{js} V_{r,i}^s - B_{ki}^{js} V_{m,i}^s \right) - \frac{P_k^j V_{r,k}^j + Q_k^j V_{m,k}^j}{(V_{r,k}^j)^2 + (V_{m,k}^j)^2},$$

$$g_2 = \sum_{s \in (a,b,c)} \sum_{i=1}^{N} \left(G_{k,i}^{js} V_{r,i}^s + B_{ki}^{js} V_{m,i}^s \right) - \frac{P_k^j V_{m,k}^j - Q_k^j V_{r,k}^j}{(V_{r,k}^j)^2 + (V_{m,k}^j)^2},$$

$$g_3 = P_k^j - P_{dg,k}^j + P_{l,k}^j,$$

$$g_4 = Q_k^j - Q_{dg,k}^j + Q_{l,k}^j,$$

with boundary conditions

$$V_{min} \le V_{r,k}^j, V_{m,k}^j \le V_{max},$$

$$P_{min} \le P_k^j \le P_{max},$$

$Q_{min} \le Q_k^j \le Q_{max}$,

$P_{dg,min} \le P_{dg,k}^j \le P_{dg,max}$,

$Q_{dg,min} \le Q_{dg,k}^j \le Q_{dg,max}$,

And,

$$I_{r,1}^s = \sum_{k\in(a,b,c)} \sum_{i=1}^{N} (G_{1,i}^{s,k} V_{r,i}^k - B_{1,i}^{s,k} V_{m,i}^k),$$

$$I_{m,1}^s = \sum_{k\in(a,b,c)} \sum_{i=1}^{N} (G_{1,i}^{s,k} V_{m,i}^k + B_{1,i}^{s,k} V_{r,i}^k),$$

where k = 1, 2, 3, 4, ….. N, j = (a, b, c), P_i and Q_i represent active and reactive injected power, and N represents the number of buses in system.

5.39 Optimal sizing of single-phase distribution generation with reactive power support for phase balancing at main transformer/grid and minimizing reactive power loss

The main objective of this problem is to minimize reactive power loss. This test function has 2 objective functions (f_{39_1}, and f_{39_2}). The mathematical model of this test function is as Mishra et al. (2019)

$$f_{39_1} = (I_{r,1}^a + I_{r,1}^b + I_{r,1}^c)^2 + (I_{m,1}^a + I_{m,1}^b + I_{m,1}^c)^2 + (I_{r,1}^a - 0.5\left(I_{r,1}^b + I_{r,1}^c\right) - 0.5\sqrt{3}\left(I_{m,1}^b - I_{m,1}^c\right))^2 + (I_{m,1}^a - 0.5\left(I_{m,1}^b + I_{m,1}^c\right) + 0.5\sqrt{3}\left(I_{r,1}^b - I_{r,1}^c\right))^2,$$

$$f_{39_2} = \sum_{i=1}^{N} \sum_{j\in(a,b,c)} Q_i^j.$$

The various equality constraints used for this objective function are

$$g_1 = \sum_{s\in(a,b,c)} \sum_{i=1}^{N} \left(G_{k,i}^{js} V_{r,i}^s - B_{ki}^{js} V_{m,i}^s\right) - \frac{P_k^j V_{r,k}^j + Q_k^j V_{m,k}^j}{(V_{r,k}^j)^2 + (V_{m,k}^j)^2},$$

$$g_2 = \sum_{s\in(a,b,c)} \sum_{i=1}^{N} \left(G_{k,i}^{js} V_{r,i}^s + B_{ki}^{js} V_{m,i}^s\right) - \frac{P_k^j V_{m,k}^j - Q_k^j V_{r,k}^j}{(V_{r,k}^j)^2 + (V_{m,k}^j)^2},$$

$g_3 = P_k^j - P_{dg,k}^j + P_{l,k}^j$,

$g_4 = Q_k^j - Q_{dg,k}^j + Q_{l,k}^j$,

with boundary conditions

$V_{min} \le V_{r,k}^j, V_{m,k}^j \le V_{max}$,

$P_{min} \le P_k^j \le P_{max}$,

$Q_{min} \le Q_k^j \le Q_{max}$,

$P_{dg,min} \le P_{dg,k}^j \le P_{dg,max}$,

$Q_{dg,min} \le Q_{dg,k}^j \le Q_{dg,max}$.

And, $I_{r,1}^s = \sum_{k\in(a,b,c)} \sum_{i=1}^{N} (G_{1,i}^{s,k} V_{r,i}^k - B_{1,i}^{s,k} V_{m,i}^k)$,

$$I_{m,1}^s = \sum_{k\in(a,b,c)} \sum_{i=1}^{N} (G_{1,i}^{s,k} V_{m,i}^k + B_{1,i}^{s,k} V_{r,i}^k),$$

where k = 1, 2, 3, 4, ….., N, j = (a, b, c), P_i and Q_i represent active and reactive injected power, and N represents the number of buses in system.

5.40 Synchronous optimal pulse-width modulation of 13-level inverters

This test function has 2 objective functions (f_{40_1}, as well as f_{40_2}), and 1 equality constraint (g_1). The mathematical model of this test function is as (Edpuganti and Rathore 2017)

$$f_{40_1} = \frac{\sqrt{\sum_k (k^{-4}) \left(\sum_{i=1}^{N} s(i) cos(kx_i)\right)^2}}{6\sqrt{\sum_k (k^{-4})}},$$

$$f_{40_2} = \left[6m - \sum_{i=1}^{N} s(i) cos(x_i)\right]^2.$$

The various constraints used for this objective function are

$g_1 = x_{i+1} - x_i - 10^{-5} > 0$,

where i = 1, 2, 3, 4, ….., N-1, k = 5, 7, 11, 13, …., 97, s = [1, 1, 1, 1, − 1, 1, − 1, − 1, − 1, 1, − 1, − 1], and $N = \left\lfloor \frac{6.f_{s,max}}{f.m} \right\rfloor$.

5.41 Optimal power flow for minimizing fuel cost, and active power loss

The main objective of this problem is to minimize fuel cost and active power loss. This test function has 2 objective functions (f_{41_1}, and f_{41_2}). The mathematical model of this test function are as (Biswas et al. 2020)

$$f_{41_1} = \sum_{i=1}^{N} (P_{g,i} - P_{l,i}),$$

$$f_{41_2} = \sum_{i=1}^{N} \left(a_i + b_i P_{g,i} + c_i P_{g,i}^2\right).$$

The various equality constraints used for this objective function are

$$g_1 = P_{g,i} - P_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\cos(\delta_i - \delta_j) + B_{i,j}\sin(\delta_i - \delta_j)),$$

$$g_2 = Q_{g,i} - Q_{l,i} - V_i \sum_{j=1}^{N} V_j (G_{i,j}\sin(\delta_i - \delta_j) - B_{i,j}\cos(\delta_i - \delta_j)),$$

with boundary conditions

$V_{min} \le V_k \le V_{max}$,

$\delta_{min} \le \delta_k \le \delta_{max}$,

$P_{min} \le P_{g,k} \le P_{max}$,

$Q_{min} \le Q_{g,k} \le Q_{max}$,

where k = 1, 2, 3, 4,…..N, i = 1, 2, 3, …….N, cost coefficient of i-th bus (a_i, b_i, and c_i), the total number of buses in the system (N), and $P_i = (P_{g,i} - P_{l,i})$ and $Q_i = (Q_{g,i} - Q_{l,i})$ represents the active and reactive injected power.

5.42 Synchronous optimal pulse-width modulation of 7-level inverters

This test function has 2 objective functions (f_{42_1}, as well as f_{42_2}), and 1 equality constraint (g_1). The mathematical model of this test function is as (Edpuganti and Rathore 2014)

$$f_{42_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)cos(kx_i)\right)^2}}{3\sqrt{\sum_k (k^{-4})}},$$

$$f_{42_2} = \left[3m - \sum_{i=1}^{N} s(i)cos(x_i)\right]^2.$$

The various constraints used for this objective function are

$g_1 = x_{i+1} - x_i - 10^{-5} > 0$,

where i = 1, 2, 3, 4, ….., N-1, k = 5, 7, 11, 13, …., 97, s = [1, 1, 1, 1, − 1, 1, − 1, − 1, − 1, 1, − 1, − 1] and $N = \left\lfloor \frac{3.f_{s,max}}{f.m} \right\rfloor$.

5.43 Synchronous optimal pulse-width modulation of 11-level inverters

This test function has 2 objective functions (f_{43_1}, as well as f_{43_2}), and 1 equality constraint (g_1). The mathematical model of this test function are as (Edpuganti and Rathore 2017)

$$f_{43_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{5\sqrt{\sum_k (k^{-4})}},$$

$$f_{43_2} = \left[5m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2.$$

The various constraints used for this objective function are

$g_1 = x_{i+1} - x_i - 10^{-5} > 0$,

with boundary conditions

$0 < x_i < \frac{\pi}{2}$.

where i = 1, 2, 3, 4,…..,N-1, k = 5, 7, 11, 13,…0.97, s = [1, 1, 1, 1, − 1, 1, − 1, − 1, − 1, 1, − 1, − 1], and $N = \left\lfloor \frac{5.f_{s,max}}{f.m} \right\rfloor$.

5.44 Synchronous optimal pulse-width modulation of 5-level inverters

This test function has 2 objective functions (f_{44_1}, and f_{44_2}). The mathematical model of this test function are as (Rathore et al. 2010a)

$$f_{44_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{2\sqrt{\sum_k (k^{-4})}},$$

$$f_{44_2} = \left[2m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2.$$

The various constraints used for this objective function are

$g_1 = x_{i+1} - x_i - 10^{-5} > 0$,

with boundary conditions

$0 < x_i < \frac{\pi}{2}$.

Here, i = 1, 2, 3, 4,….., N-1, k = 5, 7, 11, 13,…0.97, s = [1, 1, 1, 1, − 1, 1, − 1, − 1, − 1, 1, − 1, − 1], and $N = \left\lfloor \frac{2.f_{s,max}}{f.m} \right\rfloor$.

5.45 Synchronous optimal pulse-width modulation of 3-level inverters

This test function has 2 objective functions (f_{45_1}, and f_{45_2}). The mathematical model of this test function is as (Rathore et al. 2010b)

$$f_{45_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{4\sqrt{\sum_k (k^{-4})}},$$

$$f_{45_2} = \left[m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2.$$

The constraint used for this objective function is

$$g_1 = x_{i+1} - x_i - 10^{-5} > 0,$$

with boundary conditions

$$0 < x_i < \frac{\pi}{2}.$$

where i = 1, 2, 3, 4,…., N-1, k = 5, 7, 11, 13,…0.97, $s = (-1)^{i+1}$, and $N = \left\lfloor \frac{f_{s,max}}{f.m} \right\rfloor$.

5.46 Synchronous optimal pulse-width modulation of 9-level inverters

This test function has 2 objective functions (f_{46_1}, and f_{46_2}). The mathematical model of this test function is expressed as (Kumar et al. 2021b; Edpuganti et al. 2015)

$$f_{46_1} = \frac{\sqrt{\sum_k (k^{-4})\left(\sum_{i=1}^{N} s(i)\cos(kx_i)\right)^2}}{4\sqrt{\sum_k (k^{-4})}},$$

$$f_{46_2} = \left[4m - \sum_{i=1}^{N} s(i)\cos(x_i)\right]^2.$$

The various constraints used for this objective function are

$$g_1 = x_{i+1} - x_i - 10^{-5} > 0,$$

with boundary conditions

$$0 < x_i < \frac{\pi}{2}.$$

Here, i = 1, 2, 3, 4,…., N-1, k = 5, 7, 11, 13,…0.97, s = [1, 1, 1, 1,- 1, 1, -1, -1, -1, 1, -1, -1], and $N = \left\lfloor \frac{4.f_{s,max}}{f.m} \right\rfloor$.

5.47 Two-reactor problem

The TRP test function has 8 variables (x_1, x_2, x_3, x_4, x_5, x_6, x_7, as well as x_8), 4 inequality constraints (g_1, g_2, g_3, as well as g_4), and 4 equality constraints (g_5, g_6, g_7, as well as g_8). The mathematical model of this test function can be calculated as Cantú et al. (2021) and Kocis and Grossmann (1989)

$$f_{47} = 7.5x_5 + 5.5x_6 + 7x_1 + 6x_2 + 5(x_3 + x_4).$$

The various inequality constraints and equality constraints used for this objective function are

$$g_1 = x_1 - 10x_5 \le 0,$$
$$g_2 = x_2 - 10x_6 \le 0,$$
$$g_3 = x_3 - 20x_5 \le 0,$$
$$g_4 = x_4 - 20x_6 \le 0,$$
$$g_5 = x_5 + x_6 - 1,$$
$$g_6 = x_7 - 0.9x_3(1 - \exp(-0.5x_1)),$$
$$g_7 = x_8 - 0.8x_4(1 - \exp(-0.4x_2)),$$
$$g_8 = x_7 + x_8 - 10,$$

with boundary conditions

$$x_i \ge 0, i = \{1, 2, 3, 4, 5, 6, 7, 8\},$$
$$x_i = \{0, 1\}, i = \{5, 6\}.$$

5.48 Heat exchanger network

This test function has 3 objective functions (f_{48_1}, f_{48_2}, as well as f_{48_3}), 9 variables (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, as well as x_9), and 8 equality constraints (g_1, g_2, g_3, g_4, g_5, g_6, g_7, as well as g_8). The mathematical model of this test function can be calculated as Guillén-Gosálbez (2011)

$$f_{48_1} = 35x_1^{0.6} + 35x_2^{0.6},$$
$$f_{48_2} = 200x_1x_4 - x_3,$$
$$f_{48_3} = 200x_1x_6 - x_5.$$

The various equality constraints used for this objective function are

$$g_1 = 200x_1x_4 - x_3,$$
$$g_2 = 200x_2x_6 - x_5,$$
$$g_3 = x_3 - 10000(x_7 - 100),$$
$$g_4 = x_5 - 10000(300 - x_7),$$
$$g_5 = x_3 - 10000(600 - x_8),$$
$$g_6 = x_5 - 10000(900 - x_9),$$
$$g_7 = x_4\ln(x_8 - 100) - x_4\ln(600 - x_7) - x_8 + x_7 + 500,$$
$$g_8 = x_6\ln(x_9 - x_7) - x_6\ln(600 - x_9) + x_7 + 600,$$

with boundary conditions

$0 \le x_1 \le 10; 0 \le x_2 \le 200; 0 \le x_3 \le 100; 0 \le x_4 \le 200; 1000 \le x_5 \le 2000000; 0 \le x_6 \le 600; 100 \le x_7 \le 600; 100 \le x_8 \le 600; 100 \le x_9 \le 900.$

5.49 Reactor network design

Many novel MH optimization techniques were tested with this test function. This test function has 2 objective functions (f_{49_1}, as well as f_{49_2}), 6 variables (x_1, x_2, x_3, x_4, x_5, as well as x_6), and 5 constraints (g_1, g_2, g_3, g_4, as well as g_5). The mathematical model of this test function can be calculated as Gujarathi et al. (2015, 2016)

$$f_{49_1} = -x_4,$$

$$f_{49_2} = x_5^{0.5} + x_6^{0.5}.$$

The various constraints used for this objective function are

$$g_1 = p_1 x_5 x_2 + x_1 - 1,$$

$$g_2 = p_3 x_5 x_3 + x_3 + x_1 - 1,$$

$$g_3 = p_2 x_6 x_2 - x_1 + x_2,$$

$$g_4 = p_4 x_6 x_4 + x_2 - x_1 + x_4 - x_3,$$

$$g_6 = x_5^{0.5} + x_6^{0.5} \le 4,$$

with boundary conditions

$$0 \le x_4, x_3, x_2, x_1 \le 4, 0.00001 \le x_6, x_5 \le 16,$$

where $p_3 = 0.0391908, p_4 = 0.9p_3, p_1 = 0.09755988,$ and$p_2 = 0.99p_1$

This test function can be re-designed by eliminating the constraints as follows:

$$\max f_{49_3} = \frac{p_2 x_6(1 + p_3 x_5) + p_1 x_5(1 + p_2 x_6)}{(1 + p_1 x_5)(1 + p_2 x_6)(1 + p_3 x_5)(1 + p_4 x_6)}.$$

The various constraints used for this objective function are

$$g_6 = x_5^{0.5} + x_6^{0.5} \le 4,$$

with boundary conditions

$$10^{-5} \le x_5, x_6 \le 16.$$

5.50 Haverly's pooling

This test function has 2 objective functions (f_{50_1}, as well as f_{50_2}), 9 variables (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, as well as x_9), 4 equality constraints (g_1, g_2, g_3, as well as g_4), and 2 inequality constraints (g_5, as well as g_6). The description of this test function is shown as follows (Maier and Whiting 1998):

$$f_{50_1} = -9x_1 - 15x_2 + 6x_3 + 16x_4,$$

$$f_{50_2} = 10(x_5 + x_6).$$

The various equality and inequality constraints used for this objective function are

$$g_1 = x_7 + x_8 - x_4 - x_3,$$

$$g_2 = x_1 - x_5 - x_7,$$

$$g_3 = x_2 - x_6 - x_8,$$

$$g_4 = x_9 x_7 + x_9 x_8 - 3x_3 - x_4,$$

$$g_5 = x_9 x_7 + 2x_5 - 2.5x_1 \le 0,$$

$$g_6 = x_9 x_8 + 2x_6 - 1.5x_2 \le 0,$$

with boundary conditions

$$0 \le x_8, x_6, x_5, x_4, x_3, x_1 \le 100; 0 \le x_9, x_7, x_2 \le 200.$$

5.51 Multi-product batch plant

This test function has 3 objective functions (f_{51_1}, f_{51_2}, as well as f_{51_3}), 4 variables (N_i, V_j, T_{Li} as well as B_j), and 3 inequality constraints (g_1, g_2, as well as g_3). The description of this test function is shown as follows (Angira and Babu 2006):

$$f_{51_1} = \sum_{j=1}^{M} \alpha_j N_j V_j^{\beta_j},$$

$$f_{51_2} = 65\left[\frac{Q_1}{B_1} + \frac{Q_2}{B_2}\right] + 0.08Q_1 + 0.1Q_2,$$

$$f_{51_3} = Q_1 \frac{T_{L1}}{B_1} + Q_2 \frac{T_{L2}}{B_2}.$$

The various constraints used for this objective function are

$$g_1 = S_{ij} B_i - V_j \le 0,$$

$$g_2 = -H + \sum_{i=1}^{N} \frac{Q_i T_{Li}}{B_i} \le 0,$$

$$g_3 = t_{ij} - N_j T_{Li} \le 0.$$

Other values are calculated as follows:

$$T_{Li}^l = max\left[\frac{t_{ij}}{N_j^u}\right],$$

$$T_{Li}^u = max(t_{ij}),$$

$$B_j^l = \frac{Q_i T_{Li}}{H},$$

$$B_j^u = min\left(Q_i, min\left(\frac{V_j^u}{S_{ij}}\right)\right),$$

Table 10 Values of S_{ij} and t_{ij}

S_{ij}		
2	3	4
4	6	3
t_{ij}		
8	20	8
16	4	4

where $H = 6000$, $M = 3$, $N = 2$, $\alpha_j = 250$, $\beta_j = 0.6$, $V_j^l = 250$, $V_j^u = 2500$, and $N_j^u = 3$. In Table 10, given the parameter of S_{ij} and t_{ij}

With boundary conditions

$$1 \le N_i \le N_j^u; V_j^l \le V_j \le V_j^u; T_{Li}^l \le T_{Li} \le T_{Li}^u; B_j^l \le B_j \le B_j^u.$$

5.52 Process synthesis and design problem

The PSAD test function has 2 objective functions (f_{52_1} as well as f_{52_2}), 3 variables (x_1, x_2, as well as x_3) 1 equality constraints (g_1), and 1 inequality constraints (g_2). The description of this test function is shown as follows (Angira and Babu 2006):

$$f_{52_1} = x_2 - x_3 + 2x_1,$$

$$f_{52_2} = x_1x_3 - x_1^2 - x_2.$$

The equality constraints and inequality constraints used for this objective function are

$$g_1 = x_1 - 2\exp(-x_2),$$

$$g_2 = x_3 + x_2 - x_1 \le 0,$$

with boundary conditions

$$0.5 \le x_2, x_1 \le 1.4; x_3 \varepsilon \{0, 1\}.$$

5.53 Parameter estimation of solar PV models

This test function has 5 objective functions and 5 variables. The mathematical model of this test function can be calculated as Sharma and Chinnappa Naidu (2022) and Sharma et al. (2022a)

$$f_{53_1} = I_{sc} - I_{ph} + I_o\left(\exp\left(\frac{I_{sc}R_s}{V_T}\right) - 1\right) + \frac{I_{sc}R_s}{R_p},$$

$$f_{53_2} = I_{ph} - I_o\left(\exp\left(\frac{V_{0c}}{V_T}\right) - 1\right) + \frac{V_{oc}}{R_p},$$

$$f_{53_3} = I_{MP} - I_{ph} + I_o\left(\exp\left(\frac{V_{MP} + I_{MP}R_s}{V_T}\right) - 1\right) + \frac{V_{MP} + I_{MP}R_s}{R_p},$$

$$f_{53_4} = I_{MP} - (V_{MP} - I_{MP}R_s)\left(\frac{I_o}{V_T}\exp\left(\frac{V_{MP} + I_{MP}R_s}{V_T}\right) + \frac{1}{R_p}\right),$$

$$f_{53_5} = \frac{R_s}{R_p} - \frac{I_o}{V_T}\exp\left(\frac{I_{sc}R_s}{V_T}\right)(R_p - R_s),$$

with boundary conditions

$$8 \le I_{ph} \le 8.6,$$

$$7X10^{-10} \le I_o \le 1X10^{-7},$$

$$0.1 \le R_s \le 0.5,$$

$$50 \le R_p \le 5000,$$

$$0.1 \le V_T \le 2,$$

where charge (q) with a value of $1.6021764 \times 10^{-19}$, the Boltzmann constant is denoted as K, the temperature is denoted as T, the number of cells connected in parallel is denoted as Np, the number of cells connected in series is denoted as Ns, and the diode ideality factor denoted as n. Also, photocurrent is denoted as I_{Ph}, reverse saturation current is denoted as I_{o}, shunt resistance is denoted as R_{sh}, R_{s} is series resistance, and I_{Sh} is the shunt resistance current.

5.54 Wind farm layout

The objective function of this test function can be described as below and has 91 inequality constraints (g_i) (Wang et al. 2018). The mathematical model of the wind farm layout is given as

$$f_{54} = \sum_{i=1}^{N} E(P_i).$$

The various constraints used for this objective function are

$$g_i = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \ge 5Rj \ne i, \text{j} = 1,2,3,\ldots, \text{N}$$

with boundary conditions

$$x + R \le x_i \le \bar{x} - R,$$

$$y + R \le y_i \le \bar{y} - R,$$

where

$$E(P_i) = \sum_{n=1}^{h} \xi_n \left[P_r \left(exp(-\left(\frac{v_r}{c_i'\left(\frac{\theta_{n-1}+\theta_n}{2}\right)\right)^{k_{i((\frac{\theta_{n-1}+\theta_n}{2})}} \right) \right.\right.$$
$$\left. -exp(-\left(\frac{v_{co}}{c_i'\left(\frac{\theta_{n-1}+\theta_n}{2}\right)\right)^{k_{i((\frac{\theta_{n-1}+\theta_n}{2})}} \right) \right)$$
$$+ \sum_{j=1}^{s} \left(exp(-\left(\frac{v_{j-1}}{c_i'\left(\frac{\theta_{n-1}+\theta_n}{2}\right)\right)^{k_{i((\frac{\theta_{n-1}+\theta_n}{2})}} \right) \right.$$
$$\left.\left. -exp(-\left(\frac{v_j}{c_i'\left(\frac{\theta_{n-1}+\theta_n}{2}\right)\right)^{k_{i((\frac{\theta_{n-1}+\theta_n}{2})}} \right) \right) \frac{exp((v_{j-1}+v_j)/2)}{\alpha + \beta exp((v_{j-1}+v_j)/2)} \right].$$

And, ξ_n is the frequency of the interval $\{\theta_{n-1}, \theta_n\}$

5.55 10-Bar truss

This test function has 10 variables (x_i) and 3 inequality constraints (g_1, g_2, as well as g_3). The description of this test function is shown as follows (Kumar et al. 2020d):

$$f_{55} = \sum_{i=1}^{10} L_i(x_i)\rho_i A_i.$$

The 3 inequality constraints used for this objective function are

$$g_1 = \frac{7}{w_1(\overline{x})} - 1 \le 0,$$

$$g_2 = \frac{15}{w_2(\overline{x})} - 1 \le 0,$$

$$g_3 = \frac{20}{w_3(\overline{x})} - 1 \le 0,$$

with boundary conditions

$$6.45X10^{-5} \le A_i \le 5X10^{-3},$$

$$\rho = 2770,$$

$$\overline{x} = \{A_1, A_2, A_3, \ldots\ldots\ldots, A_{10} and i = 1, 2, 3, \ldots\ldots, 10.$$

5.56 Beef cattle

This test function has 59 variables and 14 inequality ($g_1 to g_{14}$) and equality constraints (g_{15}). The mathematical model of this test function can be calculated as (Uyeh et al. 2018)

$$f_{56} = \sum_{i=1}^{n} x_i c_i.$$

The 14 inequality and 1 equality constraints used for this objective function are

$$g_1 = b - \sum_{i=1}^{N} x_i CP_i \le 0,$$

$$g_2 = \sum_{i=1}^{N} x_i CP_i - c \le 0,$$

$$g_3 = d - \sum_{i=1}^{N} x_i TDN_i \le 0,$$

$$g_4 = \sum_{i=1}^{N} x_i TDN_i - e \le 0,$$

$$g_5 = f - \sum_{i=1}^{N} x_i Ca_i \le 0,$$

$$g_6 = \sum_{i=1}^{N} x_i Ca_i - g \le 0,$$

$$g_7 = h - \sum_{i=1}^{N} x_i P_i \le 0,$$

$$g_8 = \sum_{i=1}^{N} x_i P_i - j \le 0,$$

$$g_9 = k - \sum_{i=1}^{N} x_i Rhage_i \le 0,$$

$$g_{10} = \sum_{i=1}^{N} x_i Rhage_i - l \le 0,$$

$$g_{11} = m - \sum_{i=1}^{N} x_i MC_i \le 0,$$

$$g_{12} = \sum_{i=1}^{N} x_i MC_i - o \le 0,$$

$$g_{13} = p - \sum_{i=1}^{N} x_i Conc_i \le 0,$$

$$g_{14} = \sum_{i=1}^{N} x_i Conc_i - q \le 0,$$

$$g_{15} = \sum_{i=1}^{N} x_i DMI_i - a,$$

where the constraints are based on the nutrients and weights. Dry Matter Intake (DMI), Crude Protein (CP), Calcium (Ca), Total Digestible Nutrients (TGN), and Roughages (Rhage) in Kg.

5.57 Dairy cattle

This test function has 64 variables and 6 equality constraints (g_1, g_2, g_3, g_4, g_5 as well as g_6). The mathematical

model of this test function can be calculated as (Uyeh et al. 2018)

$$f_{57} = \sum_{i=1}^{n} x_i c_i.$$

The 6 equality constraints used for this objective function are

$$g_1 = \sum_{i=1}^{N} x_i MP_i - r,$$

$$g_2 = \sum_{i=1}^{N} x_i Lys_i - s,$$

$$g_3 = \sum_{i=1}^{N} x_i Ca_i - t,$$

$$g_4 = \sum_{i=1}^{N} x_i P_i - u,$$

$$g_5 = \sum_{i=1}^{N} x_i ME_i - v,$$

$$g_6 = \sum_{i=1}^{N} x_i Met_i - z,$$

where Metabolizable Protein (MP), Lysine (Lys), Calcium Phosphorous (P), Metabolizable Energy (ME), and Methionine (Met) in Kg.

6 Conclusion

This paper is based on the benchmark test function and engineering design problem analysis of research publication, author's contribution, number of publications per year, continents' contribution, and most cited papers which are obtained from the Scopus database. A total of 2998 articles were published in the area of the classical benchmark test function, and most of them were original research articles 66%, followed by conference papers 32%. While Jeng Shyang Pan from Shandong University of Science and Technology has published the highest number of articles (33), and based on the number of publications, the Ministry of Education China has the most active institute (86 publications) in that area. Also, a total of 1665 articles were published in the area of engineering design problems, and most of them were original research articles 60.3%, followed by conference papers 36.15%. While Kalyanmoy Deb from Michigan State University has published the highest number of publications (40), and based on the number of publications, Guangxi University for Nationalities has the most active institute (30 publications) in that area.

The fast-increasing trend in the number of novel optimization studies that have been proposed in the previous couple of decades has shown a lot of interest; however, there is little reason to think that there will be a paradigm shift in the future. Researchers from a range of science and engineering fields have been using MH optimization techniques to tackle challenges in their respective disciplines. In the literature, it is evident that the discussion about the evaluation of new MH has not yet concluded, because everyone uses a different set of benchmark test functions to evaluate their optimization techniques. Therefore, there is a significant need to develop a set of modified benchmark test functions that will help to evaluate the ever-increasing complexity of future MH optimization algorithms. The focal point of this review paper is to present the classical mathematical benchmark test function and real-world engineering design problem for the MH optimization techniques with MATLAB code references, which is carried out from the literature. Also, it illustrates the MATLAB code references for the CEC test suites and BBOB test suites independently. The winners of the CEC test suites are given their reference papers. This review paper provides a baseline for future benchmark test functions for novel research and development.

The future for benchmark test functions is encouraging. As the development of optimization algorithms advances, there will arise a need for novel and more demanding benchmark test functions. Future work based on this paper may include developing some novel parametric benchmark test functions for evaluating the performance of the MH optimization algorithms. Furthermore, it will help the researchers to identify the performance of newly proposed MH optimization techniques. In summation, this study stands as a meticulous exploration of benchmark test functions and engineering design problems, encapsulating pivotal trends, and presents the way for future advancements in the field.

Author contributions PS and SR have conceived the idea and converted it into the manuscript. The concept was proposed by SR for the review article on "Metaheuristic Optimization Algorithms: A Comprehensive Overview and Classification of Benchmark Test Functions" and also supervised the process. PS investigated and collected all the data, and the draft was written and converted into a review article. All authors read and approved the final manuscript.

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